

Risk Dynamics: From Tail Risk to Systemic Contagion

VaR, CVaR, Heavy Tails, Monte Carlo Estimation and Default Cascades

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1 Introduction and Motivation

Financial risk is inherently multidimensional. At the level of an individual institution or portfolio, risk is commonly associated with the distribution of potential losses and, in particular, with the behavior of its extreme tail. At the level of a financial system, however, the consequences of a loss cannot be understood solely from its magnitude or probability. Financial institutions are interconnected through contractual obligations, counterparty exposures, common assets, and other channels of dependence. Consequently, a shock that is initially local may propagate through the network and generate losses far beyond the institution at which it originated.

This distinction motivates two complementary questions. The first concerns the magnitude of extreme losses:

$$\textit{How large can losses become in the tail of the distribution?} \quad (1)$$

The second concerns their propagation:

$$\textit{What happens when an extreme loss enters an interconnected financial system?} \quad (2)$$

The first question belongs naturally to the theory of financial risk measures. Let L denote a random loss, with positive values representing financial losses. For a confidence level $\alpha \in (0, 1)$, the Value at Risk (VaR) is defined as the corresponding loss quantile,

$$\text{VaR}_\alpha(L) = \inf \{ \ell \in \mathbb{R} : \mathbb{P}(L \leq \ell) \geq \alpha \}. \quad (3)$$

VaR therefore identifies a threshold separating ordinary observations from increasingly extreme losses. Nevertheless, knowledge of this threshold alone does not describe the magnitude of losses beyond it. This limitation motivates tail-sensitive measures such as Conditional Value at Risk (CVaR), also closely related to Expected Shortfall (ES). For a continuous loss distribution, one may write

$$\text{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)]. \quad (4)$$

Thus, while VaR determines where the relevant tail begins, CVaR provides information about the severity of losses within that tail. The distinction is particularly important when extreme losses are relevant for risk management and capital allocation. The axiomatic study of risk measures also highlights important differences between quantile-based measures and coherent approaches to risk aggregation [2, 14].

A second difficulty arises from the shape of the loss distribution itself. Models with similar means and variances may imply substantially different probabilities of extreme events. In particular, Gaussian assumptions may underrepresent tail behavior when the underlying loss distribution

is heavy-tailed. A convenient model for illustrating this phenomenon is the standardized Student- t construction

$$L = \mu + \sigma \sqrt{\frac{\nu-2}{\nu}} T_\nu, \quad \nu > 2, \quad (5)$$

where T_ν denotes a Student- t random variable with ν degrees of freedom. Since

$$\text{Var}(T_\nu) = \frac{\nu}{\nu-2}, \quad (6)$$

the standardized variable

$$Z_\nu = \sqrt{\frac{\nu-2}{\nu}} T_\nu \quad (7)$$

has unit variance. This permits a direct comparison between Gaussian and heavy-tailed models at a common variance scale while preserving their different tail structures.

Even when the probability distribution is specified, extreme-risk measurement presents a statistical problem. Given a Monte Carlo sample

$$L_1, \dots, L_N \stackrel{\text{iid}}{\sim} F_L, \quad (8)$$

only a fraction approximately equal to $1 - \alpha$ belongs to the upper tail selected by a confidence level α . Hence, the expected number of observations available for tail estimation is approximately

$$\boxed{\mathbb{E}[N_{\text{tail}}] \approx N(1 - \alpha)}. \quad (9)$$

For example, at $\alpha = 0.99$, a sample of $N = 100$ contains on average only about one observation in the upper one percent of the distribution, whereas $N = 50\,000$ provides approximately 500. Estimating extreme risk is therefore not only a problem of choosing an appropriate risk measure or probability distribution; it is also a problem of information in the tail.

These considerations remain primarily concerned with the risk of an individual portfolio or institution. Systemic risk introduces a fundamentally different dimension: *interdependence*. A financial institution may be capable of absorbing a direct external shock while remaining vulnerable to losses transmitted by its counterparties. Conversely, the default of one institution may alter the financial condition of several others. The relevant object is therefore no longer only the marginal loss distribution of a single entity, but the joint evolution of losses across an interconnected system.

Let a financial network be represented by

$$G = (V, E), \quad (10)$$

where V denotes the set of institutions and E represents financial exposures among them. A weighted exposure matrix

$$W = (w_{ij}) \quad (11)$$

can then encode the losses transmitted through counterparty relationships. Models of financial clearing and contagion demonstrate that network structure can fundamentally affect how shocks are absorbed or amplified through the system. Eisenberg and Noe [7] formulate the clearing of interdependent obligations as a fixed-point problem, while Gai and Kapadia [9] study contagion in financial networks and emphasize the possibility of a “robust-yet-fragile” financial architecture: contagion may be relatively unlikely for many shocks, yet become widespread once it occurs.

This motivates the central conceptual transition developed throughout this study:

$$\boxed{\begin{array}{l} \text{Loss Distribution} \longrightarrow \text{Tail Risk} \longrightarrow \text{Statistical Estimation} \\ \longrightarrow \text{External Shock} \longrightarrow \text{Network Propagation} \longrightarrow \text{Systemic Risk.} \end{array}} \quad (12)$$

The objective is therefore not merely to compare alternative measures of financial risk, nor merely to visualize a default cascade. Rather, the aim is to develop a unified pedagogical and computational perspective connecting the statistical characterization of extreme losses with the mechanisms through which those losses may propagate across a financial network.

The analysis proceeds through four closely related themes. First, VaR and CVaR are examined as complementary descriptions of loss thresholds and tail severity. Second, Gaussian and heavy-tailed models are compared to illustrate the sensitivity of extreme-risk estimates to distributional assumptions. Third, Monte Carlo experiments are used to study the role of sample size and tail information in the estimation of VaR and CVaR. Finally, the analysis moves from individual to systemic risk by introducing interbank exposures, capital buffers, external shocks, and iterative default propagation.

The resulting framework emphasizes three distinct sources of uncertainty:

$$\boxed{\text{distributional uncertainty} + \text{estimation uncertainty} + \text{network amplification.}} \quad (13)$$

These components should not be conflated. Heavy tails concern the probabilistic structure of extreme losses; Monte Carlo uncertainty concerns our ability to estimate that structure from finite information; and network contagion concerns the endogenous propagation of losses after a shock has entered an interconnected system. Their interaction provides the central motivation for studying tail risk and systemic risk within a common framework.

2 Loss Distributions and Financial Risk

The mathematical analysis of financial risk begins with the specification of a random variable representing changes in financial value. Depending on the application, this variable may describe the return of an asset, the profit and loss of a portfolio, or the monetary loss associated with a given risk horizon. Although these representations contain closely related information, their sign conventions are different and must be stated explicitly before defining tail-risk measures.

2.1 Returns versus Losses

Let $P_t > 0$ denote the price of a financial asset at time t . The simple return over the interval $[t-1, t]$ is defined by

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1. \quad (14)$$

A positive value of R_t represents a gain, whereas a negative value represents a loss. An alternative representation is the continuously compounded, or logarithmic, return,

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right) = \log(1 + R_t). \quad (15)$$

Logarithmic returns possess the useful property of temporal additivity. For T consecutive periods,

$$\sum_{t=1}^T r_t = \log\left(\frac{P_T}{P_0}\right). \quad (16)$$

For small price changes, the first-order approximation

$$\log(1 + R_t) \approx R_t \quad (17)$$

implies that simple and logarithmic returns are numerically close.

Risk measurement is often more naturally formulated in terms of a *loss variable*. Under the convention adopted throughout this study, a normalized loss is defined as

$$\boxed{L = -R}. \quad (18)$$

Consequently,

$$R < 0 \iff L > 0, \quad (19)$$

so that increasingly severe financial losses correspond to increasingly large positive values of L .

For a portfolio with initial value V_0 , the corresponding monetary loss may instead be written as

$$L^\$ = -V_0 R, \quad (20)$$

provided that R denotes the portfolio return over the relevant horizon.

This change of sign is conceptually important. When returns are analyzed, adverse events appear in the *left tail* of the return distribution. When the same financial outcomes are represented as positive losses, adverse events appear in the *right tail* of the loss distribution:

$$\boxed{\text{left tail of returns} \iff \text{right tail of losses.}} \quad (21)$$

The loss convention is particularly convenient for Value at Risk, Expected Shortfall, and related risk measures because greater numerical values then correspond directly to more severe adverse outcomes [13].

2.2 The Loss Distribution

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let

$$L : \Omega \rightarrow \mathbb{R} \quad (22)$$

be a random variable representing financial loss over a fixed horizon.

The cumulative distribution function (CDF) of L is

$$F_L(\ell) = \mathbb{P}(L \leq \ell), \quad \ell \in \mathbb{R}. \quad (23)$$

Thus, $F_L(\ell)$ gives the probability that the realized loss does not exceed the level ℓ .

Equivalently, the probability that the loss exceeds ℓ is

$$\mathbb{P}(L > \ell) = 1 - F_L(\ell). \quad (24)$$

This quantity is described by the survival function

$$\bar{F}_L(\ell) = 1 - F_L(\ell). \quad (25)$$

For tail-risk analysis, the survival function is particularly informative because it directly measures the probability of exceeding a specified loss threshold.

If L admits a probability density function f_L , then

$$F_L(\ell) = \int_{-\infty}^{\ell} f_L(x) dx, \quad (26)$$

and therefore

$$\bar{F}_L(\ell) = \int_{\ell}^{\infty} f_L(x) dx. \quad (27)$$

The complete loss distribution contains substantially more information than its first two moments. The expectation

$$\mu_L = \mathbb{E}[L] \quad (28)$$

describes its location, while

$$\sigma_L^2 = \text{Var}(L) = \mathbb{E}[(L - \mu_L)^2] \quad (29)$$

describes dispersion around the mean. However, neither quantity uniquely determines the probability or severity of extreme losses.

Indeed, two distributions may satisfy

$$\mathbb{E}[L_1] = \mathbb{E}[L_2] \quad (30)$$

and

$$\text{Var}(L_1) = \text{Var}(L_2), \quad (31)$$

while having substantially different tail probabilities,

$$\mathbb{P}(L_1 > x) \neq \mathbb{P}(L_2 > x) \quad (32)$$

for sufficiently large thresholds x . This observation will become central when Gaussian and heavy-tailed loss models are compared in a later section.

2.3 Quantiles

A fundamental object in risk measurement is the quantile function. For a confidence level $\alpha \in (0, 1)$, the α -quantile of the loss distribution is defined by the generalized inverse

$$q_\alpha(L) = F_L^{\leftarrow}(\alpha) = \inf \{ \ell \in \mathbb{R} : F_L(\ell) \geq \alpha \}. \quad (33)$$

For a continuous and strictly increasing distribution function,

$$F_L(q_\alpha) = \alpha, \quad (34)$$

and hence

$$q_\alpha = F_L^{-1}(\alpha). \quad (35)$$

The corresponding exceedance probability is

$$\mathbb{P}(L > q_\alpha) = 1 - \alpha \quad (36)$$

under continuity.

For example, if

$$\alpha = 0.99, \quad (37)$$

then the 99% loss quantile separates approximately 99% of the distribution from the most extreme 1% of losses:

$$\mathbb{P}(L > q_{0.99}) = 0.01. \quad (38)$$

This quantile construction provides the mathematical basis for Value at Risk:

$$\text{VaR}_\alpha(L) = q_\alpha(L). \quad (39)$$

An important distinction should already be emphasized. A quantile identifies a *location* in the distribution; it does not, by itself, describe how large observations become after that threshold has been crossed. Two loss distributions may therefore possess similar values of VaR_α while exhibiting substantially different behavior beyond that VaR level.

2.4 Why Tails Matter

The central concern of tail-risk analysis is not merely whether losses occur, but how the probability and magnitude of extreme losses behave.

For a high threshold u , tail risk is governed by quantities such as

$$\mathbb{P}(L > u) = \bar{F}_L(u), \quad (40)$$

and by the conditional distribution of losses given that the threshold has already been exceeded,

$$\mathbb{P}(L - u \leq x \mid L > u). \quad (41)$$

These two objects answer different questions. The first concerns the *frequency* of extreme losses; the second concerns their *severity* after an extreme event has occurred.

This distinction is essential in finance. A model may assign a relatively small probability to a severe market movement, but the economic consequences conditional on that movement may be extremely large. Consequently, a complete analysis of tail risk requires attention both to exceedance probabilities and to the magnitude of exceedances.

Empirical financial return distributions have long been observed to depart from the Gaussian benchmark. Classical empirical work documented distributions with substantially more probability in the extremes than would be suggested by a simple normal model [8]. Modern quantitative risk management therefore treats non-normality and heavy tails as central features in the statistical analysis of financial losses [13].

A useful conceptual comparison can be expressed through two standardized random variables X and Y satisfying

$$\mathbb{E}[X] = \mathbb{E}[Y] = 0 \quad (42)$$

and

$$\text{Var}(X) = \text{Var}(Y) = 1. \quad (43)$$

Equality of the first two moments does not imply equality of their tails. It is entirely possible that

$$\mathbb{P}(Y > x) > \mathbb{P}(X > x) \quad (44)$$

for sufficiently large x . In such a case, Y presents greater right-tail risk despite having exactly the same mean and variance as X .

This is precisely why the choice of probability distribution matters. Variance measures average dispersion, whereas tail-risk measures focus on low-probability, high-severity regions of the distribution.

The distinction becomes even more important when individual financial losses interact with leverage and network exposures. A tail realization that appears rare from the perspective of a single institution may become the initiating shock of a broader contagion process. In this sense, the tail of the loss distribution constitutes the conceptual bridge between individual financial risk and the systemic-risk mechanisms developed later in this study.

The progression can be summarized as

$$\boxed{\begin{array}{l} \text{Price Dynamics} \longrightarrow \text{Returns} \longrightarrow \text{Loss Distribution} \\ \longrightarrow \text{Quantiles} \longrightarrow \text{Tail Events} \longrightarrow \text{Risk Measures.} \end{array}} \quad (45)$$

The next step is therefore to move from the statistical description of the loss distribution to explicit measures of tail risk, beginning with Value at Risk and subsequently examining measures that account for the severity of losses beyond the quantile threshold.

3 Value at Risk (VaR)

Value at Risk (VaR) is one of the most widely known quantitative measures of financial risk. Its mathematical structure is simple: VaR is a quantile of the loss distribution. Nevertheless, this simplicity requires careful interpretation. VaR identifies a loss threshold associated with a prescribed confidence level, but it does not determine the magnitude of losses beyond that threshold.

Throughout this study, L denotes a financial loss, so that positive values correspond to adverse outcomes. Let

$$F_L(\ell) = \mathbb{P}(L \leq \ell) \quad (46)$$

denote the cumulative distribution function of L .

3.1 Definition

For a confidence level $\alpha \in (0, 1)$, the Value at Risk of L is defined as the lower α -quantile of the loss distribution,

$$\boxed{\text{VaR}_\alpha(L) = \inf \{ \ell \in \mathbb{R} : F_L(\ell) \geq \alpha \} .} \quad (47)$$

Equivalently,

$$\text{VaR}_\alpha(L) = F_L^{\leftarrow}(\alpha), \quad (48)$$

where $F_L^{\leftarrow}$ denotes the generalized inverse of the distribution function.

If F_L is continuous and strictly increasing, the definition simplifies to

$$\boxed{\text{VaR}_\alpha(L) = F_L^{-1}(\alpha).} \quad (49)$$

Consequently,

$$\mathbb{P}(L \leq \text{VaR}_\alpha(L)) = \alpha, \quad (50)$$

and, under continuity,

$$\mathbb{P}(L > \text{VaR}_\alpha(L)) = 1 - \alpha. \quad (51)$$

Thus, VaR partitions the loss distribution into two regions:

$$\boxed{\underbrace{\mathbb{P}(L \leq \text{VaR}_\alpha)}_{\alpha} \quad \underbrace{\mathbb{P}(L > \text{VaR}_\alpha)}_{1-\alpha}.} \quad (52)$$

3.2 Interpretation

Suppose that the one-day 99% VaR of a portfolio is

$$\text{VaR}_{0.99}(L) = \$1\,000\,000. \quad (53)$$

Under the assumed loss model, this means that the probability of observing a one-day loss exceeding \$1 million is approximately 1%:

$$\mathbb{P}(L > \$1\,000\,000) \approx 0.01. \quad (54)$$

It does *not* mean that the portfolio cannot lose more than \$1 million.

It also does not mean that \$1 million is the maximum possible loss. VaR identifies a quantile threshold rather than a worst-case outcome.

This distinction is fundamental:

$$\boxed{\text{VaR}_\alpha = \text{loss threshold at confidence level } \alpha,} \quad (55)$$

rather than

$$\boxed{\text{VaR}_\alpha \neq \text{maximum possible loss.}} \quad (56)$$

For $\alpha = 0.99$, VaR therefore identifies the boundary of the upper 1% loss region, but provides no direct information about how severe losses become within that region.

3.3 Historical and Empirical VaR

VaR can be estimated directly from observed or simulated losses without assuming a specific parametric probability distribution.

Consider a sample

$$L_1, \dots, L_N. \quad (57)$$

The empirical distribution function is

$$\hat{F}_N(\ell) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}_{\{L_i \leq \ell\}}. \quad (58)$$

The empirical VaR estimator is therefore

$$\boxed{\widehat{\text{VaR}}_\alpha = \hat{F}_N^{\leftarrow}(\alpha).} \quad (59)$$

Let

$$L_{(1)} \leq L_{(2)} \leq \dots \leq L_{(N)} \quad (60)$$

denote the ordered losses. A simple empirical quantile convention gives

$$\widehat{\text{VaR}}_\alpha = L_{(\lceil N\alpha \rceil)}. \quad (61)$$

For example, if

$$N = 1000, \quad \alpha = 0.99, \quad (62)$$

then

$$\lceil N\alpha \rceil = 990, \quad (63)$$

so that, under this convention,

$$\widehat{\text{VaR}}_{0.99} = L_{(990)}. \quad (64)$$

Different statistical software packages may implement slightly different sample-quantile interpolation conventions. For sufficiently large samples these differences are often small, but the convention should be reported when precise empirical comparisons are required.

Historical simulation applies this empirical construction to a historical sample of market-factor changes or portfolio losses. Its principal advantage is that no explicit Gaussian or other parametric distribution needs to be imposed on the data.

Schematically,

$$\boxed{\text{Historical observations} \longrightarrow \text{empirical loss distribution} \longrightarrow \text{empirical quantile} \longrightarrow \widehat{\text{VaR}}_{\alpha}.} \quad (65)$$

However, historical VaR implicitly relies on the relevance of the historical sample for the risk horizon being assessed. If the sample does not contain market conditions representative of current or future stress, the resulting empirical quantile may provide an incomplete description of the relevant risk.

3.4 Parametric VaR

An alternative approach assumes that the loss distribution belongs to a parametric family

$$L \sim F(\cdot; \theta), \quad (66)$$

where θ denotes a vector of model parameters.

Once θ has been estimated, VaR is obtained from the corresponding parametric quantile:

$$\boxed{\text{VaR}_{\alpha}(L) = F^{-1}(\alpha; \theta).} \quad (67)$$

For a location-scale model

$$L = \mu + \sigma Z, \quad \sigma > 0, \quad (68)$$

where Z has distribution function F_Z , we have

$$F_L(\ell) = F_Z\left(\frac{\ell - \mu}{\sigma}\right). \quad (69)$$

Setting

$$F_L(\ell) = \alpha \quad (70)$$

gives

$$F_Z\left(\frac{\ell - \mu}{\sigma}\right) = \alpha. \quad (71)$$

Therefore,

$$\frac{\ell - \mu}{\sigma} = F_Z^{-1}(\alpha), \quad (72)$$

and hence

$$\boxed{\text{VaR}_{\alpha}(L) = \mu + \sigma F_Z^{-1}(\alpha).} \quad (73)$$

This formula makes explicit that parametric VaR depends on both the estimated scale of losses and the assumed shape of the standardized distribution F_Z .

3.5 Gaussian VaR

Suppose that losses are normally distributed,

$$L \sim \mathcal{N}(\mu, \sigma^2). \quad (74)$$

Then

$$Z = \frac{L - \mu}{\sigma} \sim \mathcal{N}(0, 1). \quad (75)$$

Let Φ denote the standard normal cumulative distribution function and define

$$z_\alpha = \Phi^{-1}(\alpha). \quad (76)$$

The Gaussian VaR is therefore

$$\boxed{\text{VaR}_\alpha^{\text{Gaussian}}(L) = \mu + \sigma z_\alpha.} \quad (77)$$

For example,

$$z_{0.95} \approx 1.6449, \quad (78)$$

and

$$z_{0.99} \approx 2.3263. \quad (79)$$

Thus, for a zero-mean loss distribution with daily volatility

$$\sigma = 0.02, \quad (80)$$

the Gaussian 99% VaR is

$$\text{VaR}_{0.99} = 0 + 0.02(2.3263) \quad (81)$$

$$\approx 0.04653. \quad (82)$$

Hence,

$$\boxed{\text{VaR}_{0.99} \approx 4.65\%}. \quad (83)$$

This result is conditional on the Gaussian model. If the true loss distribution has heavier tails, the corresponding extreme quantile may differ substantially even when the two distributions possess the same variance.

3.6 Time Horizon and Scaling

VaR must always be associated with a specified risk horizon. A one-day VaR and a ten-day VaR represent different random losses and therefore different quantiles.

Under restrictive assumptions—in particular, independent and identically distributed increments with finite variance and a stable volatility environment—volatility is often scaled according to the square-root-of-time rule,

$$\sigma_h = \sigma_1 \sqrt{h}, \quad (84)$$

which under an appropriate location-scale model yields the approximation

$$\text{VaR}_{\alpha, h} \approx \mu_h + z_\alpha \sigma_1 \sqrt{h}. \quad (85)$$

If the mean is negligible over the relevant horizon,

$$\boxed{\text{VaR}_{\alpha,h} \approx \text{VaR}_{\alpha,1} \sqrt{h}.} \quad (86)$$

This scaling rule should not be interpreted as universal. Serial dependence, time-varying volatility, jumps, liquidity effects, and non-Gaussian dynamics can invalidate simple square-root-of-time scaling.

3.7 Advantages of VaR

VaR has several practical advantages.

First, it summarizes a potentially complex loss distribution through a single quantity expressed in economically interpretable units.

Second, it incorporates an explicit confidence level and risk horizon,

$$\text{VaR}_{\alpha,h}, \quad (87)$$

which allows risk statements to be associated with both probability and time.

Third, VaR can be estimated using several distinct methodologies,

$$\boxed{\text{Historical} \quad \text{Parametric} \quad \text{Monte Carlo}.} \quad (88)$$

This flexibility allows the same conceptual risk measure to be used with very different models of portfolio behavior.

Finally, because VaR is a quantile, it is naturally connected to statistical estimation, back-testing, and exceedance analysis.

3.8 Limitations of VaR

Despite these advantages, VaR possesses important conceptual and mathematical limitations.

3.8.1 No Information About Tail Severity

The most immediate limitation follows directly from the quantile definition.

Suppose two loss variables L_A and L_B satisfy

$$\text{VaR}_{0.99}(L_A) = \text{VaR}_{0.99}(L_B). \quad (89)$$

This equality does not imply that their extreme losses are comparable. It is possible that

$$\mathbb{E}[L_A \mid L_A > \text{VaR}_{0.99}(L_A)] \ll \mathbb{E}[L_B \mid L_B > \text{VaR}_{0.99}(L_B)]. \quad (90)$$

Thus,

$$\boxed{\text{same VaR} \not\Rightarrow \text{same tail severity}.} \quad (91)$$

This limitation is one of the central motivations for Expected Shortfall and related tail-sensitive measures.

3.8.2 Dependence on Distributional Assumptions

Parametric VaR is only as reliable as the assumed probability model and its estimated parameters. For example,

$$L_N \sim \mathcal{N}(0, \sigma^2) \quad (92)$$

and a standardized heavy-tailed loss variable may have

$$\text{Var}(L_N) = \text{Var}(L_{HT}) = \sigma^2, \quad (93)$$

while nevertheless satisfying

$$\mathbb{P}(L_{HT} > x) > \mathbb{P}(L_N > x) \quad (94)$$

for sufficiently extreme x .

Consequently, similar volatility estimates do not guarantee similar extreme-risk estimates.

3.8.3 Sampling Error

Empirical VaR is itself an estimator. Given a finite sample,

$$\widehat{\text{VaR}}_\alpha \neq \text{VaR}_\alpha \quad (95)$$

in general.

As α approaches one, fewer observations are available near and beyond the relevant quantile. The statistical uncertainty associated with extreme-quantile estimation can therefore become substantial.

3.8.4 Failure of Subadditivity in General

A particularly important mathematical limitation concerns diversification.

For a coherent risk measure ρ , subadditivity requires

$$\rho(X + Y) \leq \rho(X) + \rho(Y). \quad (96)$$

The inequality formalizes the idea that diversification should not be penalized by the risk measure.

VaR does not satisfy this property for arbitrary loss distributions [2].

A simple example illustrates the problem. Let X and Y be independent losses satisfying

$$X = \begin{cases} 0, & \text{with probability } 0.96, \\ 100, & \text{with probability } 0.04, \end{cases} \quad (97)$$

and analogously for Y .

At the 95% confidence level,

$$\text{VaR}_{0.95}(X) = \text{VaR}_{0.95}(Y) = 0. \quad (98)$$

However,

$$\mathbb{P}(X + Y = 0) = 0.96^2 = 0.9216 < 0.95. \quad (99)$$

Furthermore,

$$\mathbb{P}(X + Y \leq 100) = 1 - (0.04)^2 = 0.9984. \quad (100)$$

Therefore,

$$\text{VaR}_{0.95}(X + Y) = 100, \quad (101)$$

while

$$\text{VaR}_{0.95}(X) + \text{VaR}_{0.95}(Y) = 0. \quad (102)$$

Hence,

$$\boxed{\text{VaR}_{0.95}(X + Y) > \text{VaR}_{0.95}(X) + \text{VaR}_{0.95}(Y).} \quad (103)$$

VaR therefore fails subadditivity in this example. This does not imply that VaR is never subadditive: under important classes of distributions, including appropriate elliptical settings, subadditivity can hold. The point is that it is not guaranteed in general.

3.9 VaR as a Threshold, Not a Description of the Tail

The preceding discussion reveals the fundamental mathematical role of VaR:

$$\boxed{\text{VaR}_\alpha = \text{location of a tail threshold.}} \quad (104)$$

VaR answers the question

At what loss level does the upper $(1 - \alpha)$ fraction of the loss distribution begin?

It does not answer

How severe are losses once that threshold has been exceeded?

The distinction can be summarized as

$$\boxed{\text{Loss Distribution} \longrightarrow \underbrace{\text{VaR}_\alpha}_{\text{tail threshold}} \longrightarrow \underbrace{\text{unobserved tail severity}}_{\text{not measured by VaR}}.} \quad (105)$$

This missing information provides the natural motivation for the next risk measure considered in this study: Conditional Value at Risk, or Expected Shortfall, which explicitly incorporates the magnitude of losses beyond the VaR threshold.

4 Conditional Value at Risk (CVaR / Expected Shortfall)

Value at Risk identifies the location of an extreme-loss threshold, but does not quantify the magnitude of losses beyond that threshold. This limitation motivates a second class of tail-risk measures designed to incorporate the severity of adverse outcomes.

Conditional Value at Risk (CVaR) and Expected Shortfall (ES) are closely related concepts. For continuous loss distributions, the commonly used representations coincide and can be interpreted as the average loss in the worst $(1 - \alpha)$ fraction of outcomes. For distributions containing atoms or discontinuities, however, greater care is required in defining the tail average. This distinction is important both mathematically and computationally [15, 1].

Throughout this section, L denotes a financial loss and

$$q_\alpha = \text{VaR}_\alpha(L) \quad (106)$$

denotes its lower α -quantile.

4.1 Definition

For an integrable loss random variable L , a robust definition of Expected Shortfall at confidence level $\alpha \in (0, 1)$ is

$$\boxed{\text{ES}_\alpha(L) = \frac{1}{1 - \alpha} \int_\alpha^1 \text{VaR}_u(L) du.} \quad (107)$$

This definition averages the loss quantiles over all confidence levels between α and 1. Thus, unlike VaR, which selects one particular quantile, Expected Shortfall aggregates information throughout the remaining upper tail.

In the terminology of Rockafellar and Uryasev, Conditional Value at Risk provides a closely related tail-loss measure and admits a particularly useful optimization representation [14, 15]. For the continuous distributions emphasized in the present simulations, CVaR and ES coincide, and we therefore write

$$\text{CVaR}_\alpha(L) = \text{ES}_\alpha(L). \quad (108)$$

This convention will be used throughout the continuous Gaussian and Student- t experiments considered below.

4.2 Tail Conditional Expectation

Suppose that L has a continuous distribution and let

$$q_\alpha = \text{VaR}_\alpha(L). \quad (109)$$

Then

$$\mathbb{P}(L > q_\alpha) = 1 - \alpha. \quad (110)$$

In this setting, Expected Shortfall can be written as the conditional expectation

$$\boxed{\text{ES}_\alpha(L) = \mathbb{E}[L \mid L > q_\alpha].} \quad (111)$$

Equivalently,

$$\text{ES}_\alpha(L) = \frac{\mathbb{E}[L \mathbf{1}_{\{L > q_\alpha\}}]}{1 - \alpha}. \quad (112)$$

If L admits a density f_L , then

$$\boxed{\text{ES}_\alpha(L) = \frac{1}{1 - \alpha} \int_{q_\alpha}^{\infty} \ell f_L(\ell) d\ell.} \quad (113)$$

The interpretation is immediate: VaR determines the boundary of the tail, whereas ES computes the average magnitude of the losses located inside that tail.

Thus,

$$\boxed{\begin{array}{ll} \text{VaR}_\alpha : & \text{Where does the tail begin?} \\ \text{ES}_\alpha : & \text{How severe are losses inside the tail?} \end{array}} \quad (114)$$

For discontinuous distributions, however, the event $\{L > q_\alpha\}$ need not have probability exactly $1 - \alpha$. Consequently, a naive conditional-expectation formula is not, in general, equivalent to the coherent Expected Shortfall definition. This is one reason why the quantile-integral representation is mathematically preferable when working with general loss distributions [1].

4.3 Integral Representation

The quantile representation

$$\text{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 \text{VaR}_u(L) du \quad (115)$$

provides an especially transparent interpretation of Expected Shortfall. VaR at confidence level α uses only one point of the quantile function,

$$u = \alpha, \quad (116)$$

whereas ES averages all quantiles from α to 1. Schematically,

$$\boxed{\text{VaR}_\alpha = q_\alpha, \quad \text{ES}_\alpha = \frac{1}{1-\alpha} \int_\alpha^1 q_u du.} \quad (117)$$

Hence ES incorporates the entire quantile structure of the selected tail rather than a single boundary point.

For a continuous loss distribution, the same quantity can be recovered from the density representation. Let

$$u = F_L(\ell). \quad (118)$$

Then

$$du = f_L(\ell) d\ell, \quad (119)$$

and

$$\ell = F_L^{-1}(u). \quad (120)$$

Therefore,

$$\text{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_{q_\alpha}^\infty \ell f_L(\ell) d\ell \quad (121)$$

$$= \frac{1}{1-\alpha} \int_\alpha^1 F_L^{-1}(u) du \quad (122)$$

$$= \frac{1}{1-\alpha} \int_\alpha^1 \text{VaR}_u(L) du. \quad (123)$$

This establishes the connection between the conditional-tail and quantile-average interpretations in the continuous case.

4.4 Gaussian Expected Shortfall

A useful closed-form expression is obtained under the Gaussian model. Suppose

$$L \sim \mathcal{N}(\mu, \sigma^2), \quad (124)$$

and let

$$z_\alpha = \Phi^{-1}(\alpha), \quad (125)$$

where Φ is the standard normal distribution function.

As shown previously,

$$\text{VaR}_\alpha(L) = \mu + \sigma z_\alpha. \quad (126)$$

Let ϕ denote the standard normal density,

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}. \quad (127)$$

Since

$$L = \mu + \sigma Z, \quad Z \sim \mathcal{N}(0, 1), \quad (128)$$

we obtain

$$\text{ES}_\alpha(L) = \mathbb{E}[\mu + \sigma Z \mid Z > z_\alpha] \quad (129)$$

$$= \mu + \sigma \mathbb{E}[Z \mid Z > z_\alpha]. \quad (130)$$

For the standard normal distribution,

$$\mathbb{E}[Z \mid Z > z_\alpha] = \frac{\phi(z_\alpha)}{1 - \alpha}. \quad (131)$$

Therefore,

$$\boxed{\text{ES}_\alpha^{\text{Gaussian}}(L) = \mu + \sigma \frac{\phi(z_\alpha)}{1 - \alpha}.} \quad (132)$$

For example, at

$$\alpha = 0.99, \quad (133)$$

we have approximately

$$z_{0.99} = 2.3263, \quad (134)$$

and

$$\frac{\phi(z_{0.99})}{0.01} \approx 2.6652. \quad (135)$$

For a zero-mean loss distribution with

$$\sigma = 0.02, \quad (136)$$

this gives

$$\text{VaR}_{0.99} \approx 0.02(2.3263) = 0.0465, \quad (137)$$

whereas

$$\text{ES}_{0.99} \approx 0.02(2.6652) = 0.0533. \quad (138)$$

Thus,

$$\boxed{\text{VaR}_{0.99} \approx 4.65\%, \quad \text{ES}_{0.99} \approx 5.33\%}. \quad (139)$$

The difference has a clear interpretation: 4.65% identifies the 99% loss threshold, while 5.33% represents the average loss conditional on being in the worst 1% of outcomes under the Gaussian model.

4.5 VaR versus CVaR / Expected Shortfall

VaR and Expected Shortfall describe different aspects of the same loss distribution.

For continuous distributions,

$$\text{VaR}_\alpha(L) = q_\alpha, \quad (140)$$

while

$$\text{ES}_\alpha(L) = \mathbb{E}[L \mid L > q_\alpha]. \quad (141)$$

Consequently, provided the relevant expectation exists,

$$\boxed{\text{ES}_\alpha(L) \geq \text{VaR}_\alpha(L)}. \quad (142)$$

For a nondegenerate continuous distribution with genuine probability mass above the VaR threshold, the inequality is typically strict.

The distinction can be summarized as follows:

Property	VaR	CVaR / ES
Object	Quantile	Tail average
Tail boundary	Yes	Yes, implicitly
Tail severity	No	Yes
Uses losses beyond VaR	No	Yes
Coherent in general	No	Yes
Subadditive in general	No	Yes

The difference becomes particularly important when two distributions have similar quantiles but very different extreme tails.

Suppose

$$\text{VaR}_\alpha(L_A) \approx \text{VaR}_\alpha(L_B), \quad (143)$$

but distribution L_B has substantially greater probability mass at very large losses. Then it is possible that

$$\text{ES}_\alpha(L_B) \gg \text{ES}_\alpha(L_A). \quad (144)$$

VaR may therefore indicate similar threshold risk while ES reveals a substantial difference in tail severity.

4.6 Coherence

The theory of coherent risk measures provides an axiomatic framework for evaluating how a risk measure behaves under aggregation, scaling, and capital adjustments [2].

Let ρ denote a risk measure defined on financial losses. Under the loss convention used here, the four standard coherence properties can be written as follows.

4.6.1 Monotonicity

If

$$X \leq Y \quad \text{almost surely}, \quad (145)$$

then

$$\boxed{\rho(X) \leq \rho(Y)}. \quad (146)$$

A position that never produces a greater loss should not receive a larger risk measure.

4.6.2 Translation Invariance

For every deterministic amount $m \in \mathbb{R}$,

$$\boxed{\rho(X + m) = \rho(X) + m.} \quad (147)$$

Under the loss convention, adding a certain loss of m increases measured risk by exactly m .

4.6.3 Positive Homogeneity

For every $\lambda \geq 0$,

$$\boxed{\rho(\lambda X) = \lambda \rho(X).} \quad (148)$$

Scaling all losses by a positive factor scales measured risk by the same factor.

4.6.4 Subadditivity

For any losses X and Y ,

$$\boxed{\rho(X + Y) \leq \rho(X) + \rho(Y).} \quad (149)$$

This property formalizes the diversification principle: combining two positions should not create a diversification penalty purely as an artifact of the risk measure.

Expected Shortfall, when properly defined, satisfies these coherence properties, whereas VaR is not subadditive for arbitrary loss distributions [1, 15].

Hence,

$$\boxed{\text{ES}_\alpha \text{ is a coherent risk measure.}} \quad (150)$$

4.7 Why CVaR Captures Tail Severity

The fundamental difference between VaR and CVaR/ES can be understood by examining what happens after the quantile threshold has been crossed.

Consider two stylized loss distributions with the same 99% VaR:

$$\text{VaR}_{0.99}(L_A) = \text{VaR}_{0.99}(L_B) = 10. \quad (151)$$

Suppose the losses in the worst 1% of outcomes for L_A remain close to the threshold, for example around

$$10, 11, 12, 13, \quad (152)$$

whereas the corresponding tail of L_B contains losses such as

$$10, 20, 50, 100. \quad (153)$$

VaR sees essentially the same boundary in both cases:

$$q_{0.99} = 10. \quad (154)$$

Expected Shortfall does not stop at this boundary. It averages the losses throughout the selected tail and therefore distinguishes the two risk profiles.

Schematically,

	VaR _{0.99}	ES _{0.99}	
L_A	10	moderate	(155)
L_B	10	large	

This is precisely the information that VaR discards.

More generally, VaR depends on a single quantile,

$$\text{VaR}_\alpha(L) = q_\alpha, \quad (156)$$

whereas ES depends on the complete upper quantile curve,

$$\text{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 q_u du. \quad (157)$$

Therefore, changes in the loss distribution beyond q_α may leave VaR approximately unchanged while substantially changing Expected Shortfall.

This property makes ES particularly informative when the principal concern is not simply whether a rare event occurs, but how damaging the event may be once it occurs.

4.8 Optimization Representation of CVaR

An additional advantage of CVaR is its useful optimization representation. For an integrable loss L , define

$$\Psi_\alpha(\xi) = \xi + \frac{1}{1-\alpha} \mathbb{E}[(L - \xi)_+], \quad (158)$$

where

$$(x)_+ = \max(x, 0). \quad (159)$$

Then

$$\text{CVaR}_\alpha(L) = \min_{\xi \in \mathbb{R}} \left\{ \xi + \frac{1}{1-\alpha} \mathbb{E}[(L - \xi)_+] \right\}. \quad (160)$$

Under suitable conditions, a VaR quantile is among the minimizers of this function. This representation is particularly important in portfolio optimization because it transforms tail-risk minimization into a form that can be handled efficiently in scenario-based optimization [14, 15].

For a finite sample

$$L_1, \dots, L_N, \quad (161)$$

the expectation can be replaced by the sample average,

$$\hat{\Psi}_\alpha(\xi) = \xi + \frac{1}{N(1-\alpha)} \sum_{i=1}^N (L_i - \xi)_+. \quad (162)$$

This formula establishes a direct bridge between Expected Shortfall, Monte Carlo simulation, and numerical optimization.

4.9 From Tail Threshold to Tail Severity

VaR and CVaR/ES should therefore not be viewed as interchangeable statistics. They answer complementary questions about the loss distribution:

$$\begin{array}{l} \text{VaR}_\alpha \longrightarrow \text{location of the extreme-loss threshold,} \\ \text{CVaR}_\alpha = \text{ES}_\alpha \longrightarrow \text{severity of losses beyond that threshold.} \end{array} \quad (163)$$

The transition from VaR to Expected Shortfall is therefore a transition from a single quantile to an integrated description of the tail:

$$\text{Loss Distribution} \longrightarrow \text{VaR}_\alpha \longrightarrow \text{ES}_\alpha \longrightarrow \text{Tail Severity.} \quad (164)$$

This distinction becomes especially important for heavy-tailed distributions. Two models calibrated to similar central behavior can produce markedly different Expected Shortfall values because ES remains sensitive to observations far beyond the VaR boundary. This provides the natural transition to the comparison between Gaussian and heavy-tailed loss models developed in the following section.

5 Gaussian Models and Heavy Tails

The choice of a probability distribution is a fundamental component of financial risk measurement. Two loss models may possess identical means and variances while assigning substantially different probabilities to extreme events. Consequently, calibration of the first two moments alone does not determine the behavior of Value at Risk, Expected Shortfall, or other tail-sensitive quantities.

This section compares two important location-scale models: the Gaussian distribution and the Student- t distribution. The comparison is performed after standardization to a common variance, allowing differences in tail risk to be attributed to distributional shape rather than to differences in volatility.

Empirical studies of financial returns provide substantial motivation for such comparisons. Return distributions frequently exhibit deviations from Gaussian behavior, including heavy tails and extreme fluctuations [8, 6]. The Gaussian model therefore provides an important benchmark, but not a universal description of financial losses.

5.1 The Normal Distribution

Let a financial loss L_G follow a Gaussian distribution,

$$L_G \sim \mathcal{N}(\mu, \sigma^2), \quad \sigma > 0. \quad (165)$$

Its probability density function is

$$f_G(\ell) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{(\ell - \mu)^2}{2\sigma^2} \right]. \quad (166)$$

Introducing the standardized variable

$$Z = \frac{L_G - \mu}{\sigma}, \quad (167)$$

we obtain

$$Z \sim \mathcal{N}(0, 1). \quad (168)$$

The standard normal density is therefore

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}. \quad (169)$$

The corresponding cumulative distribution function is

$$\Phi(z) = \int_{-\infty}^z \phi(u) du. \quad (170)$$

For an upper-tail threshold x ,

$$\mathbb{P}(Z > x) = 1 - \Phi(x). \quad (171)$$

The Gaussian density decreases exponentially in x^2 ,

$$\phi(x) \propto e^{-x^2/2}, \quad (172)$$

which produces rapidly vanishing probabilities for observations many standard deviations from the mean.

The Gaussian model is mathematically convenient and plays a central role in probability theory and financial modeling. However, its rapidly decaying tails can be restrictive when the empirical loss distribution contains substantially more extreme observations than predicted by the normal benchmark.

5.2 The Student- t Distribution

A widely used alternative for representing heavier tails is the Student- t distribution.

Let

$$T_\nu \sim t_\nu, \quad (173)$$

where $\nu > 0$ denotes the degrees of freedom.

Its density is

$$f_\nu(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi} \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}. \quad (174)$$

Like the standard Gaussian distribution, the Student- t distribution is symmetric around zero. Its tail behavior, however, is fundamentally different.

For large $|t|$,

$$f_\nu(t) \sim C_\nu |t|^{-(\nu+1)}, \quad |t| \rightarrow \infty, \quad (175)$$

for a positive constant C_ν .

Correspondingly, its survival probability behaves asymptotically as

$$\mathbb{P}(T_\nu > x) \sim K_\nu x^{-\nu}, \quad x \rightarrow \infty, \quad (176)$$

for another positive constant K_ν .

Thus the Student- t distribution has polynomially decaying tails, in contrast to the much faster Gaussian decay.

The parameter ν controls tail thickness. Smaller values of ν produce heavier tails, whereas

$$T_\nu \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } \nu \rightarrow \infty. \quad (177)$$

The Student- t family therefore provides a convenient bridge between strongly heavy-tailed models and the Gaussian benchmark.

5.3 Moment Existence

An important feature of the Student- t distribution is that the existence of its moments depends on the degrees of freedom.

For $\nu > 1$,

$$\mathbb{E}[T_\nu] = 0. \quad (178)$$

For $\nu > 2$,

$$\boxed{\text{Var}(T_\nu) = \frac{\nu}{\nu - 2}.} \quad (179)$$

More generally, absolute moments of order p are finite only when

$$\boxed{p < \nu.} \quad (180)$$

This observation is particularly important for risk modeling. A Student- t model may have a finite variance while higher-order moments remain infinite.

5.4 Variance Standardization

A direct comparison between a standard normal variable and an ordinary Student- t variable is potentially misleading because, for finite $\nu > 2$,

$$\text{Var}(T_\nu) = \frac{\nu}{\nu - 2} > 1. \quad (181)$$

Part of any observed difference would therefore arise simply because the two variables have different variances.

To isolate the effect of tail shape, define the standardized Student- t variable

$$\boxed{Z_\nu = \sqrt{\frac{\nu - 2}{\nu}} T_\nu, \quad \nu > 2.} \quad (182)$$

Then

$$\text{Var}(Z_\nu) = \frac{\nu - 2}{\nu} \text{Var}(T_\nu) \quad (183)$$

$$= \frac{\nu - 2}{\nu} \frac{\nu}{\nu - 2} \quad (184)$$

$$= 1. \quad (185)$$

Therefore,

$$\boxed{\mathbb{E}[Z_\nu] = 0, \quad \text{Var}(Z_\nu) = 1.} \quad (186)$$

We can now construct two loss models,

$$L_G = \mu + \sigma Z, \quad Z \sim \mathcal{N}(0, 1), \quad (187)$$

$$L_T = \mu + \sigma Z_\nu, \quad Z_\nu = \sqrt{\frac{\nu - 2}{\nu}} T_\nu. \quad (188)$$

Both satisfy

$$\mathbb{E}[L_G] = \mathbb{E}[L_T] = \mu \quad (189)$$

and

$$\boxed{\text{Var}(L_G) = \text{Var}(L_T) = \sigma^2.} \quad (190)$$

The two models therefore possess exactly the same mean and variance but different distributional shapes.

5.5 Tail Probabilities

Consider a threshold expressed in volatility units,

$$\ell_k = \mu + k\sigma, \quad k > 0. \quad (191)$$

For the Gaussian model,

$$\boxed{\mathbb{P}(L_G > \mu + k\sigma) = 1 - \Phi(k).} \quad (192)$$

For the variance-standardized Student- t model,

$$\mathbb{P}(L_T > \mu + k\sigma) = \mathbb{P}(Z_\nu > k) \quad (193)$$

$$= \mathbb{P}\left(T_\nu > k\sqrt{\frac{\nu}{\nu-2}}\right). \quad (194)$$

If F_{t_ν} denotes the Student- t CDF, then

$$\boxed{\mathbb{P}(L_T > \mu + k\sigma) = 1 - F_{t_\nu}\left(k\sqrt{\frac{\nu}{\nu-2}}\right).} \quad (195)$$

The essential point is asymptotic:

$$\mathbb{P}(Z > k) \quad (196)$$

decreases much faster than

$$\mathbb{P}(Z_\nu > k) \quad (197)$$

as k becomes sufficiently large.

Thus, after controlling for variance, sufficiently extreme events remain substantially more probable under the Student- t model.

5.6 A Subtle Point: Heavy Tails Do Not Mean More Probability Everywhere

It is tempting to state that a heavy-tailed distribution assigns more probability than a Gaussian distribution to every observation outside the center. This statement is not generally correct after variance standardization.

A standardized Student- t distribution typically redistributes probability mass relative to the Gaussian benchmark. Because both models have the same total probability and the same variance, their densities and tail probabilities may cross.

Consequently, for some moderate thresholds k it is possible that

$$\mathbb{P}(Z_\nu > k) < \mathbb{P}(Z > k), \quad (198)$$

whereas for sufficiently large k ,

$$\boxed{\mathbb{P}(Z_\nu > k) > \mathbb{P}(Z > k).} \quad (199)$$

The term *heavy-tailed* therefore refers to asymptotic tail decay, not to pointwise dominance at every finite threshold.

This distinction is especially important when comparing VaR values at different confidence levels.

5.7 Kurtosis

Kurtosis provides information about the fourth standardized moment of a distribution, provided that the fourth moment exists.

For a random variable X with finite fourth moment, Pearson kurtosis is

$$\text{Kurt}(X) = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4}. \quad (200)$$

The Gaussian distribution satisfies

$$\text{Kurt}(Z) = 3. \quad (201)$$

Excess kurtosis is defined as

$$\kappa(X) = \text{Kurt}(X) - 3. \quad (202)$$

Hence,

$$\kappa(Z) = 0 \quad (203)$$

for a Gaussian variable.

For a Student- t variable with

$$\nu > 4, \quad (204)$$

the excess kurtosis is

$$\kappa(T_\nu) = \frac{6}{\nu - 4}. \quad (205)$$

Since kurtosis is invariant under positive location-scale transformations,

$$\kappa(Z_\nu) = \frac{6}{\nu - 4}, \quad \nu > 4. \quad (206)$$

Therefore,

$$\kappa(Z_\nu) > 0 \quad (207)$$

for every finite $\nu > 4$, while

$$\kappa(Z_\nu) \rightarrow 0 \quad \text{as } \nu \rightarrow \infty. \quad (208)$$

However, if

$$2 < \nu \leq 4, \quad (209)$$

the variance exists but the fourth moment does not. Consequently, finite kurtosis is not defined.

This regime is particularly relevant to the numerical experiments in this study. For example, with

$$\nu = 3.2, \quad (210)$$

we have

$$\text{Var}(T_{3.2}) < \infty, \quad (211)$$

but

$$\boxed{\mathbb{E}[T_{3.2}^4] = \infty.} \quad (212)$$

Thus the model can be standardized to unit variance, while its theoretical fourth moment remains infinite.

This distinction should be preserved in empirical work: a finite sample always produces a finite numerical sample kurtosis, but this does not imply that the corresponding theoretical distribution possesses a finite fourth moment.

5.8 Kurtosis and Tail Risk Are Not Identical Concepts

Although kurtosis is frequently used as an indicator of non-Gaussian behavior, it should not be interpreted as a complete measure of tail risk.

Kurtosis is a global fourth-moment functional,

$$\mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right], \quad (213)$$

whereas financial tail risk is concerned with quantities such as

$$\mathbb{P}(L > x), \quad (214)$$

$$\text{VaR}_\alpha(L), \quad (215)$$

and

$$\text{ES}_\alpha(L). \quad (216)$$

Distributions with the same kurtosis need not have identical tails, and some heavy-tailed distributions may not possess a finite fourth moment at all.

Kurtosis is therefore a useful diagnostic of distributional shape, but it should not replace direct analysis of tail probabilities and tail-risk measures.

5.9 Implications for VaR

The difference between Gaussian and heavy-tailed models directly affects Value at Risk.

For the Gaussian model,

$$\text{VaR}_\alpha^G = \mu + \sigma \Phi^{-1}(\alpha). \quad (217)$$

For the standardized Student- t model,

$$\boxed{\text{VaR}_\alpha^t = \mu + \sigma \sqrt{\frac{\nu-2}{\nu}} t_\nu^{-1}(\alpha),} \quad (218)$$

where t_ν^{-1} denotes the quantile function of the ordinary Student- t distribution.

Hence the ratio of standardized VaR values is

$$\frac{\text{VaR}_\alpha^t - \mu}{\text{VaR}_\alpha^G - \mu} = \frac{\sqrt{\frac{\nu-2}{\nu}} t_\nu^{-1}(\alpha)}{\Phi^{-1}(\alpha)}. \quad (219)$$

At sufficiently high confidence levels, the heavy-tailed model can produce substantially larger extreme quantiles than the Gaussian benchmark.

The qualification “sufficiently high” is important because, after variance standardization, the Student- t quantile need not exceed the Gaussian quantile at every confidence level.

5.10 Implications for Expected Shortfall

Tail thickness also affects Expected Shortfall.

For a Gaussian loss,

$$\text{ES}_\alpha^G = \mu + \sigma \frac{\phi(z_\alpha)}{1 - \alpha}, \quad z_\alpha = \Phi^{-1}(\alpha). \quad (220)$$

For an ordinary Student- t variable T_ν with $\nu > 1$, let

$$q_{\alpha,\nu} = t_\nu^{-1}(\alpha). \quad (221)$$

Using the Student- t density f_ν ,

$$\mathbb{E}[T_\nu \mid T_\nu > q_{\alpha,\nu}] = \frac{\nu + q_{\alpha,\nu}^2}{(\nu - 1)(1 - \alpha)} f_\nu(q_{\alpha,\nu}). \quad (222)$$

Therefore, for the variance-standardized Student- t loss model with $\nu > 2$,

$$\boxed{\text{ES}_\alpha^t = \mu + \sigma \sqrt{\frac{\nu - 2}{\nu}} \frac{\nu + q_{\alpha,\nu}^2}{(\nu - 1)(1 - \alpha)} f_\nu(q_{\alpha,\nu}).} \quad (223)$$

Expected Shortfall is particularly informative in this comparison because it incorporates the magnitude of losses throughout the selected tail, rather than using only one quantile boundary.

5.11 Extreme-Event Implications

The practical consequences of distributional choice become increasingly important as the analysis moves deeper into the tail.

Suppose two loss variables satisfy

$$\mathbb{E}[L_G] = \mathbb{E}[L_T] = \mu \quad (224)$$

and

$$\text{Var}(L_G) = \text{Var}(L_T) = \sigma^2. \quad (225)$$

Variance alone would describe the two models as having the same average squared dispersion around the mean.

Nevertheless, for sufficiently extreme thresholds,

$$\mathbb{P}(L_T > x) \gg \mathbb{P}(L_G > x). \quad (226)$$

Consequently, the models can imply very different probabilities for extreme losses and different values of high-confidence VaR and Expected Shortfall.

The mechanism is not an increase in variance. By construction, variance is identical. The difference arises entirely from the allocation of probability mass across the distribution.

5.12 Same Variance Does Not Imply the Same Tail Risk

The central result of this comparison can therefore be expressed as

$$\boxed{\text{Var}(L_1) = \text{Var}(L_2) \not\Rightarrow \text{TailRisk}(L_1) = \text{TailRisk}(L_2).} \quad (227)$$

More concretely,

$$\text{Var}(L_G) = \text{Var}(L_T) = \sigma^2 \quad (228)$$

does not imply

$$\text{VaR}_\alpha(L_G) = \text{VaR}_\alpha(L_T), \quad (229)$$

nor does it imply

$$\text{ES}_\alpha(L_G) = \text{ES}_\alpha(L_T). \quad (230)$$

The distinction becomes increasingly important as α approaches one.

This provides a direct conceptual connection between volatility-based risk descriptions and tail-sensitive risk measures:

$$\boxed{\begin{array}{l} \text{Variance} \longrightarrow \text{overall quadratic dispersion,} \\ \text{VaR}_\alpha \longrightarrow \text{extreme-loss threshold,} \\ \text{ES}_\alpha \longrightarrow \text{extreme-loss severity.} \end{array}} \quad (231)$$

These quantities contain different information and should not be treated as interchangeable measures of financial risk.

5.13 From Distributional Risk to Estimation Risk

Heavy-tailed modeling introduces an additional statistical challenge. Extreme observations are simultaneously more consequential and relatively rare. Consequently, the estimation of high quantiles and tail averages may require substantially more information than the estimation of central distributional characteristics.

At confidence level α , a sample of size N contains approximately

$$N(1 - \alpha) \quad (232)$$

observations in the relevant upper-tail region.

Thus, at

$$\alpha = 0.99, \quad (233)$$

only approximately one percent of the sample directly populates the region used to estimate extreme tail behavior.

The analysis therefore moves naturally from a problem of *distributional specification*,

$$\text{Gaussian versus heavy-tailed models,} \quad (234)$$

to a problem of *statistical estimation*,

$$\text{How accurately can tail risk be estimated from finite samples?} \quad (235)$$

This question motivates the Monte Carlo analysis developed in the next section.

6 Monte Carlo Estimation of Tail Risk

The previous sections assumed that the probability distribution of the financial loss L was either known or specified through a parametric model. In practical risk measurement, however, quantities such as Value at Risk and Expected Shortfall frequently need to be estimated numerically.

Monte Carlo simulation provides a general framework for this purpose. Rather than requiring a closed-form expression for the complete loss distribution, the method generates repeated realizations from the underlying stochastic model and constructs an empirical approximation to the distribution of losses.

This approach is particularly useful when portfolio losses depend on multiple risk factors, nonlinear instruments, stochastic dynamics, or other mechanisms for which analytical risk measures are unavailable [10].

The central computational pipeline is

$$\boxed{\text{Stochastic Model} \longrightarrow \text{Random Scenarios} \longrightarrow \text{Portfolio Losses} \longrightarrow \hat{F}_N \longrightarrow \widehat{\text{VaR}}_\alpha, \widehat{\text{ES}}_\alpha} \quad (236)$$

Monte Carlo estimation introduces an additional source of uncertainty: even if the probability model is correctly specified, a finite simulation sample produces statistical estimation error.

6.1 Monte Carlo Sampling

Let L denote a financial loss with cumulative distribution function F_L . A basic Monte Carlo experiment generates independent and identically distributed observations

$$L_1, L_2, \dots, L_N \stackrel{\text{iid}}{\sim} F_L. \quad (237)$$

Each L_i represents one possible realization of the financial loss under the assumed stochastic model.

For a parametric model

$$L = g(Z; \theta), \quad (238)$$

where Z represents the underlying random factors, simulation proceeds by generating

$$Z_1, \dots, Z_N \quad (239)$$

and evaluating

$$L_i = g(Z_i; \theta), \quad i = 1, \dots, N. \quad (240)$$

For example, under a Gaussian loss model,

$$L_i = \mu + \sigma Z_i, \quad Z_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (241)$$

Under the variance-standardized Student- t model introduced previously,

$$L_i = \mu + \sigma \sqrt{\frac{\nu - 2}{\nu}} T_{\nu,i}, \quad \nu > 2, \quad (242)$$

with

$$T_{\nu,i} \stackrel{\text{iid}}{\sim} t_\nu. \quad (243)$$

Thus the Monte Carlo machinery itself does not determine the loss model. It propagates the assumptions of the selected stochastic model into a sample of simulated losses.

This distinction is fundamental:

$$\boxed{\text{Monte Carlo error} \neq \text{model error.}} \quad (244)$$

Increasing N can reduce simulation error, but it cannot correct an incorrectly specified probability model.

6.2 The Empirical Distribution

Given simulated losses

$$L_1, \dots, L_N, \quad (245)$$

the empirical cumulative distribution function is

$$\boxed{\hat{F}_N(\ell) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}_{\{L_i \leq \ell\}}.} \quad (246)$$

For any fixed threshold ℓ ,

$$\hat{F}_N(\ell) \quad (247)$$

is simply the fraction of simulated losses that do not exceed ℓ .

Similarly, the empirical survival function is

$$\hat{\bar{F}}_N(\ell) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}_{\{L_i > \ell\}}. \quad (248)$$

The empirical distribution is the fundamental object from which nonparametric Monte Carlo estimates of VaR and Expected Shortfall are constructed.

6.3 Convergence of the Empirical Distribution

A fundamental theoretical justification for simulation-based risk estimation is provided by the Glivenko–Cantelli theorem. Under i.i.d. sampling,

$$\boxed{\sup_{\ell \in \mathbb{R}} \left| \hat{F}_N(\ell) - F_L(\ell) \right| \xrightarrow{\text{a.s.}} 0 \quad \text{as } N \rightarrow \infty.} \quad (249)$$

Thus the empirical distribution converges uniformly almost surely to the true loss distribution.

This provides the theoretical foundation for extracting risk measures from Monte Carlo samples: as the empirical distribution becomes a better approximation of F_L , appropriately defined empirical quantiles and tail functionals converge toward their population counterparts.

Monte Carlo risk assessment based on empirical distributions and its connection with Glivenko–Cantelli convergence are discussed in the simulation literature [10, 12].

6.4 Empirical Quantiles

Let

$$L_{(1)} \leq L_{(2)} \leq \dots \leq L_{(N)} \quad (250)$$

denote the order statistics of the Monte Carlo sample.

The empirical α -quantile is

$$\boxed{\hat{q}_{\alpha, N} = \hat{F}_N^{\leftarrow}(\alpha).} \quad (251)$$

Under the simple order-statistic convention,

$$\widehat{q}_{\alpha,N} = L_{(\lceil N\alpha \rceil)}. \quad (252)$$

Consequently, the empirical quantile is itself a random variable: different Monte Carlo samples generally produce different values of $\widehat{q}_{\alpha,N}$.

This distinction is essential:

$$q_\alpha = \text{population quantity}, \quad \widehat{q}_{\alpha,N} = \text{finite-sample estimator}. \quad (253)$$

6.5 Monte Carlo VaR Estimator

Because Value at Risk is a quantile,

$$\text{VaR}_\alpha(L) = F_L^{\leftarrow}(\alpha), \quad (254)$$

its natural Monte Carlo estimator is

$$\widehat{\text{VaR}}_{\alpha,N} = \widehat{F}_N^{\leftarrow}(\alpha). \quad (255)$$

Equivalently,

$$\widehat{\text{VaR}}_{\alpha,N} = L_{(\lceil N\alpha \rceil)} \quad (256)$$

under the order-statistic convention used here.

For example, if

$$N = 10\,000, \quad \alpha = 0.99, \quad (257)$$

then

$$\lceil N\alpha \rceil = 9900, \quad (258)$$

and therefore

$$\widehat{\text{VaR}}_{0.99} = L_{(9900)}. \quad (259)$$

Only approximately 100 simulated observations lie beyond this estimated quantile.

6.6 Monte Carlo Expected Shortfall Estimator

For a continuous loss distribution, Expected Shortfall can be written as

$$\text{ES}_\alpha(L) = \mathbb{E}[L \mid L > \text{VaR}_\alpha(L)]. \quad (260)$$

A natural Monte Carlo estimator is therefore

$$\widehat{\text{ES}}_{\alpha,N} = \frac{\sum_{i=1}^N L_i \mathbf{1}_{\{L_i > \widehat{\text{VaR}}_{\alpha,N}\}}}{\sum_{i=1}^N \mathbf{1}_{\{L_i > \widehat{\text{VaR}}_{\alpha,N}\}}}. \quad (261)$$

This expression makes the statistical difficulty of Expected Shortfall immediately visible: although the simulation may contain N observations, ES uses only observations located in the extreme tail.

For an order-statistic implementation with

$$k = \lceil N\alpha \rceil, \quad (262)$$

a simple tail-average estimator is

$$\widehat{\text{ES}}_{\alpha, N} = \frac{1}{N - k} \sum_{j=k+1}^N L_{(j)}, \quad (263)$$

when $N - k > 0$.

For finite samples and arbitrary values of $N\alpha$, more refined estimators may assign fractional weight to the observation at the empirical quantile so that exactly a fraction $1 - \alpha$ of empirical probability mass contributes to the tail average. This distinction becomes important for small samples and discrete loss distributions.

6.7 Tail Sample Size

The number of observations falling beyond the population VaR threshold is itself random.

Define

$$I_i = \mathbf{1}_{\{L_i > \text{VaR}_\alpha(L)\}}. \quad (264)$$

For a continuous loss distribution,

$$\mathbb{P}(I_i = 1) = 1 - \alpha. \quad (265)$$

Therefore,

$$N_{\text{tail}} = \sum_{i=1}^N I_i \quad (266)$$

satisfies

$$N_{\text{tail}} \sim \text{Binomial}(N, 1 - \alpha). \quad (267)$$

Consequently,

$$\mathbb{E}[N_{\text{tail}}] = N(1 - \alpha), \quad (268)$$

and

$$\text{Var}(N_{\text{tail}}) = N\alpha(1 - \alpha). \quad (269)$$

Hence,

$$\text{SD}(N_{\text{tail}}) = \sqrt{N\alpha(1 - \alpha)}. \quad (270)$$

This provides a precise interpretation of the commonly used approximation

$$N_{\text{tail}} \approx N(1 - \alpha). \quad (271)$$

6.8 How Much Tail Information Is Actually Available?

Consider several confidence levels.

At $\alpha = 0.95$,

$$\mathbb{E}[N_{\text{tail}}] = 0.05N. \quad (272)$$

At $\alpha = 0.99$,

$$\mathbb{E}[N_{\text{tail}}] = 0.01N. \quad (273)$$

At $\alpha = 0.999$,

$$\mathbb{E}[N_{\text{tail}}] = 0.001N. \quad (274)$$

For $N = 10\,000$, this corresponds approximately to

α	$\mathbb{E}[N_{\text{tail}}]$
0.95	500
0.99	100
0.999	10

(275)

Thus a simulation that appears large when measured by its total number of observations can become extremely small when measured by the number of observations relevant to extreme-tail estimation.

An especially revealing quantity is the probability that no observation at all falls beyond the population VaR threshold:

$$\boxed{\mathbb{P}(N_{\text{tail}} = 0) = \alpha^N.} \quad (276)$$

For very high confidence levels and small N , this probability need not be negligible.

6.9 Consistency

Under standard regularity conditions, empirical quantiles are consistent. If F_L is continuous and strictly increasing in a neighborhood of

$$q_\alpha = F_L^{-1}(\alpha), \quad (277)$$

then

$$\boxed{\widehat{\text{VaR}}_{\alpha,N} \xrightarrow{\text{a.s.}} \text{VaR}_\alpha(L)} \quad (278)$$

as

$$N \rightarrow \infty. \quad (279)$$

Under suitable integrability and regularity assumptions, the corresponding Expected Shortfall estimator also converges:

$$\boxed{\widehat{\text{ES}}_{\alpha,N} \longrightarrow \text{ES}_\alpha(L).} \quad (280)$$

Thus the instability observed for small N is not a contradiction of Monte Carlo convergence. It is a finite-sample phenomenon.

6.10 Sampling Error of VaR

Consistency alone does not describe how quickly an estimator stabilizes.

Suppose L possesses a continuous density f_L that is positive at

$$q_\alpha = \text{VaR}_\alpha(L). \quad (281)$$

Then the empirical quantile satisfies the asymptotic normality relation

$$\sqrt{N} \left(\widehat{\text{VaR}}_{\alpha,N} - \text{VaR}_\alpha(L) \right) \xrightarrow{d} \mathcal{N} \left(0, \frac{\alpha(1-\alpha)}{f_L(q_\alpha)^2} \right). \quad (282)$$

Consequently, for sufficiently large N ,

$$\text{SE} \left(\widehat{\text{VaR}}_{\alpha,N} \right) \approx \frac{\sqrt{\alpha(1-\alpha)}}{\sqrt{N} f_L(q_\alpha)}. \quad (283)$$

This formula reveals two distinct mechanisms controlling VaR precision.

First,

$$\text{SE} \propto \frac{1}{\sqrt{N}}, \quad (284)$$

which is the familiar Monte Carlo convergence rate.

Second, precision depends on the density at the target quantile,

$$f_L(q_\alpha). \quad (285)$$

If the density is small in the extreme tail, the quantile location is statistically difficult to estimate precisely.

Thus increasing the confidence level affects estimation not only because fewer observations occupy the tail, but also because the distribution may be locally sparse near the relevant quantile.

6.11 Sampling Error and the Monte Carlo Rate

For many standard Monte Carlo estimators, statistical uncertainty decreases at the characteristic rate

$$O(N^{-1/2}). \quad (286)$$

This rate has an important computational consequence.

If the standard error is approximately

$$\frac{C}{\sqrt{N}}, \quad (287)$$

then reducing the error by a factor of two requires approximately four times as many simulations:

$$\frac{C}{\sqrt{4N}} = \frac{1}{2} \frac{C}{\sqrt{N}}. \quad (288)$$

Similarly, reducing the error by a factor of ten may require approximately one hundred times as many simulations.

Hence brute-force Monte Carlo accuracy can become computationally expensive, especially when the quantity of interest is determined by rare events.

6.12 Why Extreme Expected Shortfall Is Difficult to Estimate

Expected Shortfall presents a particularly demanding estimation problem because two statistical uncertainties interact.

First, the tail boundary is unknown and must itself be estimated:

$$q_\alpha \longrightarrow \hat{q}_{\alpha,N}. \quad (289)$$

Second, after estimating that boundary, the observations beyond it must be used to estimate the average tail severity.

Schematically,

$$\boxed{\text{ES estimation} = \underbrace{\text{quantile estimation}}_{\text{Where does the tail begin?}} + \underbrace{\text{tail-mean estimation}}_{\text{How severe is the tail?}}.} \quad (290)$$

At high confidence levels,

$$1 - \alpha \ll 1, \quad (291)$$

so only a small fraction of the simulation contributes directly to the tail average.

For example, at

$$\alpha = 0.999, \quad (292)$$

a simulation with

$$N = 100\,000 \quad (293)$$

contains on average only

$$N(1 - \alpha) = 100 \quad (294)$$

observations beyond the population threshold.

Thus

$$\boxed{100\,000 \text{ simulations} \neq 100\,000 \text{ tail observations.}} \quad (295)$$

For tail estimation, the relevant effective sample size is of order

$$N_{\text{tail}} \approx N(1 - \alpha). \quad (296)$$

6.13 Heavy Tails Make the Problem Harder

The estimation problem becomes even more challenging when the underlying loss distribution is heavy-tailed.

Conditional on exceeding the VaR threshold, the observations entering the Expected Shortfall estimator may themselves exhibit substantial variability. A small number of extremely large losses can therefore have a strong effect on the estimated tail mean.

Consider

$$Y_\alpha = L \mid L > q_\alpha. \quad (297)$$

If the conditional variance exists, then a heuristic approximation to the standard error of a tail mean based on N_{tail} observations is

$$\text{SE} \approx \frac{\sqrt{\text{Var}(Y_\alpha)}}{\sqrt{N_{\text{tail}}}}. \quad (298)$$

Using

$$N_{\text{tail}} \approx N(1 - \alpha), \quad (299)$$

this suggests

$$\boxed{\text{SE} \approx \frac{\sigma_{\text{tail}}}{\sqrt{N(1 - \alpha)}}} \quad (300)$$

This expression is heuristic because the threshold is itself estimated, but it reveals the fundamental difficulty: extreme confidence levels simultaneously reduce the effective sample size and expose the estimator to increasingly extreme observations.

6.14 An Important Case: Student- t with $\nu = 3.2$

The Student- t experiment considered in this study provides an especially instructive example. For

$$\nu = 3.2, \quad (301)$$

the first moment exists because

$$\nu > 1, \quad (302)$$

and therefore Expected Shortfall is finite.

The variance also exists because

$$\nu > 2. \quad (303)$$

However, as shown previously, the fourth moment does not exist because

$$\nu < 4. \quad (304)$$

Consequently, the model simultaneously permits a well-defined VaR, a finite Expected Shortfall, and finite variance while retaining extremely strong higher-order tail behavior.

This helps explain the visual behavior of the Monte Carlo experiment: extreme observations can produce substantial finite-sample fluctuations in the estimated tail risk even though the estimators target finite population quantities.

6.15 Running Estimates and Convergence Diagnostics

The convergence process can be visualized through running estimates.

For each intermediate sample size

$$n \leq N, \quad (305)$$

define

$$\widehat{\text{VaR}}_{\alpha,n} \quad (306)$$

and

$$\widehat{\text{ES}}_{\alpha,n}. \quad (307)$$

Plotting these quantities against n produces a convergence path.

For small n , the estimators may fluctuate substantially because only a small number of observations determine the empirical tail.

As n increases, the empirical distribution becomes more stable and the running estimates typically fluctuate within progressively narrower regions around their population values.

This is the mechanism represented in the computational visualization:

$$\begin{array}{lcl}
 N \uparrow & \implies & N_{\text{tail}} \uparrow, \\
 & \implies & \text{greater tail information,} \\
 & \implies & \text{smaller sampling fluctuations.}
 \end{array} \tag{308}$$

6.16 Monte Carlo Error versus Model Risk

A numerically stable Monte Carlo estimator should not be confused with an accurate model of financial reality.

Suppose

$$\widehat{\text{ES}}_{\alpha, N} \longrightarrow \text{ES}_{\alpha}^{\text{model}} \tag{309}$$

as

$$N \rightarrow \infty. \tag{310}$$

This convergence demonstrates that the simulation accurately estimates the Expected Shortfall implied by the selected model. It does not establish that

$$\text{ES}_{\alpha}^{\text{model}} \tag{311}$$

is an accurate representation of real-world financial risk.

The distinction can be expressed as

$$\underbrace{\widehat{\rho}_N - \rho_{\text{model}}}_{\text{Monte Carlo error}} \quad \text{versus} \quad \underbrace{\rho_{\text{model}} - \rho_{\text{true}}}_{\text{model risk}}. \tag{312}$$

Increasing N addresses the first term but not the second.

This observation connects the present section with the preceding comparison between Gaussian and heavy-tailed models. An arbitrarily large Gaussian Monte Carlo simulation can estimate Gaussian tail risk with great numerical precision while still understating risk if the Gaussian model provides an inadequate description of the relevant extreme-loss behavior.

6.17 Computational Efficiency and Rare-Event Methods

Brute-force Monte Carlo is conceptually simple but may be inefficient for very rare events. If

$$1 - \alpha \tag{313}$$

is extremely small, most simulated scenarios contribute little direct information about the tail region of interest.

This motivates variance-reduction and rare-event techniques such as importance sampling, stratification, control variates, and related simulation methods [10].

For complex derivative portfolios, risk measurement may require an additional layer of simulation. An outer simulation generates risk-factor scenarios at the risk horizon, while an inner simulation revalues the portfolio conditional on each outer scenario. This produces a nested Monte Carlo problem. Gordy and Juneja analyze the allocation of a fixed computational budget between these layers for portfolio risk measures including VaR and Expected Shortfall [11].

The distinction between ordinary and nested simulation can be summarized as

Standard Monte Carlo: scenario $\rightarrow$ loss, Nested Monte Carlo: outer scenario $\rightarrow$ inner valuation $\rightarrow$ loss.	(314)
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6.18 From Statistical Tail Risk to Financial Shocks

Monte Carlo simulation completes the statistical part of the analysis.

The preceding sections established the sequence

Loss Model $\rightarrow$ Tail Distribution $\rightarrow$ VaR_α $\rightarrow$ ES_α $\rightarrow$ Monte Carlo Estimation.	(315)
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The next question is qualitatively different.

Until this point, an extreme loss has been treated primarily as an outcome of a probability distribution. In an interconnected financial system, however, such a loss may alter an institution's balance sheet, exhaust capital buffers, trigger default, and transmit losses to counterparties.

Thus the next transition is

Estimated Tail Event $\rightarrow$ Financial Shock $\rightarrow$ Default $\rightarrow$ Network Contagion.	(316)
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This transition moves the analysis from the statistical estimation of individual risk to the endogenous propagation of losses across a financial network.

7 From Individual Risk to Systemic Risk

The preceding analysis considered financial risk primarily from the perspective of a single loss variable. Value at Risk and Expected Shortfall characterize the distribution of extreme losses, while Monte Carlo simulation provides a numerical mechanism for estimating these quantities.

Financial systems, however, are composed of interacting institutions. A loss experienced by one institution may therefore have consequences beyond its own balance sheet. If the institution is connected to other institutions through contractual obligations, interbank loans, securities holdings, or other financial exposures, deterioration in its financial condition may transmit losses to its counterparties.

This observation introduces a fundamental distinction between individual and systemic risk:

Individual risk : How large can the losses of one institution become? Systemic risk : How can losses propagate through an interconnected system?	(317)
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Network models provide a natural mathematical framework for studying this transition. Financial institutions are represented by nodes, financial relationships by directed weighted edges, and balance-sheet variables determine whether transmitted losses can be absorbed or lead to further distress.

The resulting conceptual pipeline is

External Shock $\rightarrow$ Institutional Loss $\rightarrow$ Capital Erosion $\rightarrow$ Default $\rightarrow$ Counterparty Losses $\rightarrow$ Systemic Contagion.	(318)
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This perspective underlies a broad class of models of financial contagion, clearing mechanisms, and default cascades [7, 9].

7.1 Balance-Sheet Representation

Consider a financial system containing n institutions,

$$i = 1, \dots, n. \quad (319)$$

A stylized balance sheet for institution i separates assets and liabilities into external and interbank components.

Let

$$A_i^E \quad (320)$$

denote external assets and

$$A_i^{IB} \quad (321)$$

denote interbank assets.

Total assets are therefore

$$\boxed{A_i = A_i^E + A_i^{IB}.} \quad (322)$$

Similarly, let

$$L_i^E \quad (323)$$

denote external liabilities and

$$L_i^{IB} \quad (324)$$

denote interbank liabilities.

Then total liabilities are

$$\boxed{L_i = L_i^E + L_i^{IB}.} \quad (325)$$

The institution's equity, net worth, or capital buffer is

$$\boxed{K_i = A_i - L_i.} \quad (326)$$

Equivalently,

$$K_i = A_i^E + A_i^{IB} - L_i^E - L_i^{IB}. \quad (327)$$

The balance-sheet identity may therefore be written as

$$\boxed{A_i = L_i + K_i.} \quad (328)$$

This simple identity becomes the local mechanism through which shocks are translated into financial distress.

7.2 Capital Buffers

Capital provides a first layer of protection against losses.

Suppose institution i initially has capital

$$K_i^{(0)} > 0 \quad (329)$$

and experiences a loss

$$\Delta_i \geq 0. \quad (330)$$

After the loss, its capital becomes

$$\boxed{K_i^{(1)} = K_i^{(0)} - \Delta_i.} \quad (331)$$

As long as

$$\Delta_i < K_i^{(0)}, \quad (332)$$

the loss can be absorbed by the institution's initial capital buffer.

If instead

$$\Delta_i \geq K_i^{(0)}, \quad (333)$$

the institution reaches or crosses the insolvency boundary under the simplified balance-sheet criterion.

Thus,

$$\boxed{\Delta_i \begin{cases} < K_i^{(0)} & \implies \text{capital absorbs the shock,} \\ \geq K_i^{(0)} & \implies \text{insolvency threshold is reached.} \end{cases}} \quad (334)$$

This introduces an important nonlinearity. A shock does not necessarily produce default immediately. Its effect depends on the size of the shock relative to the institution's loss-absorbing capacity.

7.3 Capital Ratios and Relative Vulnerability

Absolute capital alone does not completely describe financial vulnerability. Institutions of different sizes may possess very different balance sheets while maintaining similar relative capitalization.

A simple capital ratio is

$$\boxed{c_i = \frac{K_i}{A_i}.} \quad (335)$$

A lower value of c_i implies that a smaller loss relative to total assets is sufficient to exhaust the capital buffer.

If an external shock destroys a fraction θ_i of total assets,

$$\Delta_i = \theta_i A_i, \quad (336)$$

then the institution becomes insolvent whenever

$$\theta_i A_i \geq K_i. \quad (337)$$

Dividing by $A_i > 0$ gives

$$\boxed{\theta_i \geq c_i.} \quad (338)$$

Hence the capital ratio provides a natural normalized threshold for balance-sheet shocks.

7.4 Interbank Exposures

The transition from individual to systemic risk occurs when balance sheets are interconnected.
Let

$$W_{ij} \geq 0 \quad (339)$$

denote the exposure of institution i to institution j .
We adopt the convention

$$\boxed{W_{ij} = \text{amount owed by } j \text{ to } i.} \quad (340)$$

Thus W_{ij} is an asset of institution i and a liability of institution j .
Under this convention, the total interbank assets of institution i are

$$\boxed{A_i^{IB} = \sum_{j=1}^n W_{ij},} \quad (341)$$

whereas its total interbank liabilities are

$$\boxed{L_i^{IB} = \sum_{j=1}^n W_{ji}.} \quad (342)$$

Usually,

$$W_{ii} = 0, \quad (343)$$

since self-exposures are excluded.
The matrix

$$\boxed{W = (W_{ij})_{i,j=1}^n} \quad (344)$$

therefore contains the bilateral structure of the interbank system.

7.5 Directed Weighted Financial Networks

The exposure matrix defines a directed weighted graph

$$\boxed{\mathcal{G} = (V, E, W),} \quad (345)$$

where

$$V = \{1, \dots, n\} \quad (346)$$

is the set of financial institutions and

$$E = \{(j, i) : W_{ij} > 0\} \quad (347)$$

is the set of financial relationships under the exposure convention defined above.
Because W_{ij} represents an amount owed by j to i , the edge

$$j \longrightarrow i \quad (348)$$

represents the contractual payment direction from debtor j to creditor i .
The network is *directed* because generally

$$W_{ij} \neq W_{ji}, \quad (349)$$

and it is *weighted* because exposures may differ substantially in magnitude.

This distinction is essential. Two financial networks can possess the same unweighted topology while producing different contagion dynamics if their exposure weights differ.

7.6 Degree and Exposure Strength

Several elementary network quantities help characterize the structure of the system.

Under the convention adopted above, the number of counterparties to which institution i is exposed is

$$k_i^{\text{in}} = \sum_{j=1}^n \mathbf{1}_{\{W_{ij} > 0\}}, \quad (350)$$

when graph arrows follow the payment direction $j \rightarrow i$.
Similarly,

$$k_i^{\text{out}} = \sum_{j=1}^n \mathbf{1}_{\{W_{ji} > 0\}} \quad (351)$$

counts institutions exposed to i through claims on it.

For weighted networks, the corresponding exposure magnitudes are often more informative:

$$s_i^{\text{assets}} = \sum_{j=1}^n W_{ij} = A_i^{IB}, \quad (352)$$

and

$$s_i^{\text{liabilities}} = \sum_{j=1}^n W_{ji} = L_i^{IB}. \quad (353)$$

Connectivity therefore has both a topological and a balance-sheet dimension.

7.7 External Shocks

Interbank exposures determine how distress may propagate, but a cascade must generally be initiated by some disturbance.

Let

$$S_i \geq 0 \quad (354)$$

denote an exogenous loss imposed on institution i .

A convenient vector representation is

$$\mathbf{S} = (S_1, \dots, S_n)^\top. \quad (355)$$

If the shock acts directly on external assets, then

$$A_i^{E,(1)} = A_i^{E,(0)} - S_i. \quad (356)$$

Before any interbank contagion occurs, capital becomes

$$K_i^{(1)} = K_i^{(0)} - S_i. \quad (357)$$

An institution is initially insolvent under the simplified balance-sheet criterion if

$$S_i \geq K_i^{(0)}. \quad (358)$$

7.8 Proportional External-Asset Shocks

Instead of specifying an absolute loss, an external shock can be expressed as a fraction of external assets.

Let

$$\theta_i \in [0, 1] \quad (359)$$

denote the fraction of external assets lost by institution i .

Then

$$S_i = \theta_i A_i^E. \quad (360)$$

The post-shock capital is

$$K_i^{(1)} = K_i^{(0)} - \theta_i A_i^E. \quad (361)$$

Initial default occurs when

$$\theta_i A_i^E \geq K_i^{(0)}. \quad (362)$$

Equivalently, if $A_i^E > 0$,

$$\theta_i \geq \frac{K_i^{(0)}}{A_i^E}. \quad (363)$$

This defines an institution-specific critical shock threshold.

7.9 Single and Multiple Shocks

The shock vector allows several stress scenarios to be represented within the same framework.

A single-institution shock to institution k may be written as

$$\mathbf{S} = S \mathbf{e}_k, \quad (364)$$

where $\mathbf{e}_k$ is the k th canonical basis vector.

A simultaneous shock to several institutions can be represented as

$$\mathbf{S} = \sum_{k \in \mathcal{S}} S_k \mathbf{e}_k, \quad (365)$$

where $\mathcal{S}$ denotes the initially shocked set.

A common proportional shock can similarly be expressed as

$$S_i = \theta A_i^E, \quad (366)$$

for all institutions belonging to the shocked set.

These alternatives distinguish idiosyncratic disturbances from common or system-wide stress events.

7.10 From External Losses to Counterparty Losses

Suppose institution j becomes unable to honor its interbank obligations.

Because institution i holds a claim of size

$$W_{ij} \quad (367)$$

on institution j , default of j may impose a loss on i .

Let

$$R_j \in [0, 1] \quad (368)$$

denote the recovery fraction associated with the obligations of defaulting institution j .

Then a simple loss-given-default specification gives

$$\Delta_{i \leftarrow j} = (1 - R_j)W_{ij}. \quad (369)$$

Defining

$$LGD_j = 1 - R_j, \quad (370)$$

we equivalently obtain

$$\Delta_{i \leftarrow j} = LGD_j W_{ij}. \quad (371)$$

If several counterparties default, the direct credit losses received by institution i can be represented as

$$\Delta_i^{IB} = \sum_{j \in \mathcal{D}} (1 - R_j)W_{ij}, \quad (372)$$

where $\mathcal{D}$ denotes the set of defaulted institutions.

7.11 Capital as the Transmission Threshold

Combining the external shock and interbank losses gives a simplified post-loss capital position

$$K_i^{\text{post}} = K_i^{(0)} - S_i - \Delta_i^{IB}. \quad (373)$$

Substituting the counterparty-loss expression,

$$K_i^{\text{post}} = K_i^{(0)} - S_i - \sum_{j \in \mathcal{D}} (1 - R_j)W_{ij}. \quad (374)$$

The simplified default condition becomes

$$S_i + \sum_{j \in \mathcal{D}} (1 - R_j)W_{ij} \geq K_i^{(0)}. \quad (375)$$

This equation contains the essential mechanism of a default cascade.

An institution may survive the initial external shock,

$$S_i < K_i^{(0)}, \quad (376)$$

but fail after receiving losses from defaulted counterparties:

$$S_i + \Delta_i^{IB} \geq K_i^{(0)}. \quad (377)$$

Thus,

$$\boxed{\text{individually survivable shock} + \text{network losses} \implies \text{default.}} \quad (378)$$

7.12 Why Connectivity Can Both Absorb and Transmit Risk

Interbank connectivity does not have a universally stabilizing or destabilizing effect.

Diversification across many counterparties can reduce dependence on a single borrower. At the same time, interconnections create channels through which distress can propagate across institutions.

The systemic consequences therefore depend not only on the number of links, but also on quantities such as

$$W_{ij}, \quad K_i, \quad R_i, \quad (379)$$

and on the topology of the network itself.

This motivates the distinction

$$\boxed{\text{network structure} + \text{exposure size} + \text{capitalization} + \text{shock magnitude} \longrightarrow \text{systemic outcome.}} \quad (380)$$

Network topology alone is therefore insufficient to characterize financial stability.

7.13 From Individual Tail Risk to Network Vulnerability

The connection with the preceding sections can now be made explicit.

Suppose institution i has a stochastic external loss

$$L_i. \quad (381)$$

Its individual tail risk may be characterized by

$$\text{VaR}_\alpha(L_i) \quad (382)$$

or

$$\text{ES}_\alpha(L_i). \quad (383)$$

However, whether a realized loss produces systemic consequences depends on the institution's balance sheet and network position.

The relevant comparison is therefore not merely

$$L_i \text{ versus } \text{VaR}_\alpha(L_i), \quad (384)$$

but also

$$\boxed{L_i \text{ versus } K_i.} \quad (385)$$

Once

$$L_i \geq K_i, \quad (386)$$

the institution may become a source of losses to other nodes through its interbank liabilities. This gives the conceptual transition

$$\begin{array}{ccccccc}
\underbrace{L_i}_{\text{individual loss}} & \longrightarrow & \underbrace{K_i - L_i}_{\text{balance-sheet impact}} & \longrightarrow & \underbrace{\text{default of } i}_{\text{threshold event}} & \longrightarrow & \underbrace{W_{ji}}_{\text{network transmission}} \longrightarrow \underbrace{\text{losses of counterparties.}}_{\text{systemic effect}}
\end{array}
\tag{387}$$

Individual tail risk and systemic risk are therefore related but distinct. A large individual loss becomes systemically relevant when it interacts with insufficient loss-absorbing capacity and a network capable of transmitting distress.

7.14 A Minimal Network Contagion State

The system can now be summarized by three fundamental objects.

First, the vector of capital buffers,

$$\mathbf{K} = (K_1, \dots, K_n)^\top. \tag{388}$$

Second, the matrix of bilateral exposures,

$$W = (W_{ij})_{n \times n}. \tag{389}$$

Third, the vector of external shocks,

$$\mathbf{S} = (S_1, \dots, S_n)^\top. \tag{390}$$

Thus a minimal representation of the financial system is

$$\mathcal{S} = (\mathbf{K}, W, \mathbf{S}). \tag{391}$$

Given a recovery specification, these quantities determine the direct balance-sheet mechanism through which an initial disturbance may generate subsequent defaults.

7.15 Transition to Default Cascades

The ingredients required for contagion have now been established:

Nodes	$\longleftrightarrow$	financial institutions,
Edges	$\longleftrightarrow$	interbank obligations,
Weights	$\longleftrightarrow$	exposure magnitudes,
Capital	$\longleftrightarrow$	loss-absorption capacity,
External shocks	$\longleftrightarrow$	initial disturbances.

(392)

The remaining question is dynamic.

If an initial set of institutions defaults at time $t = 0$, which counterparties fail at $t = 1$? Which institutions subsequently fail at $t = 2$? Does the process terminate after a small number of defaults, or does it generate a large systemic cascade?

This produces the transition

$$\text{External Shock} \rightarrow \mathcal{D}_0 \rightarrow \mathcal{D}_1 \rightarrow \mathcal{D}_2 \rightarrow \dots \rightarrow \mathcal{D}_\infty, \tag{393}$$

where $\mathcal{D}_t$ denotes the set of institutions that have defaulted by cascade round t .

The problem has therefore changed fundamentally. We are no longer asking only how probable an extreme loss is. We are asking how the financial system transforms an initial loss into an endogenous sequence of defaults.

This is the starting point for the default-cascade dynamics developed in the following section.

8 Default Cascades

The previous section introduced the balance-sheet and network structure required for systemic loss transmission. We now turn to the dynamic mechanism through which an initial set of defaults may generate subsequent defaults across the financial network.

Consider a system of n financial institutions. Let

$$\mathbf{K} = (K_1, \dots, K_n)^\top \quad (394)$$

denote the vector of initial capital buffers,

$$\mathbf{S} = (S_1, \dots, S_n)^\top \quad (395)$$

the vector of external losses, and

$$W = (W_{ij})_{n \times n} \quad (396)$$

the interbank exposure matrix.

As in the previous section, the convention

$$\boxed{W_{ij} = \text{claim of institution } i \text{ on institution } j} \quad (397)$$

is maintained throughout.

Thus, if institution j defaults, institution i may suffer a loss on the claim W_{ij} .

The fundamental mechanism is

$$\boxed{\text{Initial Shock} \longrightarrow \text{Initial Defaults} \longrightarrow \text{Counterparty Losses} \longrightarrow \text{New Defaults} \longrightarrow \dots} \quad (398)$$

The process terminates when an iteration produces no additional defaults.

8.1 Default Indicator

For each institution i , define the cumulative default indicator at cascade generation t by

$$\boxed{D_i^{(t)} = \begin{cases} 1, & \text{if institution } i \text{ has defaulted by generation } t, \\ 0, & \text{otherwise.} \end{cases}} \quad (399)$$

Collecting the indicators gives the state vector

$$\boxed{\mathbf{D}^{(t)} = (D_1^{(t)}, \dots, D_n^{(t)})^\top \in \{0, 1\}^n.} \quad (400)$$

The corresponding cumulative set of defaulted institutions is

$$\boxed{\mathcal{D}_t = \{i : D_i^{(t)} = 1\}.} \quad (401)$$

Because default is treated as absorbing in the present cascade model,

$$\boxed{D_i^{(t+1)} \geq D_i^{(t)}} \quad (402)$$

for every institution and every generation.

Equivalently,

$$\mathcal{D}_t \subseteq \mathcal{D}_{t+1}. \quad (403)$$

8.2 Initial Defaults

Before contagion begins, institutions are affected only by the external shock vector.

Institution i initially defaults if its external loss exhausts its capital buffer:

$$S_i \geq K_i. \quad (404)$$

Thus,

$$D_i^{(0)} = \mathbf{1}_{\{S_i \geq K_i\}}. \quad (405)$$

The initially defaulted set is therefore

$$\mathcal{D}_0 = \{i : S_i \geq K_i\}. \quad (406)$$

This set constitutes the seed of the contagion process.

8.3 Loss Propagation

Suppose institution j has defaulted.

Let

$$R_j \in [0, 1] \quad (407)$$

denote its recovery fraction and define

$$LGD_j = 1 - R_j. \quad (408)$$

The counterparty loss transmitted from j to creditor i is

$$\Delta_{i \leftarrow j} = (1 - R_j)W_{ij} = LGD_j W_{ij}. \quad (409)$$

Given the cumulative default state at generation t , the total interbank loss received by institution i is

$$C_i^{(t)} = \sum_{j=1}^n (1 - R_j)W_{ij}D_j^{(t)}. \quad (410)$$

The total loss affecting institution i is therefore

$$\Lambda_i^{(t)} = S_i + C_i^{(t)}. \quad (411)$$

Hence,

$$\Lambda_i^{(t)} = S_i + \sum_{j=1}^n (1 - R_j)W_{ij}D_j^{(t)}. \quad (412)$$

8.4 Matrix Representation

Let

$$\mathbf{r} = (R_1, \dots, R_n)^\top \quad (413)$$

and define the vector of loss-given-default values

$$\boldsymbol{\lambda} = \mathbf{1} - \mathbf{r}. \quad (414)$$

Let

$$\Lambda_R = \text{diag}(1 - R_1, \dots, 1 - R_n). \quad (415)$$

Then the vector of cumulative interbank losses can be written compactly as

$$\mathbf{C}^{(t)} = W \Lambda_R \mathbf{D}^{(t)}. \quad (416)$$

Consequently,

$$\mathbf{\Lambda}^{(t)} = \mathbf{S} + W \Lambda_R \mathbf{D}^{(t)}. \quad (417)$$

This representation makes explicit the three ingredients governing direct default contagion:

$$\boxed{\text{shock vector} + \text{network exposures} + \text{loss given default.}} \quad (418)$$

8.5 Iterative Contagion

At each cascade generation, the total accumulated loss is compared with the institution's capital buffer.

Institution i is in default at generation $t + 1$ whenever

$$\Lambda_i^{(t)} \geq K_i. \quad (419)$$

Therefore,

$$D_i^{(t+1)} = \mathbf{1} \left\{ S_i + \sum_{j=1}^n (1 - R_j) W_{ij} D_j^{(t)} \geq K_i \right\}. \quad (420)$$

Because default is absorbing, an equivalent explicit formulation is

$$D_i^{(t+1)} = \max \left\{ D_i^{(t)}, \mathbf{1} \left[S_i + \sum_{j=1}^n (1 - R_j) W_{ij} D_j^{(t)} \geq K_i \right] \right\}. \quad (421)$$

Under the present nonnegative-exposure specification, the cumulative-loss formulation is monotone, so once an institution defaults it cannot return to the solvent state.

8.6 The Contagion Operator

It is useful to express the cascade as an operator acting on the binary default state.

Define

$$\mathcal{T} : \{0, 1\}^n \longrightarrow \{0, 1\}^n \quad (422)$$

componentwise by

$$\mathcal{T}_i(\mathbf{D}) = \mathbf{1} \left\{ S_i + \sum_{j=1}^n (1 - R_j) W_{ij} D_j \geq K_i \right\}. \quad (423)$$

The cascade dynamics are therefore

$$\mathbf{D}^{(t+1)} = \mathcal{T}(\mathbf{D}^{(t)}). \quad (424)$$

With the absorbing-state convention, one may instead define

$$\tilde{\mathcal{T}}(\mathbf{D}) = \mathbf{D} \vee \mathcal{T}(\mathbf{D}), \quad (425)$$

where $\vee$ denotes the componentwise maximum, giving

$$\boxed{\mathbf{D}^{(t+1)} = \tilde{\mathcal{T}}(\mathbf{D}^{(t)})}. \quad (426)$$

The network contagion problem has thus been transformed into the iteration of a nonlinear threshold operator on a finite state space.

8.7 Cascade Generations

The cumulative default set

$$\mathcal{D}_t \quad (427)$$

should be distinguished from the set of institutions that default *for the first time* at generation t .

Define

$$\boxed{\mathcal{G}_0 = \mathcal{D}_0}, \quad (428)$$

and, for $t \geq 1$,

$$\boxed{\mathcal{G}_t = \mathcal{D}_t \setminus \mathcal{D}_{t-1}}. \quad (429)$$

Thus,

$$\mathcal{G}_t \quad (430)$$

contains precisely the institutions newly defaulting during contagion round t .
The cumulative set satisfies

$$\boxed{\mathcal{D}_t = \bigcup_{s=0}^t \mathcal{G}_s}. \quad (431)$$

This gives a natural temporal decomposition of the cascade:

$$\boxed{\mathcal{G}_0 \rightarrow \mathcal{G}_1 \rightarrow \mathcal{G}_2 \rightarrow \cdots \rightarrow \mathcal{G}_T}. \quad (432)$$

Here,

- $\mathcal{G}_0$ contains the institutions directly affected by the initial shock;
- $\mathcal{G}_1$ contains defaults caused by losses from $\mathcal{G}_0$;
- $\mathcal{G}_2$ contains subsequent defaults;
- and the process continues until no new defaults occur.

8.8 Incremental versus Cumulative Loss Accounting

The cumulative formulation above charges each defaulted counterparty's loss through the cumulative state $\mathbf{D}^{(t)}$.

An equivalent implementation may propagate only newly generated losses.

Define the new-default indicator

$$\Delta D_j^{(t)} = D_j^{(t)} - D_j^{(t-1)}, \quad (433)$$

for $t \geq 1$, with

$$\Delta D_j^{(0)} = D_j^{(0)}. \quad (434)$$

The losses newly transmitted during generation t are

$$\Delta C_i^{(t)} = \sum_{j=1}^n (1 - R_j) W_{ij} \Delta D_j^{(t)}. \quad (435)$$

Cumulative interbank losses then satisfy

$$C_i^{(t)} = \sum_{s=0}^t \Delta C_i^{(s)}. \quad (436)$$

This formulation is particularly useful computationally because each default transmits its loss exactly once.

8.9 A Simple Three-Generation Cascade

To illustrate the mechanism, suppose an external shock initially causes institution 1 to fail:

$$\mathcal{G}_0 = \{1\}. \quad (437)$$

Its default generates losses on institutions 2 and 3.

Suppose institution 2 remains solvent but institution 3 crosses its capital threshold. Then

$$\mathcal{G}_1 = \{3\}. \quad (438)$$

The failure of institution 3 may subsequently transmit losses to institution 5. If these losses exhaust the capital of institution 5,

$$\mathcal{G}_2 = \{5\}. \quad (439)$$

If institution 5 produces no additional defaults, then

$$\mathcal{G}_3 = \emptyset. \quad (440)$$

The terminal cumulative default set is

$$\mathcal{D}_\infty = \{1, 3, 5\}. \quad (441)$$

This elementary example illustrates how a single exogenous disturbance can generate several endogenous rounds of distress.

8.10 Finite Termination

Because default is absorbing,

$$\mathcal{D}_t \subseteq \mathcal{D}_{t+1}. \quad (442)$$

At every nonterminal iteration, at least one new institution must default:

$$\mathcal{D}_{t+1} \supsetneq \mathcal{D}_t. \quad (443)$$

Since the financial system contains only n institutions, there can be at most n default additions.

Therefore the iteration reaches a terminal state after finitely many rounds.

More precisely, if $\mathcal{D}_0$ is nonempty, the number of subsequent nonempty generations cannot exceed

$$n - |\mathcal{D}_0|. \quad (444)$$

Thus there exists a finite T such that

$$\boxed{\mathbf{D}^{(T+1)} = \mathbf{D}^{(T)}}. \quad (445)$$

8.11 Fixed Point

The terminal default state is a fixed point of the contagion operator.

Let

$$\mathbf{D}^* \quad (446)$$

denote the final default vector.

Then

$$\boxed{\mathbf{D}^* = \tilde{\mathcal{T}}(\mathbf{D}^*)}. \quad (447)$$

Equivalently,

$$\mathcal{D}^* = \mathcal{D}_\infty \quad (448)$$

is the state for which applying another contagion round creates no new defaults.

Hence,

$$\boxed{\mathcal{G}_{T+1} = \emptyset}. \quad (449)$$

The cascade has then exhausted all defaults implied by the initial shock, capital configuration, exposure network, and recovery assumptions of the model.

8.12 Least Fixed Point Reached from the Initial Shock

Because the contagion operator is monotone under nonnegative exposures and loss-given-default values, the iteration starting from the initial default state generates an increasing sequence

$$\mathbf{D}^{(0)} \leq \mathbf{D}^{(1)} \leq \mathbf{D}^{(2)} \leq \dots. \quad (450)$$

The terminal state reached by this procedure is therefore the smallest absorbing fixed point containing the initial default configuration.

Symbolically,

$$\mathbf{D}^* = \lim_{t \rightarrow \infty} \mathbf{D}^{(t)}. \quad (451)$$

Since the state space is finite, this limit is attained after finitely many iterations rather than only asymptotically.

8.13 Default Fraction

A basic measure of cascade magnitude is the fraction of institutions that have defaulted by generation t :

$$\phi_t = \frac{1}{n} \sum_{i=1}^n D_i^{(t)} = \frac{|\mathcal{D}_t|}{n}. \quad (452)$$

By construction,

$$0 \leq \phi_t \leq 1 \quad (453)$$

and

$$\phi_{t+1} \geq \phi_t. \quad (454)$$

The terminal default fraction is

$$\phi_\infty = \frac{|\mathcal{D}_\infty|}{n}. \quad (455)$$

This quantity measures the final systemic extent of the cascade in terms of the number of failed institutions.

8.14 Endogenous Cascade Size

If the objective is to distinguish externally triggered defaults from defaults caused by network propagation, define

$$C = |\mathcal{D}_\infty| - |\mathcal{D}_0|. \quad (456)$$

The normalized endogenous cascade size is

$$c = \frac{|\mathcal{D}_\infty| - |\mathcal{D}_0|}{n}. \quad (457)$$

Thus,

$$C = 0 \quad (458)$$

means that the initial shock produces no secondary defaults, whereas

$$C > 0 \quad (459)$$

indicates endogenous amplification through the financial network.

This distinction separates the direct impact of the shock from the additional damage generated by interconnectedness.

8.15 Cascade Amplification

Another useful quantity compares the final number of defaults with the number of initially defaulted institutions.

If

$$|\mathcal{D}_0| > 0, \quad (460)$$

define the default amplification factor

$$A_D = \frac{|\mathcal{D}_\infty|}{|\mathcal{D}_0|}. \quad (461)$$

Then

$$A_D = 1 \quad (462)$$

corresponds to no contagion beyond the initial defaults, while

$$A_D > 1 \quad (463)$$

indicates network amplification.

This statistic should be interpreted jointly with the absolute and normalized cascade sizes, since the same amplification factor can correspond to very different numbers of affected institutions.

8.16 Cascade Depth

Cascade size and cascade depth measure different aspects of systemic contagion.

Define the cascade depth as the index of the last nonempty generation:

$$T_{\text{depth}} = \max\{t : \mathcal{G}_t \neq \emptyset\}. \quad (464)$$

Under this convention, the initial shock-induced defaults belong to generation zero. Therefore,

$$T_{\text{depth}} = 0 \quad (465)$$

means that no secondary default occurs.

If

$$T_{\text{depth}} = 1, \quad (466)$$

there is one generation of secondary defaults.

More generally,

$$T_{\text{depth}} = k \quad (467)$$

means that the cascade propagates through k endogenous rounds after the initial default generation.

Thus,

$$\begin{array}{l} \phi_\infty \longrightarrow \text{final cascade magnitude,} \\ T_{\text{depth}} \longrightarrow \text{propagation depth.} \end{array} \quad (468)$$

8.17 Cascade Width

The generation decomposition also permits the definition of cascade width.

Let

$$w_t = |\mathcal{G}_t| \quad (469)$$

denote the number of newly defaulting institutions in generation t .

The maximum cascade width is then

$$W_{\text{cascade}} = \max_t |\mathcal{G}_t|. \quad (470)$$

A cascade may therefore be:

- shallow and wide, with many institutions failing almost simultaneously;
- deep and narrow, with distress propagating through several successive generations;
- or both deep and wide.

The final default fraction alone cannot distinguish these propagation patterns.

8.18 A Compact Cascade Algorithm

The complete mechanism can be summarized as follows.

1. Initialize the system using the external shock:

$$D_i^{(0)} = \mathbf{1}_{\{S_i \geq K_i\}}. \quad (471)$$

2. Compute cumulative losses generated by currently defaulted counterparties:

$$C_i^{(t)} = \sum_j (1 - R_j) W_{ij} D_j^{(t)}. \quad (472)$$

3. Update the default state:

$$D_i^{(t+1)} = \max \left\{ D_i^{(t)}, \mathbf{1}_{\{S_i + C_i^{(t)} \geq K_i\}} \right\}. \quad (473)$$

4. Identify the newly defaulted generation:

$$\mathcal{G}_{t+1} = \mathcal{D}_{t+1} \setminus \mathcal{D}_t. \quad (474)$$

5. Stop when

$$\mathcal{G}_{t+1} = \emptyset. \quad (475)$$

The output is the terminal state

$$\mathbf{D}^*, \quad (476)$$

from which the final default fraction, endogenous cascade size, cascade depth, and other systemic-risk diagnostics can be computed.

8.19 Deterministic and Stochastic Cascade Experiments

For a fixed network W , fixed capital vector $\mathbf{K}$, fixed recovery rates, and fixed external shock $\mathbf{S}$, the threshold mechanism above produces a deterministic cascade.

In a stochastic experiment, one or more of these objects may instead vary across Monte Carlo realizations. For simulation index

$$m = 1, \dots, M, \quad (477)$$

one obtains a terminal default fraction

$$\phi_{\infty}^{(m)}. \quad (478)$$

The expected final default fraction can then be estimated by

$$\widehat{\mathbb{E}[\phi_{\infty}]} = \frac{1}{M} \sum_{m=1}^M \phi_{\infty}^{(m)}. \quad (479)$$

Likewise, the expected number of defaults is estimated by

$$\widehat{\mathbb{E}[N_D]} = \frac{1}{M} \sum_{m=1}^M |\mathcal{D}_{\infty}^{(m)}|. \quad (480)$$

This provides a direct connection between the Monte Carlo methodology of the previous sections and network-based systemic-risk analysis.

8.20 Default Cascades as Endogenous Risk Amplification

The distinction between the initial shock and the terminal cascade is central to systemic-risk analysis.

The initial state

$$\mathcal{D}_0 \quad (481)$$

is generated directly by the exogenous disturbance.

The additional defaults

$$\mathcal{D}_{\infty} \setminus \mathcal{D}_0 \quad (482)$$

are generated endogenously through financial interconnectedness.

Thus,

$$\underbrace{\text{External Shock}}_{\text{exogenous}} \longrightarrow \underbrace{\mathcal{D}_0}_{\text{initial failure}} \longrightarrow \underbrace{W}_{\text{transmission}} \longrightarrow \underbrace{\mathcal{D}_{\infty} \setminus \mathcal{D}_0}_{\text{endogenous failures}}. \quad (483)$$

The systemic effect is therefore not determined by shock magnitude alone. It emerges from the interaction between

$$\boxed{\mathbf{S}, \quad \mathbf{K}, \quad W, \quad \mathbf{R}.} \quad (484)$$

The same external shock can produce substantially different outcomes under different network topologies, exposure magnitudes, capital buffers, or recovery assumptions.

8.21 From Tail Events to Systemic Cascades

The full sequence developed thus far can now be expressed as

$$\begin{array}{l}
 \text{Loss Distribution} \longrightarrow \text{VaR}_\alpha, \text{ES}_\alpha \\
 \longrightarrow \text{Extreme Loss} \\
 \longrightarrow \text{Capital Depletion} \\
 \longrightarrow \mathcal{G}_0 \\
 \longrightarrow \mathcal{G}_1 \longrightarrow \mathcal{G}_2 \longrightarrow \dots \\
 \longrightarrow \mathcal{D}_\infty.
 \end{array} \tag{485}$$

This establishes the conceptual bridge between statistical tail risk and network-based systemic risk.

Value at Risk and Expected Shortfall describe the magnitude and severity of extreme losses. Default-cascade models address the subsequent question: once such losses impair financial institutions, how does the architecture of the financial system amplify or absorb the resulting distress?

The next stage is therefore to investigate how network structure controls this propagation mechanism.

9 Gai–Kapadia Framework

The network-contagion mechanism developed in the previous sections is closely related to the framework introduced by Gai and Kapadia [9]. Their model provides an analytical description of contagion in financial networks with arbitrary network structure and studies how systemic vulnerability depends on the interaction between balance sheets, counterparty exposures, shocks, and connectivity.

The central insight is that financial stability cannot be inferred from the condition of individual institutions alone. Banks are linked through a network of claims and obligations, so the failure of one institution changes the value of assets held by its counterparties. If these losses are sufficiently large relative to their capital buffers, additional institutions fail and contagion propagates through the network.

Schematically,

$$\text{Network Structure} + \text{Balance Sheets} + \text{Initial Shock} \longrightarrow \text{Default} \longrightarrow \text{Counterparty Losses} \longrightarrow \text{Contagion}. \tag{486}$$

A particularly important conclusion of the framework is that connectivity has two competing effects. Greater diversification of exposures can make the system more resistant to the failure of an individual institution, while the same dense network can provide many channels through which distress propagates once contagion begins. This mechanism produces the "robust-yet-fragile" behavior emphasized by Gai and Kapadia.

9.1 Network Interpretation

Consider a financial system represented by a directed network

$$\mathcal{G} = (V, E), \tag{487}$$

where

$$V = \{1, \dots, n\} \tag{488}$$

is the set of financial institutions.

A directed connection between institutions represents an interbank claim or obligation. The topology of the network therefore describes the channels through which losses can potentially propagate.

Let

$$a_{ij} = \begin{cases} 1, & \text{if a financial exposure exists,} \\ 0, & \text{otherwise,} \end{cases} \quad (489)$$

denote the adjacency matrix.

The degree of institution i measures the number of financial counterparties connected to it. In a directed network, one may distinguish between in-degree and out-degree according to the adopted orientation of financial claims.

The network is not merely a visualization of the banking system. It is a component of the balance-sheet mechanism itself:

$$\boxed{\text{Topology} \longrightarrow \text{Distribution of Exposures} \longrightarrow \text{Distribution of Losses.}} \quad (490)$$

9.2 Counterparty Exposures

Let

$$W_{ij} \geq 0 \quad (491)$$

denote the exposure of institution i to institution j , following the convention adopted throughout this study:

$$\boxed{W_{ij} = \text{claim of institution } i \text{ on institution } j.} \quad (492)$$

Hence, if institution j fails, institution i may suffer a loss on W_{ij} .

The total interbank assets of institution i are

$$\boxed{A_i^{IB} = \sum_{j=1}^n W_{ij},} \quad (493)$$

while its interbank liabilities are

$$\boxed{L_i^{IB} = \sum_{j=1}^n W_{ji}.} \quad (494)$$

The weighted exposure matrix

$$\boxed{W = (W_{ij})_{n \times n}} \quad (495)$$

therefore contains both the topology and the financial magnitude of bilateral relationships.

This distinction is important because contagion depends not only on whether two institutions are connected, but also on the magnitude of the corresponding exposure relative to the creditor's capacity to absorb losses.

9.3 Balance Sheets and Capital Absorption

The transmission of losses must be considered jointly with the balance sheet of each institution.
Let

$$A_i = A_i^E + A_i^{IB} \quad (496)$$

denote total assets and

$$L_i = L_i^E + L_i^{IB} \quad (497)$$

total liabilities.

The capital buffer is

$$\boxed{K_i = A_i - L_i.} \quad (498)$$

Capital determines the ability of institution i to absorb losses without defaulting.
If the institution experiences a total loss Λ_i , its post-loss capital becomes

$$\boxed{K_i^{\text{post}} = K_i - \Lambda_i.} \quad (499)$$

Under the simplified insolvency criterion used in the present implementation,

$$\boxed{\Lambda_i \geq K_i \implies \text{default of institution } i.} \quad (500)$$

Thus, a counterparty exposure is systemically relevant only relative to the loss-absorbing capacity of the institution receiving the loss.

9.4 Initial Shock

Contagion begins with an exogenous disturbance.

Let

$$S_i \geq 0 \quad (501)$$

denote the external loss experienced by institution i .

The initial default condition is

$$\boxed{S_i \geq K_i.} \quad (502)$$

An institution satisfying this condition becomes an initial source of contagion.

The corresponding initial default set is

$$\boxed{\mathcal{D}_0 = \{i : S_i \geq K_i\}.} \quad (503)$$

The subsequent evolution is endogenous: further defaults arise not directly from the original shock, but from losses transmitted through financial connections.

9.5 Counterparty Loss Transmission

Suppose institution j defaults.

If its creditors recover a fraction R_j of their claims, the loss-given-default is

$$LGD_j = 1 - R_j. \quad (504)$$

Institution i then suffers the counterparty loss

$$\Delta_{i \leftarrow j} = (1 - R_j)W_{ij}. \quad (505)$$

If several counterparties have defaulted, the accumulated interbank loss of institution i is

$$C_i = \sum_{j \in \mathcal{D}} (1 - R_j)W_{ij}. \quad (506)$$

The institution's total loss is therefore

$$\Lambda_i = S_i + C_i. \quad (507)$$

A secondary default occurs whenever

$$S_i + \sum_{j \in \mathcal{D}} (1 - R_j)W_{ij} \geq K_i. \quad (508)$$

This threshold mechanism converts a local counterparty failure into a potential systemic cascade.

9.6 Contagion Mechanism

The contagion mechanism can be represented recursively.

Let

$$D_i^{(t)} \in \{0, 1\} \quad (509)$$

denote whether institution i has defaulted by generation t .

Then a simplified direct-counterparty implementation is

$$D_i^{(t+1)} = \max \left\{ D_i^{(t)}, \mathbf{1} \left[S_i + \sum_{j=1}^n (1 - R_j)W_{ij}D_j^{(t)} \geq K_i \right] \right\}. \quad (510)$$

The sequence

$$\mathbf{D}^{(0)}, \mathbf{D}^{(1)}, \mathbf{D}^{(2)}, \dots \quad (511)$$

describes successive rounds of contagion.

Because defaults are absorbing,

$$\mathbf{D}^{(t+1)} \geq \mathbf{D}^{(t)} \quad (512)$$

componentwise.

The process terminates when

$$\mathbf{D}^{(t+1)} = \mathbf{D}^{(t)}. \quad (513)$$

The resulting terminal state describes the final extent of direct counterparty contagion under the assumed network and balance-sheet configuration.

9.7 Connectivity and Risk Sharing

One of the central insights of Gai and Kapadia is that connectivity has competing effects on systemic stability.

Consider an institution with a fixed total amount of interbank exposure

$$A_i^{IB}. \quad (514)$$

If this exposure is distributed approximately equally across k_i counterparties, a representative bilateral exposure has magnitude of order

$$W_{ij} \sim \frac{A_i^{IB}}{k_i}. \quad (515)$$

As connectivity increases,

$$k_i \uparrow, \quad (516)$$

the loss associated with the failure of any single counterparty can therefore become smaller:

$$W_{ij} \downarrow. \quad (517)$$

This diversification effect makes it easier for capital buffers to absorb an isolated counterparty failure.

Schematically,

$$k \uparrow \implies \text{exposure per counterparty} \downarrow \implies \text{greater absorption of isolated defaults.} \quad (518)$$

9.8 Connectivity and Contagion Reach

The same connectivity that diversifies individual exposures also creates additional channels for distress propagation.

If contagion survives the first round, a highly connected institution may be exposed to several counterparties that have already defaulted.

Its cumulative loss may then become

$$C_i = \sum_{j \in \mathcal{D}} (1 - R_j) W_{ij}, \quad (519)$$

and even individually small exposures can become dangerous when several defaulting counterparties contribute simultaneously.

Thus,

$$\text{more links} \implies \begin{cases} \text{greater diversification,} \\ \text{more propagation channels.} \end{cases} \quad (520)$$

This is the source of the non-monotonic relationship between connectivity and systemic vulnerability emphasized in the Gai–Kapadia framework.

9.9 Robust-yet-Fragile Financial Networks

The competing effects of diversification and propagation lead to the “robust-yet-fragile” characterization of financial networks [9].

As connectivity increases, the probability that an isolated default initiates a systemic cascade may decline because individual losses are distributed across more counterparties.

However, conditional on contagion successfully propagating beyond its initial stage, a highly connected network provides many pathways through which distress can spread.

Hence,

higher connectivity $\Rightarrow$ greater resistance to some isolated failures, but higher connectivity $\Rightarrow$ potentially broader propagation if contagion occurs. (521)
--

This distinction separates two different systemic-risk questions:

$P(\text{cascade}) = \text{probability that contagion develops,}$ $P(\text{large cascade} \mid \text{cascade}) = \text{severity conditional on contagion.}$	(522)
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A system can therefore exhibit relatively low contagion frequency while remaining vulnerable to very large cascades when contagion does occur.

9.10 Vulnerable Institutions

The interaction between exposure magnitude and capital can be expressed through a simple vulnerability condition.

Suppose institution i is exposed to a single defaulting counterparty j . Then i is vulnerable to the failure of j whenever

$(1 - R_j)W_{ij} \geq K_i - S_i.$	(523)
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In the absence of an external loss to institution i ,

$$S_i = 0, \tag{524}$$

this reduces to

$(1 - R_j)W_{ij} \geq K_i.$	(525)
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The relevant object is therefore not exposure alone, but exposure relative to available capital:

$\frac{(1 - R_j)W_{ij}}{K_i}.$	(526)
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If this ratio exceeds one, the corresponding counterparty failure is sufficient, under this simplified mechanism, to exhaust the institution’s capital buffer.

9.11 Multiple-Default Vulnerability

An institution that survives every individual counterparty default may nevertheless fail after several counterparties default.

For a set of defaulted counterparties $\mathcal{D}$,

$$\boxed{\sum_{j \in \mathcal{D}} (1 - R_j) W_{ij} \geq K_i - S_i} \quad (527)$$

is sufficient for default in the simplified threshold model.

This provides an important mechanism for later cascade generations.

An institution may satisfy

$$(1 - R_j) W_{ij} < K_i - S_i \quad (528)$$

for every individual j , while nevertheless satisfying

$$\sum_{j \in \mathcal{D}} (1 - R_j) W_{ij} \geq K_i - S_i. \quad (529)$$

Thus diversification against individual counterparty failures does not necessarily guarantee resilience against correlated or cascading counterparty defaults.

9.12 Connectivity as a Nonlinear Stability Mechanism

The preceding arguments imply that systemic vulnerability need not vary monotonically with network density.

Let

$$p \quad (530)$$

denote a connection probability in a random-network representation.

The expected degree is approximately

$$\boxed{z \simeq p(n - 1)}. \quad (531)$$

Changing p simultaneously changes

$$\begin{aligned} &\text{the number of counterparties,} \\ &\text{the distribution of bilateral exposures,} \\ &\text{the number of available contagion pathways.} \end{aligned} \quad (532)$$

Therefore the map

$$\boxed{p \longmapsto \text{systemic vulnerability}} \quad (533)$$

is generally nonlinear.

The same network parameter can influence both the probability that a cascade begins and the extent of contagion conditional on its initiation.

9.13 Direct and Indirect Contagion

The Gai–Kapadia framework also emphasizes that counterparty exposures are not the only possible channel of systemic propagation.

Direct contagion operates through bilateral claims:

$$\boxed{\text{default} \rightarrow \text{counterparty loss} \rightarrow \text{capital erosion}.} \quad (534)$$

An additional indirect mechanism may arise through common asset holdings and market illiquidity. Distressed institutions may liquidate assets, placing downward pressure on prices and imposing mark-to-market losses on other institutions.

Schematically,

$$\boxed{\text{distress} \rightarrow \text{asset sales} \rightarrow \text{price decline} \rightarrow \text{valuation losses} \rightarrow \text{additional distress.}} \quad (535)$$

The present simulation focuses primarily on the direct counterparty default channel. Consequently, its results should be interpreted as a controlled implementation of one central contagion mechanism rather than as a numerical reproduction of every channel contained in the full Gai–Kapadia framework.

9.14 Relation with the Present Simulation

The computational experiment developed in this study adopts the central network logic of the Gai–Kapadia framework while implementing it as a discrete cascade algorithm.

The simulation contains:

$$\begin{array}{ll} \text{financial institutions} & \longleftrightarrow \text{network nodes,} \\ \text{counterparty exposures} & \longleftrightarrow W_{ij}, \\ \text{capital buffers} & \longleftrightarrow K_i, \\ \text{external shocks} & \longleftrightarrow S_i, \\ \text{recovery assumptions} & \longleftrightarrow R_i, \\ \text{default states} & \longleftrightarrow D_i^{(t)}. \end{array} \quad (536)$$

At the beginning of each experiment, an external shock generates an initial set of defaults,

$$\mathcal{D}_0. \quad (537)$$

These institutions transmit counterparty losses through the exposure matrix.

Institutions whose cumulative losses exceed their available capital then enter the next default generation:

$$\mathcal{G}_1. \quad (538)$$

The process continues as

$$\boxed{\mathcal{G}_0 \rightarrow \mathcal{G}_1 \rightarrow \mathcal{G}_2 \rightarrow \cdots \rightarrow \mathcal{G}_T,} \quad (539)$$

until

$$\mathcal{G}_{T+1} = \emptyset. \quad (540)$$

9.15 Deterministic Experiment

For a fixed realization of

$$(W, \mathbf{K}, \mathbf{S}, \mathbf{R}), \quad (541)$$

the simulation produces a deterministic cascade path.

The principal outputs include

$$|\mathcal{D}_\infty|, \quad (542)$$

the final number of defaults,

$$\phi_\infty = \frac{|\mathcal{D}_\infty|}{n}, \quad (543)$$

the final default fraction, and

$$T_{\text{depth}}, \quad (544)$$

the cascade depth.

Thus the experiment maps

$$(W, \mathbf{K}, \mathbf{S}, \mathbf{R}) \mapsto (\phi_\infty, T_{\text{depth}}). \quad (545)$$

9.16 Monte Carlo Extension

The network experiment can then be embedded in a Monte Carlo procedure.

For simulation

$$m = 1, \dots, M, \quad (546)$$

one or more components of the system may vary:

$$W^{(m)}, \quad \mathbf{K}^{(m)}, \quad \mathbf{S}^{(m)}. \quad (547)$$

Each realization generates a terminal cascade

$$\mathcal{D}_\infty^{(m)} \quad (548)$$

and corresponding default fraction

$$\phi_\infty^{(m)}. \quad (549)$$

The mean systemic impact is estimated by

$$\widehat{\mathbb{E}}[\phi_\infty] = \frac{1}{M} \sum_{m=1}^M \phi_\infty^{(m)}. \quad (550)$$

Similarly,

$$\widehat{\mathbb{E}}[N_D] = \frac{1}{M} \sum_{m=1}^M |\mathcal{D}_\infty^{(m)}| \quad (551)$$

estimates the expected number of failed institutions.

9.17 From VaR and Expected Shortfall to Gai–Kapadia

The preceding sections and the network model can now be assembled into a single risk-analysis pipeline.

At the individual level,

$$L_i \quad (552)$$

describes the loss of institution i .

Tail-risk measures characterize extreme realizations:

$$\text{VaR}_\alpha(L_i), \quad \text{ES}_\alpha(L_i). \quad (553)$$

Monte Carlo simulation estimates these quantities and generates possible loss scenarios. A sufficiently severe realization can then be interpreted as an external balance-sheet shock:

$$L_i \longrightarrow S_i. \quad (554)$$

The network model determines what happens after that shock. Thus,

Loss Distribution $\longrightarrow$ VaR / ES $\longrightarrow$ Monte Carlo Tail Event $\longrightarrow$ External Balance-Sheet Shock $\longrightarrow$ Initial Default $\longrightarrow$ Counterparty Losses $\longrightarrow$ Default Cascade $\longrightarrow$ Systemic Outcome.	(555)
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The first part of this sequence quantifies the statistical severity of extreme losses. The second part determines whether those losses remain local or are amplified through financial interconnectedness.

9.18 Interpretation

The Gai–Kapadia framework highlights why systemic risk cannot be reduced to the sum of individual risks.

Two systems may contain institutions with identical individual balance sheets and identical initial capital buffers, yet produce different systemic outcomes because their network structures differ.

Similarly, the same network may respond very differently to shocks of similar magnitude depending on which institution is initially affected.

Therefore,

$\text{Systemic Risk} \neq \sum_i \text{Individual Risk}_i.$	(556)
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Rather, systemic risk emerges from the interaction

$\text{Shocks} \times \text{Balance Sheets} \times \text{Network Structure}.$	(557)
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This interaction explains why network structure is not merely an additional descriptive variable. It is part of the mechanism generating system-wide financial outcomes.

10 Connecting Tail Risk and Network Risk

The previous sections developed two complementary perspectives on financial risk.

The first is probabilistic. A random loss variable L_i describes the uncertainty faced by institution i , while Value at Risk and Expected Shortfall characterize different aspects of its tail:

$$\text{VaR}_\alpha(L_i), \quad \text{ES}_\alpha(L_i). \quad (558)$$

The second perspective is systemic. Once financial institutions are linked through bilateral exposures, the consequences of a realized loss depend not only on its magnitude, but also on capital buffers, counterparty exposures, recovery assumptions, and network topology.

The connection between these two perspectives can be represented as

$$\begin{array}{c}
\boxed{\text{Loss Distribution} \longrightarrow \text{Tail Event} \longrightarrow \text{Balance-Sheet Shock} \\
\longrightarrow \text{Initial Default} \longrightarrow \text{Counterparty Losses} \longrightarrow \text{Default Cascade.}}
\end{array} \tag{559}$$

This sequence provides a conceptual bridge from market and statistical risk measurement to network-based systemic risk.

10.1 Tail Events as Realized Losses

Let

$$L_i : \Omega \rightarrow \mathbb{R} \tag{560}$$

denote the random loss of institution i .

Before uncertainty is resolved, L_i is characterized by its probability distribution

$$F_i(\ell) = \mathbb{P}(L_i \leq \ell). \tag{561}$$

For a confidence level $\alpha \in (0, 1)$, the Value at Risk is

$$\boxed{\text{VaR}_\alpha(L_i) = F_i^{\leftarrow}(\alpha)}, \tag{562}$$

whereas Expected Shortfall may be represented, under appropriate regularity conditions, as

$$\boxed{\text{ES}_\alpha(L_i) = \frac{1}{1 - \alpha} \int_\alpha^1 \text{VaR}_u(L_i) du.} \tag{563}$$

These quantities describe the distribution of possible losses. They should not themselves be interpreted as realized shocks.

For a particular state of the world $\omega \in \Omega$, the realized loss is

$$\boxed{\ell_i = L_i(\omega)}. \tag{564}$$

A tail event at confidence level α may then be represented by

$$\boxed{L_i(\omega) > \text{VaR}_\alpha(L_i)}. \tag{565}$$

The event

$$\{L_i > \text{VaR}_\alpha(L_i)\} \tag{566}$$

identifies a realization located beyond the selected quantile threshold.

10.2 From Tail Event to Initial Shock

To connect statistical losses with the balance-sheet network model, the realized loss must be translated into an external shock.

Let

$$S_i \tag{567}$$

denote the balance-sheet loss imposed on institution i .

The simplest identification is

$$\boxed{S_i = L_i(\omega)}. \tag{568}$$

More generally, a mapping

$$S_i = \Psi_i(L_i(\omega)) \quad (569)$$

may be introduced when the simulated financial loss and the balance-sheet shock are expressed in different units or represent different economic objects.

For the entire financial system,

$$\mathbf{L}(\omega) = (L_1(\omega), \dots, L_n(\omega))^\top \quad (570)$$

can therefore generate the shock vector

$$\mathbf{S} = \Psi(\mathbf{L}(\omega)). \quad (571)$$

This mapping is the first bridge between tail-risk simulation and systemic network analysis.

10.3 Tail Severity and Shock Severity

The exceedance above the VaR threshold provides a useful measure of the severity of a realized tail event.

Define

$$X_{i,\alpha} = (L_i - \text{VaR}_\alpha(L_i))_+, \quad (572)$$

where

$$(x)_+ = \max(x, 0). \quad (573)$$

Then

$$X_{i,\alpha} > 0 \quad (574)$$

if and only if the loss exceeds the VaR threshold.

Expected Shortfall contains information about the severity of losses in this region rather than only identifying its boundary.

Thus,

$$\text{VaR}_\alpha \longrightarrow \text{tail threshold}, \quad (575)$$

whereas

$$\text{ES}_\alpha \longrightarrow \text{average tail severity}. \quad (576)$$

Neither quantity alone determines whether a financial institution defaults. Default additionally depends on its balance-sheet capacity to absorb the realized loss.

10.4 From Initial Shock to Default

Let

$$K_i > 0 \quad (577)$$

denote the initial capital buffer of institution i .

Following the simplified insolvency criterion adopted in this study, the post-shock capital position is

$$K_i^{(1)} = K_i - S_i. \quad (578)$$

The institution initially defaults whenever

$$S_i \geq K_i. \quad (579)$$

Hence the initial default indicator is

$$D_i^{(0)} = \mathbf{1}_{\{S_i \geq K_i\}}. \quad (580)$$

The initial default set is

$$\mathcal{D}_0 = \{i : S_i \geq K_i\}. \quad (581)$$

This equation highlights an important distinction:

$$\text{extreme loss} \neq \text{default}. \quad (582)$$

A loss can lie deep in the statistical tail and still be absorbed by a well-capitalized institution. Conversely, an institution with a sufficiently small capital buffer may default under a loss that is not exceptionally extreme relative to the loss distribution.

10.5 A Capital-Normalized Shock

A useful quantity for connecting loss severity with balance-sheet vulnerability is the capital-normalized shock

$$\chi_i = \frac{S_i}{K_i}. \quad (583)$$

Under the present threshold model,

$$\chi_i \begin{cases} < 1, & \text{initial capital survives the shock,} \\ \geq 1, & \text{initial default occurs.} \end{cases} \quad (584)$$

The same absolute loss can therefore have very different implications for different institutions. If

$$S_i = S_j \quad (585)$$

but

$$K_i \gg K_j, \quad (586)$$

then

$$\chi_i \ll \chi_j. \quad (587)$$

Systemic analysis therefore requires information that is absent from the loss distribution alone.

10.6 From Default to Counterparty Losses

Once institution j defaults, its financial connections become relevant.

Following the exposure convention used throughout this study,

$$W_{ij} \quad (588)$$

denotes the claim of creditor i on debtor j .

If R_j is the recovery fraction on the obligations of j , then the loss transmitted to institution i is

$$\Delta_{i \leftarrow j} = (1 - R_j)W_{ij}. \quad (589)$$

Thus a realized loss that initially affects only institution j may become a loss for several institutions through the liability structure of the network.

Schematically,

$$L_j(\omega) \rightarrow S_j \rightarrow D_j^{(0)} = 1 \rightarrow \{(1 - R_j)W_{ij}\}_{i=1}^n. \quad (590)$$

At this point, the original exogenous loss has generated endogenous losses elsewhere in the system.

10.7 Counterparty Losses and Secondary Defaults

Consider an institution i that survived the original external shock:

$$S_i < K_i. \quad (591)$$

Its residual capital immediately after the external shock is

$$B_i = K_i - S_i. \quad (592)$$

The quantity B_i is the remaining loss-absorption capacity available to withstand network losses.

If the set of currently defaulted institutions is $\mathcal{D}_t$, then institution i receives cumulative counterparty losses

$$C_i^{(t)} = \sum_{j \in \mathcal{D}_t} (1 - R_j)W_{ij}. \quad (593)$$

A secondary default occurs whenever

$$C_i^{(t)} \geq B_i. \quad (594)$$

Equivalently,

$$S_i + \sum_{j \in \mathcal{D}_t} (1 - R_j)W_{ij} \geq K_i. \quad (595)$$

This inequality provides the central mathematical bridge between an external loss and endogenous financial contagion.

10.8 A Network Vulnerability Ratio

The preceding condition motivates the definition of a dynamic capital-relative vulnerability ratio:

$$\Gamma_i^{(t)} = \frac{S_i + \sum_{j \in \mathcal{D}_t} (1 - R_j) W_{ij}}{K_i}. \quad (596)$$

Under the simplified default criterion,

$$\Gamma_i^{(t)} \begin{cases} < 1, & \text{institution } i \text{ remains solvent,} \\ \geq 1, & \text{institution } i \text{ defaults.} \end{cases} \quad (597)$$

The decomposition

$$\Gamma_i^{(t)} = \underbrace{\frac{S_i}{K_i}}_{\text{external component}} + \underbrace{\frac{\sum_{j \in \mathcal{D}_t} (1 - R_j) W_{ij}}{K_i}}_{\text{network component}} \quad (598)$$

makes the interaction between individual and systemic vulnerability explicit.

10.9 From Counterparty Losses to a Cascade

When a secondary institution defaults, it becomes a new source of counterparty losses.

Thus the process recursively generates

$$\mathcal{D}_0 \subseteq \mathcal{D}_1 \subseteq \mathcal{D}_2 \subseteq \dots \quad (599)$$

The update rule is

$$D_i^{(t+1)} = \max \left\{ D_i^{(t)}, \mathbf{1} \left[S_i + \sum_{j=1}^n (1 - R_j) W_{ij} D_j^{(t)} \geq K_i \right] \right\}. \quad (600)$$

The terminal state

$$\mathbf{D}^* \quad (601)$$

satisfies

$$\mathbf{D}^* = \tilde{\mathcal{T}}(\mathbf{D}^*), \quad (602)$$

where $\tilde{\mathcal{T}}$ denotes the absorbing contagion operator introduced previously.

The final default fraction is

$$\phi_\infty = \frac{1}{n} \sum_{i=1}^n D_i^*. \quad (603)$$

The map from an initial loss realization to systemic damage can therefore be represented as

$$\mathbf{L}(\omega) \longrightarrow \mathbf{S} \longrightarrow \mathbf{D}^{(0)} \longrightarrow \mathbf{D}^* \longrightarrow \phi_\infty. \quad (604)$$

10.10 Exogenous and Endogenous Losses

The framework naturally separates two sources of financial loss.

The first is exogenous:

$$S_i. \quad (605)$$

It originates outside the contagion mechanism and provides the initial disturbance.

The second is endogenous:

$$C_i^{(\infty)} = \sum_{j \in \mathcal{D}_\infty} (1 - R_j) W_{ij}. \quad (606)$$

It is generated internally by defaults elsewhere in the network.

Thus total loss at the terminal state is

$$\Lambda_i^{(\infty)} = S_i + C_i^{(\infty)}. \quad (607)$$

This decomposition is fundamental:

$$\boxed{\text{Total Damage} = \text{External Shock} + \text{Endogenous Network Amplification.}} \quad (608)$$

The second component is precisely what cannot be recovered from an individual loss distribution alone.

10.11 Individual Risk Is Not Systemic Risk

Individual risk measures describe properties of the marginal loss distribution of an institution.

For institution i , these include

$$\text{VaR}_\alpha(L_i) \quad (609)$$

and

$$\text{ES}_\alpha(L_i). \quad (610)$$

Systemic risk depends on additional objects:

$$\mathbf{K}, \quad W, \quad \mathbf{R}, \quad \mathbf{S}, \quad (611)$$

as well as on the location of the initially affected institutions in the network.

Consequently,

$$\boxed{\text{Individual Tail Risk} \neq \text{Systemic Risk.}} \quad (612)$$

Nor, in general, can systemic risk be represented as a simple sum of stand-alone risk measures:

$$\boxed{\text{Systemic Risk} \neq \sum_{i=1}^n \text{Risk}(L_i).} \quad (613)$$

The missing ingredient is interaction.

10.12 Identical Individual Risk, Different Systemic Outcomes

Consider two financial systems, $\mathcal{A}$ and $\mathcal{B}$, with the same institutions, marginal loss distributions, and capital buffers:

$$F_i^{\mathcal{A}} = F_i^{\mathcal{B}}, \quad (614)$$

and

$$K_i^{\mathcal{A}} = K_i^{\mathcal{B}}. \quad (615)$$

Consequently,

$$\text{VaR}_{\alpha}^{\mathcal{A}}(L_i) = \text{VaR}_{\alpha}^{\mathcal{B}}(L_i) \quad (616)$$

and

$$\text{ES}_{\alpha}^{\mathcal{A}}(L_i) = \text{ES}_{\alpha}^{\mathcal{B}}(L_i). \quad (617)$$

Suppose, however, that their exposure matrices differ:

$$\boxed{W^{\mathcal{A}} \neq W^{\mathcal{B}}.} \quad (618)$$

The same initial loss realization may then produce

$$\boxed{\phi_{\infty}^{\mathcal{A}} \neq \phi_{\infty}^{\mathcal{B}}.} \quad (619)$$

Hence identical stand-alone tail-risk measures do not imply identical systemic vulnerability. The network architecture changes the mapping from individual losses to system-wide outcomes.

10.13 Identical Shock Magnitude, Different Shock Location

Systemic consequences may also depend on *where* the shock enters the network.

Consider two shocks with equal magnitude,

$$S_a = S_b = S, \quad (620)$$

applied separately to institutions a and b .

Even if both institutions initially default,

$$D_a^{(0)} = D_b^{(0)} = 1, \quad (621)$$

the resulting cascades need not be equal:

$$\boxed{\phi_{\infty}^{(a)} \neq \phi_{\infty}^{(b)}.} \quad (622)$$

This may occur because institutions a and b have different creditors, different exposure magnitudes, or different positions within the network.

Thus systemic impact depends on both

$$\boxed{\text{shock magnitude} \quad \text{and} \quad \text{shock location}.} \quad (623)$$

10.14 Monte Carlo Coupling of Tail and Network Risk

The statistical and network layers can be integrated directly within a Monte Carlo experiment.
For simulation

$$m = 1, \dots, M, \quad (624)$$

generate a loss vector

$$\mathbf{L}^{(m)} \sim F_{\mathbf{L}}. \quad (625)$$

The joint distribution $F_{\mathbf{L}}$ may contain dependence between institutions and therefore need not factor into independent marginal distributions.

The realized losses are mapped into external shocks:

$$\mathbf{S}^{(m)} = \Psi(\mathbf{L}^{(m)}). \quad (626)$$

These shocks determine the initial defaults,

$$D_i^{(0,m)} = \mathbf{1}_{\{S_i^{(m)} \geq K_i\}}. \quad (627)$$

The contagion engine then produces the terminal state

$$\mathbf{D}^{(*,m)} \quad (628)$$

and final default fraction

$$\phi_{\infty}^{(m)} = \frac{1}{n} \sum_{i=1}^n D_i^{(*,m)}. \quad (629)$$

Thus each Monte Carlo realization generates not merely a financial loss, but an entire systemic outcome:

$$\mathbf{L}^{(m)} \rightarrow \mathbf{S}^{(m)} \rightarrow \mathbf{D}^{(0,m)} \rightarrow \mathbf{D}^{(*,m)} \rightarrow \phi_{\infty}^{(m)}. \quad (630)$$

10.15 Distribution of Systemic Outcomes

Once the tail-risk and network layers are coupled, systemic damage itself becomes a random variable.

Define

$$\Phi = \phi_{\infty}(\mathbf{L}). \quad (631)$$

The distribution of Φ describes the uncertainty associated with the final fraction of defaulted institutions.

Monte Carlo simulation produces observations

$$\Phi^{(1)}, \dots, \Phi^{(M)}, \quad (632)$$

from which quantities such as

$$\hat{\mathbb{E}}[\Phi] = \frac{1}{M} \sum_{m=1}^M \Phi^{(m)} \quad (633)$$

can be estimated.

Similarly, for a systemic threshold $\gamma \in (0, 1]$, one may estimate

$$\mathbb{P}(\Phi \geq \gamma) \approx \frac{1}{M} \sum_{m=1}^M \mathbf{1}_{\{\Phi^{(m)} \geq \gamma\}}. \quad (634)$$

This moves the analysis beyond the probability of an individual extreme loss toward the distribution of system-wide consequences.

10.16 Conditional Systemic Amplification

A useful decomposition separates initial failures from failures generated through contagion.

For realization m , define

$$N_0^{(m)} = |\mathcal{D}_0^{(m)}| \quad (635)$$

and

$$N_\infty^{(m)} = |\mathcal{D}_\infty^{(m)}|. \quad (636)$$

The number of endogenous defaults is

$$N_{\text{endo}}^{(m)} = N_\infty^{(m)} - N_0^{(m)}. \quad (637)$$

Thus the total systemic outcome decomposes into

$$N_\infty = N_0 + N_{\text{endo}}. \quad (638)$$

The first term measures the direct effect of the external loss realization. The second measures the amplification generated by the financial network.

10.17 The Complete Risk-Transmission Map

The entire framework developed in this study can now be summarized through three mathematical layers.

The first layer is probabilistic:

$$F_{\mathbf{L}} \longrightarrow \mathbf{L}(\omega). \quad (639)$$

The second layer is balance-sheet based:

$$\mathbf{L}(\omega) \longrightarrow \mathbf{S} \longrightarrow \mathbf{D}^{(0)}. \quad (640)$$

The third layer is network based:

$$\mathbf{D}^{(0)} \longrightarrow \mathbf{D}^* \longrightarrow \phi_\infty. \quad (641)$$

Combining the three gives

$$F_{\mathbf{L}} \longrightarrow \mathbf{L}(\omega) \longrightarrow \mathbf{S} \longrightarrow \mathbf{D}^{(0)} \longrightarrow \mathbf{D}^* \longrightarrow \phi_\infty. \quad (642)$$

The sequence distinguishes the probability of an adverse event from the financial system's response to that event.

10.18 Interpretation

Tail risk and network risk answer fundamentally different questions.

Tail-risk measures ask:

$$\boxed{\textit{How severe can an institution's loss become?}} \quad (643)$$

Network-risk models ask:

$$\boxed{\textit{What happens to the system after losses impair financial institutions?}} \quad (644)$$

The first question concerns the distribution of shocks.

The second concerns the transmission and amplification of those shocks.

Consequently, a comprehensive systemic-risk analysis requires both:

$$\boxed{\underbrace{\text{probability and severity of losses}}_{\text{tail-risk layer}} + \underbrace{\text{propagation and amplification}}_{\text{network layer}}.} \quad (645)$$

This distinction provides the central connection between the statistical risk measures studied in the first part of this work and the default cascade mechanisms studied in the second.

11 Comparative Analysis

The preceding sections introduced several complementary approaches to financial risk. Their differences are not merely computational. Each method addresses a different component of the risk-generation and risk-transmission process.

This section compares five central distinctions:

VaR	versus	CVaR/Expected Shortfall,	(646)
Gaussian	versus	Student- t ,	
Analytical	versus	Monte Carlo,	
Individual	versus	Systemic Risk,	
Direct Loss	versus	Contagion Loss.	

These comparisons reveal a common principle: no single statistic, distributional assumption, or computational method provides a complete description of financial vulnerability.

11.1 VaR versus CVaR / Expected Shortfall

Let L denote a loss random variable with distribution function F_L . For confidence level $\alpha \in (0, 1)$, Value at Risk is defined by

$$\boxed{\text{VaR}_\alpha(L) = F_L^{\leftarrow}(\alpha).} \quad (647)$$

VaR therefore identifies a quantile of the loss distribution.

For a continuous distribution,

$$\mathbb{P}(L \leq \text{VaR}_\alpha(L)) = \alpha. \quad (648)$$

Expected Shortfall, also referred to as CVaR under common continuous-loss conventions, measures the average severity of losses in the tail beyond the VaR threshold:

$$\boxed{\text{ES}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)]} \quad (649)$$

for an appropriate continuous loss distribution.

More generally, Expected Shortfall can be represented through the quantile function as

$$\boxed{\text{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 \text{VaR}_u(L) du.} \quad (650)$$

The distinction is fundamental:

$$\boxed{\begin{array}{l} \text{VaR}_\alpha \longrightarrow \text{location of a tail threshold,} \\ \text{ES}_\alpha \longrightarrow \text{severity of losses beyond the threshold.} \end{array}} \quad (651)$$

Two loss distributions can therefore have similar VaR values while exhibiting substantially different Expected Shortfall values.

This occurs because VaR is determined locally by a quantile, whereas Expected Shortfall depends on the entire region beyond that quantile.

11.2 What VaR Does Not Measure

Suppose two loss variables L_A and L_B satisfy

$$\text{VaR}_\alpha(L_A) = \text{VaR}_\alpha(L_B). \quad (652)$$

This equality does not imply

$$\text{ES}_\alpha(L_A) = \text{ES}_\alpha(L_B). \quad (653)$$

Indeed, the distributions may be nearly identical around their α -quantiles while differing substantially farther into the tail.

Hence,

$$\boxed{\text{same VaR} \not\Rightarrow \text{same tail severity.}} \quad (654)$$

This distinction becomes particularly important for heavy-tailed loss distributions.

11.3 Coherence: VaR versus Expected Shortfall

Expected Shortfall also differs from VaR from an axiomatic perspective.

A coherent risk measure satisfies the properties of monotonicity, translation invariance, positive homogeneity, and subadditivity [2].

Expected Shortfall is coherent under the standard framework, whereas VaR is not subadditive in general.

For a coherent risk measure ρ ,

$$\boxed{\rho(X + Y) \leq \rho(X) + \rho(Y).} \quad (655)$$

Subadditivity formalizes the principle that diversification should not increase measured risk.

The absence of this property for VaR in general provides an additional motivation for the use of Expected Shortfall in tail-risk analysis.

11.4 Gaussian versus Student- t Loss Models

The choice of probability distribution strongly affects tail-risk measurement.

Consider first a Gaussian return variable

$$R_G \sim \mathcal{N}(\mu, \sigma^2). \quad (656)$$

Its standardized form is

$$Z_G = \frac{R_G - \mu}{\sigma} \sim \mathcal{N}(0, 1). \quad (657)$$

The normal density is

$$f_G(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right). \quad (658)$$

Now consider a Student- t variable with ν degrees of freedom:

$$T_\nu \sim t_\nu. \quad (659)$$

Its density is

$$f_\nu(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}. \quad (660)$$

For $\nu > 2$,

$$\text{Var}(T_\nu) = \frac{\nu}{\nu - 2}. \quad (661)$$

A variance-standardized Student- t variable can therefore be defined by

$$Z_\nu = T_\nu \sqrt{\frac{\nu - 2}{\nu}}. \quad (662)$$

Then

$$\text{Var}(Z_\nu) = 1. \quad (663)$$

This standardization permits a meaningful comparison with a standard Gaussian variable.

11.5 Same Variance Does Not Mean Same Tail Risk

Suppose

$$\text{Var}(Z_G) = \text{Var}(Z_\nu) = 1. \quad (664)$$

The equality of variances does not imply equality of tail probabilities.

For sufficiently large thresholds x ,

$$\mathbb{P}(|Z_\nu| > x) > \mathbb{P}(|Z_G| > x). \quad (665)$$

The Gaussian tail decays exponentially in x^2 , while the Student- t distribution exhibits polynomial tail decay.

Consequently,

$$\text{same mean} + \text{same variance} \not\Rightarrow \text{same extreme-loss behavior}. \quad (666)$$

This is precisely why volatility alone cannot completely characterize extreme financial risk.

11.6 Kurtosis and Tail Thickness

For a Student- t distribution with $\nu > 4$, the excess kurtosis is

$$\kappa_{\text{excess}} = \frac{6}{\nu - 4}. \quad (667)$$

For the Gaussian distribution,

$$\kappa_{\text{excess}} = 0. \quad (668)$$

Thus, for finite $\nu > 4$, the Student- t distribution has positive excess kurtosis. When

$$2 < \nu \leq 4, \quad (669)$$

the variance exists but the fourth moment does not.

This provides an even stronger illustration of why equal variance alone does not determine extreme-event behavior.

11.7 Implications for VaR and Expected Shortfall

For a loss model with equal mean and variance, a heavy-tailed distribution can generate substantially larger extreme quantiles and tail conditional losses than a Gaussian model at sufficiently high confidence levels.

Schematically,

$$\text{heavier tails} \longrightarrow \text{greater extreme-loss probability} \longrightarrow \text{larger high-confidence tail-risk measures.} \quad (670)$$

Expected Shortfall is especially informative in this comparison because it depends on losses beyond the selected quantile rather than only on the quantile itself.

11.8 Analytical versus Monte Carlo Methods

Risk measures may sometimes be obtained analytically.

For example, if

$$L \sim \mathcal{N}(\mu_L, \sigma_L^2), \quad (671)$$

then

$$\text{VaR}_\alpha(L) = \mu_L + \sigma_L \Phi^{-1}(\alpha), \quad (672)$$

where Φ^{-1} is the inverse standard normal distribution function.

For a Gaussian loss variable, Expected Shortfall has the analytical form

$$\text{ES}_\alpha(L) = \mu_L + \sigma_L \frac{\varphi(\Phi^{-1}(\alpha))}{1 - \alpha}, \quad (673)$$

where φ denotes the standard normal density.

These expressions are computationally efficient and, conditional on the assumed model, exact.

11.9 Monte Carlo Estimation

When analytical expressions are unavailable or the model is sufficiently complex, Monte Carlo simulation provides a flexible alternative.

Generate

$$L^{(1)}, \dots, L^{(N)} \quad (674)$$

from the assumed loss model.

Let

$$L_{(1)} \leq L_{(2)} \leq \dots \leq L_{(N)} \quad (675)$$

denote the ordered sample.

An empirical VaR estimator is

$$\widehat{\text{VaR}}_\alpha = L_{(\lceil N\alpha \rceil)}. \quad (676)$$

An empirical Expected Shortfall estimator can be written approximately as

$$\widehat{\text{ES}}_\alpha = \frac{1}{N_{\text{tail}}} \sum_{m: L^{(m)} \geq \widehat{\text{VaR}}_\alpha} L^{(m)}. \quad (677)$$

The expected number of observations available in the tail is approximately

$$N_{\text{tail}} \approx N(1 - \alpha). \quad (678)$$

This relation explains why extreme Expected Shortfall estimation can require very large simulation samples.

11.10 Strengths and Limitations of Analytical Methods

Analytical methods provide several advantages:

- high computational efficiency;
- no Monte Carlo sampling error;
- transparent dependence on model parameters;
- useful benchmarks for validating numerical simulations.

Their main limitation is model dependence.

Closed-form expressions generally require assumptions sufficiently simple to permit mathematical derivation.

Thus,

$$\boxed{\text{analytical tractability} \quad \text{often requires} \quad \text{structural simplification.}} \quad (679)$$

11.11 Strengths and Limitations of Monte Carlo Methods

Monte Carlo methods can accommodate substantially more complex models, including

$$\begin{aligned} &\text{non-Gaussian distributions,} \\ &\text{nonlinear portfolios,} \\ &\text{dependent risk factors,} \\ &\text{random network configurations,} \\ &\text{multi-stage default cascades.} \end{aligned} \tag{680}$$

However, the estimators contain sampling error.

If $\hat{\theta}_N$ estimates a quantity θ , then under standard Monte Carlo conditions its error often decreases at the characteristic rate

$$\boxed{\hat{\theta}_N - \theta = O_p(N^{-1/2})}. \tag{681}$$

Consequently, reducing ordinary Monte Carlo error by a factor of two typically requires approximately four times as many simulations.

For extreme tail quantities, the problem is more severe because only a small fraction of the simulated observations contributes directly to the tail estimator.

11.12 Analytical and Monte Carlo Methods Are Complementary

Analytical and Monte Carlo approaches should not be viewed as mutually exclusive.

When an analytical solution exists, it provides a natural benchmark for testing a simulation.

One may examine the convergence

$$\boxed{\widehat{\text{VaR}}_{\alpha,N} \longrightarrow \text{VaR}_{\alpha}} \tag{682}$$

and

$$\boxed{\widehat{\text{ES}}_{\alpha,N} \longrightarrow \text{ES}_{\alpha}} \tag{683}$$

as

$$N \rightarrow \infty. \tag{684}$$

After validating the numerical engine in a tractable case, the same simulation architecture can be extended to models for which no convenient closed-form solution is available.

Hence,

$$\boxed{\text{Analytical Benchmark} \longrightarrow \text{Monte Carlo Validation} \longrightarrow \text{Complex Simulation.}} \tag{685}$$

11.13 Individual versus Systemic Risk

Individual risk concerns the loss distribution of a particular institution.

For institution i , one may compute

$$\text{VaR}_{\alpha}(L_i), \quad \text{ES}_{\alpha}(L_i). \tag{686}$$

These quantities contain important information about stand-alone loss severity.

They do not, however, encode the complete structure of financial interdependence.

Systemic outcomes additionally depend on objects such as

$$\boxed{W, \quad \mathbf{K}, \quad \mathbf{R}, \quad \mathbf{S}.} \quad (687)$$

The exposure matrix W determines the channels through which losses may propagate, $\mathbf{K}$ determines loss-absorption capacity, $\mathbf{R}$ determines recovery, and $\mathbf{S}$ determines the initial balance-sheet disturbance.

11.14 The Non-Additivity of Systemic Risk

In general,

$$\boxed{\text{Systemic Risk} \neq \sum_{i=1}^n \text{Individual Risk}_i.} \quad (688)$$

The reason is that the system contains interaction terms.

Consider two institutions i and j . The loss of j can alter the financial position of i through

$$W_{ij}. \quad (689)$$

Thus the state of institution i cannot necessarily be determined from its stand-alone loss variable alone.

Schematically,

$$\boxed{\text{Individual Risk} + \text{Financial Interdependence} \longrightarrow \text{Systemic Risk}.} \quad (690)$$

11.15 Same Individual Risk, Different Systemic Risk

Suppose two systems $\mathcal{A}$ and $\mathcal{B}$ have identical marginal loss distributions:

$$F_i^{\mathcal{A}} = F_i^{\mathcal{B}} \quad (691)$$

for every institution i .

Their individual VaR and Expected Shortfall values may therefore be identical:

$$\text{VaR}_\alpha^{\mathcal{A}}(L_i) = \text{VaR}_\alpha^{\mathcal{B}}(L_i), \quad (692)$$

$$\text{ES}_\alpha^{\mathcal{A}}(L_i) = \text{ES}_\alpha^{\mathcal{B}}(L_i). \quad (693)$$

But if

$$\boxed{W^{\mathcal{A}} \neq W^{\mathcal{B}},} \quad (694)$$

the terminal cascade fractions may differ:

$$\boxed{\phi_\infty^{\mathcal{A}} \neq \phi_\infty^{\mathcal{B}}.} \quad (695)$$

Network architecture therefore introduces information that is invisible to stand-alone risk measures.

11.16 Direct Loss versus Contagion Loss

The distinction between direct and contagion losses provides perhaps the clearest decomposition of the full framework.

Let

$$S_i \tag{696}$$

denote the external loss directly affecting institution i .

This loss is exogenous to the network cascade.

Now let

$$C_i^{(t)} = \sum_{j=1}^n (1 - R_j) W_{ij} D_j^{(t)} \tag{697}$$

denote cumulative losses generated by defaulted counterparties.

This component is endogenous to the network.

The total loss is therefore

$$\Lambda_i^{(t)} = \underbrace{S_i}_{\text{direct loss}} + \underbrace{C_i^{(t)}}_{\text{contagion loss}}. \tag{698}$$

11.17 Direct Default versus Contagion-Induced Default

The decomposition also distinguishes two mechanisms of failure.

A direct default occurs when

$$S_i \geq K_i. \tag{699}$$

A contagion-induced default may occur when

$$S_i < K_i \tag{700}$$

but

$$S_i + C_i^{(t)} \geq K_i. \tag{701}$$

Hence,

$$\begin{aligned} S_i \geq K_i &\Rightarrow \text{direct default,} \\ S_i < K_i \text{ and } S_i + C_i^{(t)} \geq K_i &\Rightarrow \text{contagion-induced default.} \end{aligned} \tag{702}$$

This distinction isolates the amplification attributable specifically to financial interconnectedness.

11.18 Network Amplification

Define the initial default set by

$$\mathcal{D}_0 \tag{703}$$

and the terminal default set by

$$\mathcal{D}_\infty. \tag{704}$$

The number of defaults generated endogenously through contagion is

$$N_{\text{endo}} = |\mathcal{D}_{\infty}| - |\mathcal{D}_0|. \quad (705)$$

If

$$|\mathcal{D}_0| > 0, \quad (706)$$

a simple amplification ratio is

$$A_D = \frac{|\mathcal{D}_{\infty}|}{|\mathcal{D}_0|}. \quad (707)$$

When

$$A_D = 1, \quad (708)$$

the initial defaults generate no additional failures.

When

$$A_D > 1, \quad (709)$$

the financial network has amplified the initial disturbance in terms of the number of defaults.

11.19 Comparative Summary

The principal differences can be summarized as follows.

Table 1: Comparison of the principal concepts used in the framework.

Comparison	First concept	Second concept
VaR vs ES	Quantile threshold of the loss distribution	Average severity across the upper tail
Gaussian vs Student- t	Light-tailed benchmark with exponentially decaying tails	Heavy-tailed model with polynomial tail decay
Analytical vs Monte Carlo	Closed-form or numerically exact model-based calculation	Simulation-based approximation with sampling error
Individual vs systemic risk	Risk associated with a stand-alone institution	Risk arising from shocks, balance sheets, and interactions
Direct vs contagion loss	Exogenous loss imposed directly on an institution	Endogenous loss transmitted by defaulted counterparties

11.20 A Unified Interpretation

The five comparisons operate at different layers of the same problem.

The distributional layer asks

$$\boxed{\text{What is the statistical structure of losses?}} \quad (710)$$

The tail-risk layer asks

$$\boxed{\text{Where does the extreme-loss region begin, and how severe is it?}} \quad (711)$$

The computational layer asks

$$\boxed{\text{How should the relevant quantities be evaluated?}} \quad (712)$$

The balance-sheet layer asks

$$\boxed{\text{Can the institution absorb the realized loss?}} \quad (713)$$

Finally, the network layer asks

$$\boxed{\text{Can the resulting distress propagate to other institutions?}} \quad (714)$$

These layers combine into the sequence

$$\boxed{\begin{array}{l} \text{Distribution} \longrightarrow \text{Tail Risk} \\ \longrightarrow \text{Realized Loss} \\ \longrightarrow \text{Balance-Sheet Impact} \\ \longrightarrow \text{Initial Default} \\ \longrightarrow \text{Network Transmission} \\ \longrightarrow \text{Systemic Outcome.} \end{array}} \quad (715)$$

11.21 Main Comparative Insight

The comparison reveals that the different methodologies are complementary rather than interchangeable.

VaR identifies a tail threshold, while Expected Shortfall characterizes loss severity beyond that threshold.

Gaussian and Student- t models demonstrate that equal volatility does not imply equal tail risk.

Analytical methods provide precision and efficiency when tractable solutions exist, while Monte Carlo methods provide flexibility for complex models at the cost of sampling uncertainty.

Individual risk measures characterize stand-alone vulnerability, whereas systemic-risk models incorporate financial interactions.

Finally, direct losses represent the original disturbance, while contagion losses measure the endogenous amplification generated by the financial network.

The complete framework can therefore be summarized as

$$\boxed{\begin{array}{ccccccc} \underbrace{\text{Distribution}} & \longrightarrow & \underbrace{\text{VaR / ES}} & \longrightarrow & \underbrace{\text{Monte Carlo}} & & \\ \text{risk environment} & & \text{tail measurement} & & \text{numerical estimation} & & \\ & & & & & & \\ & \longrightarrow & \underbrace{\text{Direct Shock}} & \longrightarrow & \underbrace{\text{Network Contagion}} & \longrightarrow & \underbrace{\text{Systemic Risk}} \\ & & \text{initial impact} & & \text{endogenous amplification} & & \text{system-wide outcome} \end{array}} \quad (716)$$

No individual component of this sequence replaces the others. Together, they describe different stages through which financial uncertainty can be transformed into system-wide instability.

12 Applications

The framework developed in the previous sections has applications across several areas of quantitative finance and financial stability. Its main advantage is that it connects three levels of analysis that are often studied separately:

$$\boxed{\text{Statistical Loss Modeling} \longrightarrow \text{Institution-Level Risk} \longrightarrow \text{Network-Level Risk.}} \quad (717)$$

At the statistical level, loss distributions, heavy tails, Value at Risk, Expected Shortfall, and Monte Carlo simulation characterize uncertainty and extreme events.

At the institutional level, realized losses interact with capital buffers and determine whether financial institutions remain solvent.

At the systemic level, defaults may transmit losses through counterparty exposures and generate endogenous amplification.

This architecture makes the framework applicable to market risk, stress testing, capital analysis, portfolio risk, counterparty risk, systemic-risk monitoring, and broader financial-stability analysis.

12.1 Market Risk

Market risk concerns losses generated by adverse movements in financial prices and risk factors, including equity prices, interest rates, exchange rates, credit spreads, and other market variables.

Let

$$V_t \quad (718)$$

denote the value of a portfolio at time t . Over a risk horizon Δt , a loss variable may be defined as

$$\boxed{L_{t,\Delta t} = -(V_{t+\Delta t} - V_t).} \quad (719)$$

Positive values of $L_{t,\Delta t}$ therefore correspond to financial losses.

Market-risk analysis seeks to characterize the distribution

$$F_L(\ell) = \mathbb{P}(L \leq \ell). \quad (720)$$

Value at Risk provides a quantile-based measure,

$$\boxed{\text{VaR}_\alpha(L) = F_L^{\leftarrow}(\alpha),} \quad (721)$$

while Expected Shortfall characterizes losses beyond the selected quantile:

$$\boxed{\text{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 \text{VaR}_u(L) du.} \quad (722)$$

The distinction is particularly relevant during periods of market stress, when the severity of losses beyond a quantile threshold may become more important than the threshold itself.

The comparison between Gaussian and heavy-tailed models developed earlier also has a direct market-risk interpretation:

$$\boxed{\text{similar volatility} \not\Rightarrow \text{similar extreme-loss probability.}} \quad (723)$$

Consequently, distributional assumptions can materially affect estimated tail risk.

12.2 Stress Testing

Stress testing reverses the usual probabilistic question.

Rather than asking only

$$\text{“How probable is a given loss?”}, \quad (724)$$

one asks

$$\boxed{\text{“What happens if a severe scenario occurs?”}} \quad (725)$$

Let

$$\mathbf{S} = (S_1, \dots, S_n)^\top \quad (726)$$

denote a vector of stressed losses imposed on financial institutions.

A stress scenario may represent simultaneous movements in several risk factors,

$$\boxed{\mathbf{z}^{\text{stress}} = (z_1^{\text{stress}}, \dots, z_m^{\text{stress}})}, \quad (727)$$

which are translated into portfolio or balance-sheet losses through a valuation mapping

$$\boxed{\mathbf{S} = \Psi(\mathbf{z}^{\text{stress}})}. \quad (728)$$

The resulting losses can then be compared with capital buffers:

$$\boxed{K_i^{\text{post}} = K_i - S_i}. \quad (729)$$

In a network setting, however, this is only the first stage.

If some institutions default under the stress scenario, their failures may generate additional counterparty losses:

$$S_i \longrightarrow D_i^{(0)} \longrightarrow C_j^{(1)} \longrightarrow D_j^{(1)} \longrightarrow \dots \quad (730)$$

Thus a network stress test can distinguish between

$$\boxed{\underbrace{\text{first-round losses}}_{\text{direct stress}} \quad \text{and} \quad \underbrace{\text{later-round losses}}_{\text{contagion}}}. \quad (731)$$

12.3 Capital Adequacy

Capital provides a loss-absorption buffer between asset losses and insolvency.

Let

$$K_i \quad (732)$$

denote the capital available to institution i .

For an external loss S_i , the remaining buffer is

$$\boxed{B_i = K_i - S_i}. \quad (733)$$

The capital-normalized shock introduced previously,

$$\boxed{\chi_i = \frac{S_i}{K_i}}, \quad (734)$$

provides a simple measure of the severity of the initial disturbance relative to available capital. Under the simplified default rule adopted in this study,

$$\chi_i < 1 \quad (735)$$

indicates survival of the direct shock, whereas

$$\chi_i \geq 1 \quad (736)$$

implies exhaustion of the initial capital buffer.

In a network, however, capital must absorb both direct and transmitted losses:

$$\boxed{K_i^{(t)} = K_i - S_i - C_i^{(t)}} \quad (737)$$

Therefore an institution may appear sufficiently capitalized against its stand-alone shock but become insolvent after receiving losses from defaulted counterparties.

This distinction motivates a systemic interpretation of capital adequacy:

$$\boxed{\text{capital adequacy} \neq \text{protection against direct losses alone.}} \quad (738)$$

12.4 Portfolio Risk

For a portfolio containing m assets, let

$$\mathbf{w} = (w_1, \dots, w_m)^\top \quad (739)$$

denote portfolio weights and

$$\mathbf{R} = (R_1, \dots, R_m)^\top \quad (740)$$

the vector of asset returns.

The portfolio return is

$$\boxed{R_P = \mathbf{w}^\top \mathbf{R}.} \quad (741)$$

A corresponding loss variable may be written as

$$\boxed{L_P = -\mathbf{w}^\top \mathbf{R}.} \quad (742)$$

If the covariance matrix of returns is

$$\Sigma = \text{Cov}(\mathbf{R}), \quad (743)$$

then portfolio variance is

$$\boxed{\sigma_P^2 = \mathbf{w}^\top \Sigma \mathbf{w}.} \quad (744)$$

This equation makes clear that portfolio risk depends not only on the volatility of individual assets but also on their dependence structure.

For example,

$$\sigma_P^2 = \sum_i w_i^2 \sigma_i^2 + 2 \sum_{i < j} w_i w_j \text{Cov}(R_i, R_j). \quad (745)$$

Hence,

$$\boxed{\text{Portfolio Risk} \neq \sum_i \text{Stand-Alone Asset Risk}_i.} \quad (746)$$

This provides a useful conceptual precursor to systemic risk: dependence already matters at the portfolio level, while financial networks introduce an additional mechanism in which losses may be transmitted dynamically between institutions.

12.5 Portfolio Tail Risk

Portfolio VaR is

$$\boxed{\text{VaR}_\alpha(L_P),} \quad (747)$$

and portfolio Expected Shortfall is

$$\boxed{\text{ES}_\alpha(L_P).} \quad (748)$$

When portfolio returns are approximately Gaussian, analytical expressions may be available.

For nonlinear portfolios, non-Gaussian factors, heavy-tailed returns, or complex dependence structures, Monte Carlo simulation becomes particularly useful:

$$\boxed{\mathbf{R}^{(m)} \longrightarrow L_P^{(m)} \longrightarrow \widehat{\text{VaR}}_\alpha, \widehat{\text{ES}}_\alpha.} \quad (749)$$

This illustrates the practical complementarity between analytical and simulation-based approaches discussed in the previous section.

12.6 Counterparty Risk

Counterparty risk concerns losses generated when a financial counterparty fails to satisfy its obligations.

Within the network notation adopted in this study,

$$W_{ij} \quad (750)$$

denotes the exposure of institution i to institution j .

If institution j defaults and R_j denotes its recovery fraction, the loss transmitted to i is

$$\boxed{\Delta_{i \leftarrow j} = (1 - R_j)W_{ij}.} \quad (751)$$

The exposure becomes especially important relative to the creditor's capital buffer.

Define

$$\boxed{\eta_{ij} = \frac{(1 - R_j)W_{ij}}{K_i}.} \quad (752)$$

This ratio measures the potential counterparty loss from j relative to the initial capital of i . Under the simplified threshold model,

$$\eta_{ij} \geq 1 \quad (753)$$

implies that the default of j alone is sufficient to exhaust the initial capital buffer of i , in the absence of other adjustments.

Multiple counterparty failures may instead generate

$$\boxed{\eta_i(\mathcal{D}) = \frac{\sum_{j \in \mathcal{D}} (1 - R_j)W_{ij}}{K_i}.} \quad (754)$$

Thus even individually manageable exposures may become dangerous when several counterparties default simultaneously or sequentially.

12.7 Counterparty Concentration

Two institutions with equal total interbank assets can have very different counterparty-risk profiles.

Let

$$A_i^{IB} = \sum_j W_{ij}. \quad (755)$$

Define normalized exposure weights

$$q_{ij} = \frac{W_{ij}}{A_i^{IB}}. \quad (756)$$

A simple concentration measure is

$$H_i = \sum_j q_{ij}^2. \quad (757)$$

A larger value of H_i indicates that exposures are concentrated among fewer counterparties, whereas a smaller value corresponds to a more distributed exposure structure.

Consequently, total exposure alone does not determine counterparty vulnerability:

$$\boxed{\text{counterparty risk} = \text{exposure magnitude} + \text{concentration} + \text{recovery} + \text{capital capacity}.} \quad (758)$$

12.8 Systemic-Risk Monitoring

Systemic-risk monitoring shifts attention from individual institutions to the state of the financial network.

Let

$$\Phi = \frac{1}{n} \sum_{i=1}^n D_i^* \quad (759)$$

denote the terminal fraction of defaulted institutions.

Under stochastic shocks, Φ becomes a random variable.

Monte Carlo simulation can then estimate

$$\boxed{\mathbb{E}[\Phi],} \quad (760)$$

as well as the probability that the cascade exceeds a systemic threshold γ :

$$\boxed{\mathbb{P}(\Phi \geq \gamma).} \quad (761)$$

This provides a natural distinction between the expected extent of contagion and the probability of a particularly severe systemic event.

12.9 Monitoring Cascade Amplification

Let

$$N_0 = |\mathcal{D}_0| \quad (762)$$

denote the number of initial defaults and

$$N_\infty = |\mathcal{D}_\infty| \quad (763)$$

the final number of defaults.

The endogenous number of failures is

$$N_{\text{endo}} = N_\infty - N_0. \quad (764)$$

When $N_0 > 0$, the default amplification factor

$$A_D = \frac{N_\infty}{N_0} \quad (765)$$

provides a simple diagnostic of network amplification.

The cascade depth

$$T_{\text{depth}} \quad (766)$$

provides complementary information.

Two scenarios may produce the same final number of defaults but have very different propagation structures:

$$N_\infty^{(A)} = N_\infty^{(B)} \not\Rightarrow T_{\text{depth}}^{(A)} = T_{\text{depth}}^{(B)}. \quad (767)$$

Therefore systemic monitoring benefits from considering both cascade size and cascade dynamics.

12.10 Institution-Level Systemic Vulnerability

The network vulnerability ratio introduced earlier provides an institution-level monitoring variable:

$$\Gamma_i^{(t)} = \frac{S_i + \sum_j (1 - R_j) W_{ij} D_j^{(t)}}{K_i}. \quad (768)$$

This ratio decomposes naturally as

$$\Gamma_i^{(t)} = \underbrace{\frac{S_i}{K_i}}_{\text{direct vulnerability}} + \underbrace{\frac{\sum_j (1 - R_j) W_{ij} D_j^{(t)}}{K_i}}_{\text{network vulnerability}}. \quad (769)$$

Monitoring $\Gamma_i^{(t)}$ therefore provides information about both the institution's direct exposure to the original disturbance and its endogenous exposure to distress elsewhere in the system.

12.11 Financial Stability

Financial stability is fundamentally a system-level concept.

A financial system may contain institutions that appear individually resilient while nevertheless remaining vulnerable to common shocks, concentrated exposures, correlated losses, or network amplification.

Conversely, an isolated institution may experience a substantial loss without generating a systemic event if its losses are absorbed without significant transmission to other institutions.

This distinction can be expressed schematically as

$$\boxed{\text{Institutional Distress} \not\equiv \text{Systemic Instability.}} \quad (770)$$

Whether institutional distress becomes systemic depends on the interaction between

$$\boxed{\text{Shock Severity} \times \text{Capital Buffers} \times \text{Exposure Structure} \times \text{Network Topology.}} \quad (771)$$

This interaction motivates the joint use of statistical tail-risk measures, balance-sheet analysis, stress scenarios, and network-contagion models.

12.12 A Multi-Layer View of Financial Stability

The complete framework can be interpreted as a sequence of diagnostic layers.

The first layer concerns market uncertainty:

$$\boxed{F_{\mathbf{L}} \longrightarrow \text{distribution of potential losses.}} \quad (772)$$

The second concerns extreme-event severity:

$$\boxed{F_{\mathbf{L}} \longrightarrow \text{VaR}_{\alpha}, \text{ES}_{\alpha}.} \quad (773)$$

The third concerns scenario generation:

$$\boxed{\mathbf{L}^{(m)} \longrightarrow \mathbf{S}^{(m)}.} \quad (774)$$

The fourth concerns institutional resilience:

$$\boxed{\mathbf{S}^{(m)} + \mathbf{K} \longrightarrow \mathbf{D}^{(0,m)}.} \quad (775)$$

Finally, the network layer determines systemic amplification:

$$\boxed{\mathbf{D}^{(0,m)} + \mathbf{W} + \mathbf{R} \longrightarrow \mathbf{D}^{(*,m)}.} \quad (776)$$

Combining these layers gives

$$\boxed{\underbrace{F_{\mathbf{L}}}_{\text{risk environment}} \rightarrow \underbrace{\text{VaR} / \text{ES}}_{\text{tail diagnostics}} \rightarrow \underbrace{\mathbf{S}}_{\text{stress}} \rightarrow \underbrace{\mathbf{D}^{(0)}}_{\text{initial failures}} \rightarrow \underbrace{\mathbf{W}}_{\text{financial network}} \rightarrow \underbrace{\mathbf{D}^*}_{\text{systemic outcome}}.} \quad (777)$$

12.13 Application Matrix

The relationship between the principal methods and their applications can be summarized as follows.

12.14 Practical Interpretation

The applications considered above answer progressively broader questions.

Market-risk analysis asks

$$\boxed{\text{How large can financial losses become?}} \quad (778)$$

Portfolio analysis asks

$$\boxed{\text{How does dependence among positions affect those losses?}} \quad (779)$$

Capital analysis asks

Table 2: Main applications of the quantitative framework.

Application	Primary Object	Representative Output
Market risk	Loss distribution	VaR and Expected Shortfall
Stress testing	Severe scenario	Post-stress losses and capital
Capital adequacy	Loss-absorption capacity	Remaining capital buffer
Portfolio risk	Joint asset distribution	Portfolio loss distribution
Counterparty risk	Bilateral exposures	Loss given counterparty default
Systemic-risk monitoring	Financial network	Cascade size and amplification
Financial stability	Integrated financial system	Distribution of system-wide outcomes

$$\boxed{\text{Can the institution absorb the realized loss?}} \quad (780)$$

Counterparty analysis asks

$$\boxed{\text{Who is exposed if the institution fails?}} \quad (781)$$

Systemic-risk analysis asks

$$\boxed{\text{Can the resulting distress propagate through the financial system?}} \quad (782)$$

Financial-stability analysis combines all of these questions.

Consequently, the framework progresses from measuring the statistical properties of losses to studying their potential amplification through financial interdependence:

$$\boxed{\text{Risk Measurement} \longrightarrow \text{Loss Absorption} \longrightarrow \text{Risk Transmission} \longrightarrow \text{Systemic Stability.}} \quad (783)$$

13 Limitations and Extensions

The framework developed in this study provides a transparent connection between tail-risk measurement, balance-sheet shocks, counterparty exposures, and default cascades. Its transparency, however, relies on a number of simplifying assumptions.

In the baseline contagion mechanism, financial institutions are connected through a directed weighted exposure matrix W , losses are transmitted through counterparty defaults, capital buffers determine default thresholds, and the process evolves iteratively until no additional defaults occur.

This structure isolates the mechanism of direct counterparty contagion, but it does not represent every channel through which financial distress may propagate.

The principal limitations and natural extensions can be summarized as

$$\boxed{\begin{aligned} \text{Direct Counterparty Contagion} &\longrightarrow \text{Endogenous Recovery} \\ &\longrightarrow \text{Common-Asset Exposure} \\ &\longrightarrow \text{Fire-Sale Contagion} \\ &\longrightarrow \text{Dynamic Networks} \\ &\longrightarrow \text{Multi-Layer Systemic Risk.} \end{aligned}} \quad (784)$$

13.1 Baseline Simplifications

The baseline model can be represented schematically by

$$S_i + \sum_{j=1}^n (1 - R_j) W_{ij} D_j^{(t)} \geq K_i, \quad (785)$$

where S_i denotes the external shock, K_i the capital buffer, W_{ij} the exposure of institution i to institution j , R_j the recovery rate associated with the default of j , and $D_j^{(t)}$ its default indicator.

This representation captures an important mechanism:

$$\text{default} \rightarrow \text{creditor loss} \rightarrow \text{capital erosion} \rightarrow \text{additional default}. \quad (786)$$

Nevertheless, several simplifying assumptions remain.

In particular, depending on the simulation specification, the model may treat recovery rates as exogenous, hold the network fixed during the cascade, omit endogenous asset-price dynamics, and represent default as a discrete state rather than through a full clearing mechanism.

These assumptions should be interpreted as deliberate simplifications that allow the counterparty-contagion channel to be studied separately.

13.2 Fire Sales

A first important extension concerns fire sales.

A distressed institution may need to liquidate assets in order to obtain liquidity, reduce leverage, or satisfy financial constraints. If many institutions attempt to sell similar assets simultaneously, market prices may decline.

Let

$$q_a^{(t)} \quad (787)$$

denote the price of asset a at time t , and let

$$z_a^{(t)} \quad (788)$$

denote aggregate sales of that asset.

An inverse-demand function can be represented abstractly as

$$q_a^{(t+1)} = f_a(z_a^{(t)}), \quad f'_a(\cdot) < 0. \quad (789)$$

Thus larger aggregate liquidations generate lower market prices.

If institution i holds H_{ia} units of asset a , the mark-to-market loss associated with the price movement is

$$L_{i,a}^{\text{FS},(t)} = H_{ia} (q_a^{(0)} - q_a^{(t)}). \quad (790)$$

Across m assets,

$$L_i^{\text{FS},(t)} = \sum_{a=1}^m H_{ia} (q_a^{(0)} - q_a^{(t)}). \quad (791)$$

The default condition would then become

$$S_i + C_i^{(t)} + L_i^{\text{FS},(t)} \geq K_i, \quad (792)$$

where $C_i^{(t)}$ denotes direct counterparty contagion losses.
This introduces a second endogenous propagation mechanism:

$$\boxed{\text{distress} \rightarrow \text{asset sales} \rightarrow \text{price decline} \rightarrow \text{mark-to-market losses} \rightarrow \text{further distress.}} \quad (793)$$

This mechanism is central to models such as Cifuentes–Ferrucci–Shin [5].

13.3 Common Asset Holdings

Fire-sale contagion requires information about asset holdings in addition to interbank exposures.
Let

$$\boxed{H = [H_{ia}] \in \mathbb{R}_+^{n \times m}} \quad (794)$$

denote the institution–asset holdings matrix.

The matrix W describes direct financial claims between institutions,

$$W_{ij} = \text{exposure of institution } i \text{ to institution } j, \quad (795)$$

whereas H describes indirect connections through common assets.

Two institutions i and j may have

$$W_{ij} = W_{ji} = 0 \quad (796)$$

and still be financially connected whenever they hold the same assets.

For example,

$$\boxed{\sum_{a=1}^m H_{ia} H_{ja} > 0} \quad (797)$$

indicates some degree of portfolio overlap.

A normalized overlap measure may be defined as

$$\boxed{O_{ij} = \frac{\sum_a H_{ia} H_{ja}}{\sqrt{\sum_a H_{ia}^2} \sqrt{\sum_a H_{ja}^2}},} \quad (798)$$

provided the corresponding portfolios are nonzero.

This creates an indirect network:

$$\boxed{\text{Institution} \rightarrow \text{Asset} \rightarrow \text{Institution.}} \quad (799)$$

Hence systemic connectivity is not restricted to direct contractual relationships.

13.4 Multiple Shocks

The framework can also be extended from a single initial disturbance to multiple simultaneous or sequential shocks.

Instead of

$$\mathbf{S} = (0, \dots, S_k, \dots, 0)^\top, \quad (800)$$

consider

$$\boxed{\mathbf{S} = (S_1, \dots, S_n)^\top.} \quad (801)$$

The initial default set becomes

$$\mathcal{D}_0 = \{i : S_i \geq K_i\}. \quad (802)$$

More generally, shocks may evolve through time:

$$\mathbf{S}^{(0)}, \mathbf{S}^{(1)}, \dots, \mathbf{S}^{(T)}. \quad (803)$$

This permits the analysis of common market shocks, correlated shocks, sector-specific disturbances, and repeated stress events.

Within a Monte Carlo framework, one may draw

$$\mathbf{S}^{(m)} \sim F_{\mathbf{S}}, \quad (804)$$

where the joint distribution $F_{\mathbf{S}}$ incorporates dependence between institutions.

This is important because

$$\mathbb{P}(S_i > s_i, S_j > s_j) \quad (805)$$

may differ substantially from

$$\mathbb{P}(S_i > s_i)\mathbb{P}(S_j > s_j). \quad (806)$$

Consequently, systemic stress may arise not only from contagion after a single default, but also from correlated initial distress.

13.5 Recovery Rates

The baseline model can assign an exogenous recovery rate R_j to a defaulted institution j .

The corresponding creditor loss is

$$\Delta_{i \leftarrow j} = (1 - R_j)W_{ij}. \quad (807)$$

This assumption is transparent but restrictive.

In practice, recovery may depend on the financial condition of the defaulting institution, collateral, seniority, liquidation values, and the state of the system.

A natural extension is therefore

$$R_j = \mathcal{R}_j(A_j, L_j, \mathbf{q}, \mathcal{D}, \dots), \quad (808)$$

where recovery becomes endogenous.

The transmitted loss is then

$$\Delta_{i \leftarrow j} = [1 - \mathcal{R}_j(\cdot)] W_{ij}. \quad (809)$$

This creates an additional feedback channel because severe systemic states may simultaneously increase defaults and reduce recovery values.

13.6 Heterogeneous Balance Sheets

Another extension concerns institutional heterogeneity.

A simplified model may characterize institutions primarily through their capital buffers and interbank exposures. A richer representation would distinguish several balance-sheet components.

For institution i , write

$$\boxed{A_i^E + A_i^{IB} = L_i^{IB} + L_i^O + K_i,} \quad (810)$$

where

$$A_i^E = \text{external assets}, \quad (811)$$

$$A_i^{IB} = \text{interbank assets}, \quad (812)$$

$$L_i^{IB} = \text{interbank liabilities}, \quad (813)$$

$$L_i^O = \text{other liabilities}, \quad (814)$$

$$K_i = \text{equity capital}. \quad (815)$$

Institutions may differ in leverage, size, portfolio composition, connectivity, funding structure, and loss-absorption capacity.

For example, leverage can be represented as

$$\boxed{\lambda_i = \frac{A_i}{K_i}.} \quad (816)$$

Two institutions exposed to the same proportional asset shock may therefore experience very different equity effects.

If asset values decline proportionally by δ , then approximately

$$\Delta A_i = -\delta A_i, \quad (817)$$

and relative to capital,

$$\boxed{\frac{|\Delta A_i|}{K_i} = \delta \lambda_i.} \quad (818)$$

Hence heterogeneity in leverage itself becomes a source of heterogeneous vulnerability.

13.7 Dynamic Networks

The baseline network is naturally represented by a fixed matrix

$$W. \quad (819)$$

A more general model allows exposures to evolve:

$$\boxed{W \longrightarrow W(t).} \quad (820)$$

The contagion process then depends jointly on financial states and network evolution.

For example,

$$\boxed{W_{ij}(t+1) = \mathcal{G}_{ij}(W(t), \mathbf{K}(t), \mathbf{D}(t), \mathbf{q}(t)).} \quad (821)$$

This extension may represent deleveraging, withdrawal of funding, counterparty substitution, maturity effects, or endogenous rewiring.

The system is no longer described only by

$$\mathbf{D}^{(t+1)} = \mathcal{T} \left(\mathbf{D}^{(t)} \right), \quad (822)$$

but rather by a coupled dynamical system:

$$\boxed{\begin{aligned} \mathbf{D}^{(t+1)} &= \mathcal{T} \left(\mathbf{D}^{(t)}, W^{(t)}, \mathbf{q}^{(t)} \right), \\ W^{(t+1)} &= \mathcal{G} \left(W^{(t)}, \mathbf{D}^{(t)}, \mathbf{q}^{(t)} \right). \end{aligned}} \quad (823)$$

Network topology therefore becomes endogenous rather than merely an input.

13.8 Eisenberg–Noe Clearing Mechanism

A major extension replaces the binary threshold cascade with a clearing mechanism.

In the Eisenberg–Noe framework [7], institutions have nominal liabilities to one another. Let

$$\bar{p}_i \quad (824)$$

denote the total nominal liabilities of institution i , and define the relative liability matrix

$$\boxed{\Pi_{ij} = \frac{L_{ij}}{\bar{p}_i}}, \quad (825)$$

when $\bar{p}_i > 0$.

Let

$$x_i \quad (826)$$

denote external assets available to institution i .

The clearing payment satisfies the fixed-point relation

$$\boxed{p_i = \min \left\{ \bar{p}_i, x_i + \sum_{j=1}^n \Pi_{ji} p_j \right\}}. \quad (827)$$

In vector notation,

$$\boxed{\mathbf{p} = \bar{\mathbf{p}} \wedge \left(\mathbf{x} + \Pi^\top \mathbf{p} \right)}, \quad (828)$$

where $\wedge$ denotes the componentwise minimum.

This differs fundamentally from a binary cascade model.

Instead of assuming only

$$D_i \in \{0, 1\}, \quad (829)$$

the clearing system determines the amount that each institution can actually pay.

The payment fraction

$$\boxed{r_i = \frac{p_i}{\bar{p}_i}} \quad (830)$$

acts as an endogenous recovery fraction on nominal liabilities whenever $\bar{p}_i > 0$.

Thus,

$$\boxed{\text{threshold cascade} \longrightarrow \text{endogenous clearing payments}} \quad (831)$$

constitutes a substantial extension of the baseline framework.

13.9 Binary Default versus Clearing

The distinction can be summarized as follows.

In the threshold model,

$$D_i = \mathbf{1}[\text{loss}_i \geq K_i]. \quad (832)$$

In the clearing framework,

$$p_i = \min \left\{ \bar{p}_i, x_i + \sum_j \Pi_{ji} p_j \right\}. \quad (833)$$

The former asks whether a capital threshold has been crossed.

The latter asks how much of the institution's liabilities can be paid consistently with payments received from the rest of the network.

These approaches therefore describe related but mathematically distinct mechanisms of financial distress.

13.10 Cifuentes–Ferrucci–Shin Extension

The Cifuentes–Ferrucci–Shin framework [5] introduces an additional market-mediated contagion mechanism.

The essential feedback is

$$\begin{aligned} \text{balance-sheet pressure} &\longrightarrow \text{asset liquidation} \longrightarrow \text{price decline} \\ &\longrightarrow \text{mark-to-market losses} \longrightarrow \text{renewed balance-sheet pressure.} \end{aligned} \quad (834)$$

This is fundamentally different from pure counterparty contagion.

With direct counterparty contagion,

$$i \rightarrow j \quad (835)$$

through a bilateral financial claim.

With common-asset contagion,

$$i \rightarrow a \rightarrow j, \quad (836)$$

where a denotes an asset held by both institutions.

A combined loss equation may therefore take the form

$$\Lambda_i^{(t)} = S_i + \underbrace{\sum_j (1 - R_j) W_{ij} D_j^{(t)}}_{\text{counterparty channel}} + \underbrace{\sum_a H_{ia} (q_a^{(0)} - q_a^{(t)})}_{\text{market channel}}. \quad (837)$$

This equation provides a natural bridge from direct network contagion to market-mediated systemic risk.

13.11 Multi-Layer Contagion

The previous extensions naturally lead to a multi-layer representation of the financial system.

Instead of one network W , consider

$$\mathcal{W} = \{W^{[1]}, W^{[2]}, \dots, W^{[L]}\}, \quad (838)$$

where each layer represents a different financial relationship.

For example,

$$\begin{aligned} W^{[1]} &= \text{interbank credit,} \\ W^{[2]} &= \text{funding relationships,} \\ W^{[3]} &= \text{common-asset exposure,} \\ W^{[4]} &= \text{other contractual exposures.} \end{aligned} \quad (839)$$

Loss transmission through layer ℓ may be represented by

$$C_i^{[\ell],(t)} = \sum_j \Lambda_{ij}^{[\ell]} D_j^{(t)}, \quad (840)$$

where $\Lambda_{ij}^{[\ell]}$ denotes the effective loss transmitted through that layer.

Total network loss becomes

$$C_i^{(t)} = \sum_{\ell=1}^L C_i^{[\ell],(t)}. \quad (841)$$

A corresponding default condition is

$$S_i + \sum_{\ell=1}^L C_i^{[\ell],(t)} \geq K_i. \quad (842)$$

However, a fully coupled multi-layer model may contain interactions between layers, so the total effect need not reduce to a simple additive sum.

13.12 Cross-Layer Feedback

The most important feature of a multi-layer system is the possibility of feedback between propagation mechanisms.

For example,

$$\begin{aligned} &\text{credit loss} \rightarrow \text{capital erosion} \\ &\quad \rightarrow \text{asset liquidation} \\ &\quad \rightarrow \text{price decline} \\ &\quad \rightarrow \text{losses at other institutions} \\ &\quad \rightarrow \text{additional defaults} \\ &\quad \rightarrow \text{new credit losses.} \end{aligned} \quad (843)$$

The system then contains interacting feedback loops rather than a single contagion channel.

Abstractly, if the state of the financial system is

$$X_t = (\mathbf{K}_t, \mathbf{p}_t, \mathbf{q}_t, W_t, \mathbf{D}_t), \quad (844)$$

a general dynamic model may be written as

$$\boxed{X_{t+1} = \mathcal{F}(X_t, \varepsilon_{t+1})}, \quad (845)$$

where ε_{t+1} represents new exogenous shocks.

This formulation provides a general mathematical architecture within which several contagion mechanisms can interact.

13.13 A Possible Integrated Extension

Combining the mechanisms discussed above suggests an extended systemic-risk architecture:

Heavy-Tailed Shocks	→	Balance-Sheet Losses
	→	Clearing Payments
	→	Counterparty Losses
	→	Asset Liquidations
	→	Endogenous Prices
	→	Mark-to-Market Losses
	→	Network Reconfiguration
	→	Systemic Outcome.

(846)

In such a framework, several quantities become endogenous simultaneously:

$$\boxed{\mathbf{p}, \quad \mathbf{q}, \quad \mathbf{R}, \quad W, \quad \mathbf{D}.} \quad (847)$$

The mathematical problem therefore changes from the propagation of a shock through a fixed network into the evolution of an interacting financial system.

13.14 Model Hierarchy

The models discussed in this study may consequently be interpreted as a hierarchy of increasing structural complexity:

Level I:	Tail-Risk Measurement,
Level II:	Threshold Default Cascades,
Level III:	Clearing Networks,
Level IV:	Common Assets and Fire Sales,
Level V:	Dynamic Multi-Layer Networks.

(848)

Each level introduces additional endogenous mechanisms.

The objective is therefore not to interpret the simpler framework as a complete representation of a financial system, but as a controlled model that isolates specific mechanisms before additional channels are introduced.

13.15 Directions for Further Research

Several extensions arise naturally from this hierarchy.

First, the stochastic loss generator may be coupled to heterogeneous institutional balance sheets rather than uniform or stylized capital structures.

Second, recovery rates may be determined endogenously through clearing payments or liquidation values.

Third, the institution–institution exposure network may be combined with an institution–asset bipartite network to represent both direct and indirect contagion.

Fourth, asset prices may become endogenous through inverse-demand functions, allowing fire-sale externalities to interact with counterparty defaults.

Fifth, the financial network itself may evolve during the stress episode, producing a dynamic topology.

Finally, these mechanisms may be integrated into a multi-layer system in which credit, liquidity, market, and funding channels interact.

The resulting research program can be summarized as

$$\boxed{\text{Tail Risk} \rightarrow \text{Default Risk} \rightarrow \text{Network Contagion} \rightarrow \text{Market Contagion} \rightarrow \text{Multi-Layer Systemic Risk.}} \quad (849)$$

13.16 Final Perspective

The limitations identified above do not invalidate the baseline model. Rather, they define its scope.

The current framework isolates the transition

$$\boxed{\text{external loss} \rightarrow \text{capital impairment} \rightarrow \text{default} \rightarrow \text{counterparty loss} \rightarrow \text{cascade.}} \quad (850)$$

More elaborate models add further feedback mechanisms:

$$\boxed{\begin{array}{l} \text{counterparty channel} + \text{clearing channel} \\ \quad + \text{common-asset channel} \\ \quad + \text{fire-sale channel} \\ \quad + \text{dynamic-network channel.} \end{array}} \quad (851)$$

The central challenge for future work is therefore not merely to add more variables, but to understand how these mechanisms interact and whether their interaction amplifies, dampens, or qualitatively changes the propagation of financial distress.

14 Mathematical Summary

This section summarizes the principal mathematical objects developed throughout the study. The complete framework connects probability distributions, tail-risk measures, Monte Carlo estimation, balance-sheet losses, network exposures, and default cascades.

The central architecture is

$$\boxed{F_{\mathbf{L}} \rightarrow \mathbf{L}(\omega) \rightarrow \mathbf{S} \rightarrow \mathbf{D}^{(0)} \rightarrow \mathbf{D}^* \rightarrow \Phi.} \quad (852)$$

Each arrow represents a distinct mathematical layer of the risk transmission mechanism.

14.1 Loss Random Variable

Let

$$L : \Omega \rightarrow \mathbb{R} \quad (853)$$

denote a financial loss random variable, with positive values corresponding to losses.

Its cumulative distribution function is

$$F_L(\ell) = \mathbb{P}(L \leq \ell). \quad (854)$$

For a portfolio with value V_t , a loss over horizon Δt may be defined as

$$L_{t,\Delta t} = -(V_{t+\Delta t} - V_t). \quad (855)$$

14.2 Quantiles and Value at Risk

For confidence level $\alpha \in (0, 1)$, the generalized quantile function is

$$F_L^{\leftarrow}(\alpha) = \inf \{ \ell \in \mathbb{R} : F_L(\ell) \geq \alpha \}. \quad (856)$$

Value at Risk is therefore

$$\text{VaR}_\alpha(L) = F_L^{\leftarrow}(\alpha). \quad (857)$$

Thus VaR identifies the loss threshold associated with a selected confidence level.

14.3 Expected Shortfall

Expected Shortfall is represented by

$$\text{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 \text{VaR}_u(L) du. \quad (858)$$

For an appropriate continuous loss distribution,

$$\text{ES}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)]. \quad (859)$$

Hence,

$$\begin{aligned} \text{VaR}_\alpha &\longrightarrow \text{tail threshold,} \\ \text{ES}_\alpha &\longrightarrow \text{tail severity.} \end{aligned} \quad (860)$$

14.4 Coherent Risk Measures

A coherent risk measure ρ satisfies

$$X \leq Y \implies \rho(X) \leq \rho(Y), \quad \text{monotonicity,} \quad (861)$$

$$\rho(X + c) = \rho(X) + c, \quad \text{translation invariance,} \quad (862)$$

$$\rho(\lambda X) = \lambda \rho(X), \quad \lambda \geq 0, \quad \text{positive homogeneity,} \quad (863)$$

$$\rho(X + Y) \leq \rho(X) + \rho(Y), \quad \text{subadditivity.} \quad (864)$$

Expected Shortfall satisfies the coherence properties under the standard loss convention, whereas VaR is not subadditive in general [2].

14.5 Gaussian Loss Model

For

$$L \sim \mathcal{N}(\mu, \sigma^2), \quad (865)$$

define

$$z_\alpha = \Phi^{-1}(\alpha). \quad (866)$$

Then

$$\boxed{\text{VaR}_\alpha(L) = \mu + \sigma z_\alpha.} \quad (867)$$

The corresponding Expected Shortfall is

$$\boxed{\text{ES}_\alpha(L) = \mu + \sigma \frac{\varphi(z_\alpha)}{1 - \alpha},} \quad (868)$$

where Φ and φ denote the standard normal distribution and density functions, respectively.

14.6 Student- t Model

Let

$$T_\nu \sim t_\nu. \quad (869)$$

Its density is

$$\boxed{f_\nu(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}.} \quad (870)$$

For $\nu > 2$,

$$\text{Var}(T_\nu) = \frac{\nu}{\nu - 2}. \quad (871)$$

A variance-standardized Student- t variable is therefore

$$\boxed{Z_\nu = T_\nu \sqrt{\frac{\nu - 2}{\nu}},} \quad (872)$$

for which

$$\text{Var}(Z_\nu) = 1. \quad (873)$$

For $\nu > 4$, its excess kurtosis is

$$\boxed{\kappa_{\text{excess}} = \frac{6}{\nu - 4}.} \quad (874)$$

The fundamental comparison is

$$\boxed{\text{same variance} \not\Rightarrow \text{same tail risk.}} \quad (875)$$

14.7 Monte Carlo Estimation

Generate independent or appropriately dependent simulated losses

$$L^{(1)}, \dots, L^{(N)}. \quad (876)$$

Let

$$L_{(1)} \leq \dots \leq L_{(N)} \quad (877)$$

denote the ordered sample.

The empirical VaR estimator is

$$\widehat{\text{VaR}}_{\alpha, N} = L_{(\lceil N\alpha \rceil)}. \quad (878)$$

A simple empirical Expected Shortfall estimator is

$$\widehat{\text{ES}}_{\alpha, N} = \frac{1}{N_{\text{tail}}} \sum_{m: L^{(m)} \geq \widehat{\text{VaR}}_{\alpha, N}} L^{(m)}. \quad (879)$$

The expected number of tail observations is approximately

$$N_{\text{tail}} \approx N(1 - \alpha). \quad (880)$$

Under standard Monte Carlo conditions, ordinary sampling error typically decreases at the rate

$$O_p(N^{-1/2}). \quad (881)$$

Thus extreme tail estimation becomes computationally demanding as $\alpha \rightarrow 1$.

14.8 Portfolio Loss

For asset-return vector

$$\mathbf{R} = (R_1, \dots, R_m)^\top \quad (882)$$

and portfolio weights

$$\mathbf{w} = (w_1, \dots, w_m)^\top, \quad (883)$$

portfolio return is

$$R_P = \mathbf{w}^\top \mathbf{R}, \quad (884)$$

and portfolio loss is

$$L_P = -\mathbf{w}^\top \mathbf{R}. \quad (885)$$

For covariance matrix Σ ,

$$\sigma_P^2 = \mathbf{w}^\top \Sigma \mathbf{w}. \quad (886)$$

14.9 From Random Losses to Balance-Sheet Shocks

For institution i , let

$$L_i(\omega) \quad (887)$$

denote its realized stochastic loss.

The corresponding balance-sheet shock may be defined through

$$S_i = \Psi_i(L_i(\omega)). \quad (888)$$

At system level,

$$\mathbf{S} = \Psi(\mathbf{L}(\omega)). \quad (889)$$

The simplest specification is

$$S_i = L_i(\omega). \quad (890)$$

14.10 Capital and Initial Default

Let $K_i > 0$ denote the initial capital buffer.

After the external shock,

$$K_i^{(1)} = K_i - S_i. \quad (891)$$

The capital-normalized shock is

$$\chi_i = \frac{S_i}{K_i}. \quad (892)$$

Under the simplified threshold model,

$$D_i^{(0)} = \mathbf{1}_{\{S_i \geq K_i\}}. \quad (893)$$

Hence,

$$\mathcal{D}_0 = \{i : S_i \geq K_i\}. \quad (894)$$

14.11 Financial Network

Let

$$W = [W_{ij}] \in \mathbb{R}_+^{n \times n} \quad (895)$$

denote the directed weighted exposure matrix, where W_{ij} represents the claim of institution i on institution j .

If institution j defaults with recovery rate R_j , the loss transmitted to creditor i is

$$\Delta_{i \leftarrow j} = (1 - R_j)W_{ij}. \quad (896)$$

14.12 Contagion Loss

At cascade step t , cumulative counterparty loss is

$$C_i^{(t)} = \sum_{j=1}^n (1 - R_j) W_{ij} D_j^{(t)}. \quad (897)$$

Total accumulated loss is therefore

$$\Lambda_i^{(t)} = \underbrace{S_i}_{\text{direct loss}} + \underbrace{C_i^{(t)}}_{\text{contagion loss}}. \quad (898)$$

The institution defaults whenever

$$\Lambda_i^{(t)} \geq K_i. \quad (899)$$

14.13 Iterative Default Cascade

The absorbing default update is

$$D_i^{(t+1)} = \max \left\{ D_i^{(t)}, \mathbf{1} \left[S_i + \sum_{j=1}^n (1 - R_j) W_{ij} D_j^{(t)} \geq K_i \right] \right\}. \quad (900)$$

Equivalently,

$$\mathbf{D}^{(t+1)} = \tilde{\mathcal{T}}(\mathbf{D}^{(t)}). \quad (901)$$

Since defaults are absorbing,

$$\mathcal{D}_0 \subseteq \mathcal{D}_1 \subseteq \cdots \subseteq \mathcal{D}_\infty. \quad (902)$$

For a finite network, the cascade terminates after finitely many additions to the default set.

14.14 Fixed Point

At the terminal state,

$$\mathbf{D}^* = \tilde{\mathcal{T}}(\mathbf{D}^*). \quad (903)$$

The final default fraction is

$$\Phi = \phi_\infty = \frac{1}{n} \sum_{i=1}^n D_i^*. \quad (904)$$

The number of contagion-induced defaults is

$$N_{\text{endo}} = |\mathcal{D}_\infty| - |\mathcal{D}_0|. \quad (905)$$

For $|\mathcal{D}_0| > 0$, the default amplification factor is

$$A_D = \frac{|\mathcal{D}_\infty|}{|\mathcal{D}_0|}. \quad (906)$$

14.15 Systemic Vulnerability

A capital-normalized systemic vulnerability measure is

$$\Gamma_i^{(t)} = \frac{S_i + \sum_j (1 - R_j) W_{ij} D_j^{(t)}}{K_i}. \quad (907)$$

It decomposes into

$$\Gamma_i^{(t)} = \underbrace{\frac{S_i}{K_i}}_{\text{direct component}} + \underbrace{\frac{\sum_j (1 - R_j) W_{ij} D_j^{(t)}}{K_i}}_{\text{network component}}. \quad (908)$$

The default threshold is

$$\Gamma_i^{(t)} \geq 1. \quad (909)$$

14.16 Monte Carlo Systemic Risk

The probabilistic and network components may be integrated by sampling

$$\mathbf{L}^{(m)} \sim F_{\mathbf{L}}, \quad m = 1, \dots, M. \quad (910)$$

For every realization,

$$\mathbf{L}^{(m)} \rightarrow \mathbf{S}^{(m)} \rightarrow \mathbf{D}^{(0,m)} \rightarrow \mathbf{D}^{(*,m)} \rightarrow \Phi^{(m)}. \quad (911)$$

The expected systemic damage can be estimated by

$$\hat{\mathbb{E}}[\Phi] = \frac{1}{M} \sum_{m=1}^M \Phi^{(m)}. \quad (912)$$

For a systemic threshold γ ,

$$\mathbb{P}(\widehat{\Phi} \geq \gamma) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}_{\{\Phi^{(m)} \geq \gamma\}}. \quad (913)$$

14.17 Common-Asset Extension

Let

$$H = [H_{ia}] \quad (914)$$

denote the institution–asset holdings matrix.

If $q_a^{(0)}$ and $q_a^{(t)}$ denote initial and current prices, respectively, the mark-to-market fire-sale loss is

$$L_i^{\text{FS},(t)} = \sum_a H_{ia} (q_a^{(0)} - q_a^{(t)}). \quad (915)$$

The extended loss equation becomes

$$\Lambda_i^{(t)} = S_i + \underbrace{\sum_j (1 - R_j) W_{ij} D_j^{(t)}}_{\text{counterparty contagion}} + \underbrace{\sum_a H_{ia} (q_a^{(0)} - q_a^{(t)})}_{\text{market contagion}}. \quad (916)$$

14.18 Eisenberg–Noe Clearing

Let $\bar{p}_i$ denote the total nominal liabilities of institution i , x_i its external assets, and Π the relative-liability matrix.

The clearing vector satisfies

$$\mathbf{p} = \bar{\mathbf{p}} \wedge (\mathbf{x} + \Pi^\top \mathbf{p}). \quad (917)$$

Componentwise,

$$p_i = \min \left\{ \bar{p}_i, x_i + \sum_j \Pi_{ji} p_j \right\}. \quad (918)$$

This replaces a purely binary default mechanism by endogenous clearing payments [7].

14.19 Multi-Layer Extension

A multi-layer financial system may be represented by

$$\mathcal{W} = \{W^{[1]}, \dots, W^{[L]}\}. \quad (919)$$

If

$$C_i^{[\ell],(t)} \quad (920)$$

denotes losses transmitted through layer ℓ , then in an additive representation

$$C_i^{(t)} = \sum_{\ell=1}^L C_i^{[\ell],(t)}. \quad (921)$$

The corresponding threshold becomes

$$S_i + \sum_{\ell=1}^L C_i^{[\ell],(t)} \geq K_i. \quad (922)$$

More general models may contain nonlinear cross-layer interactions rather than simple additive aggregation.

14.20 Complete Mathematical Architecture

The complete hierarchy can now be summarized as

$$\boxed{
\begin{aligned}
F_{\mathbf{L}} &\longrightarrow \mathbf{L}(\omega) \\
&\longrightarrow \text{VaR}_{\alpha}, \text{ES}_{\alpha} \\
&\longrightarrow \mathbf{S} \\
&\longrightarrow \mathbf{D}^{(0)} \\
&\longrightarrow \mathbf{C}^{(t)} \\
&\longrightarrow \mathbf{D}^* \\
&\longrightarrow \Phi.
\end{aligned}
} \tag{923}$$

The statistical layer is characterized by

$$\boxed{F_{\mathbf{L}}, \quad \text{VaR}_{\alpha}, \quad \text{ES}_{\alpha}.} \tag{924}$$

The balance-sheet layer is characterized by

$$\boxed{\mathbf{S}, \quad \mathbf{K}.} \tag{925}$$

The network layer is characterized by

$$\boxed{W, \quad \mathbf{R}, \quad \mathbf{D}^{(t)}.} \tag{926}$$

The systemic output is characterized by

$$\boxed{\Phi = \frac{|\mathcal{D}_{\infty}|}{n}.} \tag{927}$$

Thus the mathematical framework distinguishes three fundamentally different questions:

Tail risk:	How extreme can losses become?	(928)
Default risk:	Can capital absorb the realized losses?	
Systemic risk:	Can financial distress propagate through the network?	

14.21 Final Mathematical Synthesis

The complete mechanism can be condensed into a single expression:

$$\boxed{
\begin{array}{ccccccc}
\underbrace{F_{\mathbf{L}}}_{\text{uncertainty}} & \longrightarrow & \underbrace{\mathbf{L}(\omega)}_{\text{realized losses}} & \longrightarrow & \underbrace{\mathbf{S}}_{\text{external shocks}} & \longrightarrow & \underbrace{\mathbf{D}^{(0)}}_{\text{initial defaults}} \\
& & \longrightarrow & & \underbrace{W}_{\text{transmission structure}} & \longrightarrow & \underbrace{\mathbf{D}^*}_{\text{cascade equilibrium}} & \longrightarrow & \underbrace{\Phi}_{\text{systemic outcome}}.
\end{array}
} \tag{929}$$

The central distinction is therefore

$$\boxed{\text{risk of an extreme loss} \neq \text{risk of default} \neq \text{risk of systemic contagion}.} \tag{930}$$

The first is determined primarily by the probability distribution of losses, the second by the interaction between losses and balance-sheet capacity, and the third by the interaction between distressed institutions and the structure of the financial system.

15 Conclusions

This study developed an integrated mathematical framework connecting financial loss distributions, tail-risk measures, Monte Carlo estimation, balance-sheet vulnerability, and network-driven default cascades.

The analysis began at the level of individual financial losses. Letting L denote a loss random variable, the distribution

$$F_L(\ell) = \mathbb{P}(L \leq \ell) \quad (931)$$

provides the probabilistic foundation for the measurement of financial risk. Within this setting, Value at Risk and Expected Shortfall describe two complementary aspects of the loss distribution:

$$\boxed{\text{VaR}_\alpha(L) = F_L^{\leftarrow}(\alpha)} \quad (932)$$

identifies a tail threshold, whereas

$$\boxed{\text{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 \text{VaR}_u(L) du} \quad (933)$$

characterizes the average severity of losses beyond the selected confidence level.

This distinction is fundamental. VaR identifies where the relevant tail region begins, but does not by itself describe the magnitude of losses deeper inside that region. Expected Shortfall therefore provides additional information precisely where extreme-loss severity becomes important.

15.1 Distributional Assumptions and Tail Risk

The comparison between Gaussian and heavy-tailed distributions illustrates that volatility alone is insufficient to characterize financial risk.

Even when two distributions are standardized to have the same variance,

$$\boxed{\text{Var}(L_1) = \text{Var}(L_2)} \quad (934)$$

it does not follow that

$$\boxed{\mathbb{P}(L_1 > x) = \mathbb{P}(L_2 > x)} \quad (935)$$

for large x .

Consequently,

$$\boxed{\text{same variance} \not\Rightarrow \text{same tail risk.}} \quad (936)$$

The Gaussian–Student- t comparison makes this distinction explicit. Heavy-tailed models assign greater probability to sufficiently extreme events and may therefore generate materially different estimates of high-confidence tail risk.

The implication is broader than the comparison of two particular distributions: risk measurement depends not only on the scale of fluctuations but also on the shape of the distribution generating them.

15.2 Monte Carlo and Numerical Tail Estimation

Monte Carlo simulation provides a natural bridge between probabilistic models and numerical risk estimation.

Given simulated losses

$$L^{(1)}, \dots, L^{(N)}, \quad (937)$$

empirical quantiles provide estimates of VaR, while observations exceeding the estimated quantile provide information about Expected Shortfall.

The effective amount of information available in the tail is approximately

$$\boxed{N_{\text{tail}} \approx N(1 - \alpha)}. \quad (938)$$

This simple relation explains an important numerical difficulty. As the confidence level approaches one, an increasingly small fraction of the simulation sample contributes directly to extreme-tail estimation.

Thus,

$$\boxed{\alpha \uparrow 1 \implies N_{\text{tail}} \downarrow} \quad (939)$$

for fixed N .

Reliable estimation of extreme Expected Shortfall may therefore require large simulation samples, particularly under heavy-tailed distributions.

Monte Carlo should consequently be interpreted not as a separate notion of risk, but as a numerical methodology capable of operating across the different layers of the framework.

15.3 From Tail Risk to Default Risk

The analysis then moved from the distribution of losses to their balance-sheet consequences.

A large loss does not automatically imply default. Default depends on the relationship between the realized loss and the institution's capacity to absorb it.

For institution i , this relationship is summarized by

$$\boxed{\chi_i = \frac{S_i}{K_i}}, \quad (940)$$

where S_i denotes the external shock and K_i the capital buffer.

Under the simplified threshold mechanism,

$$\boxed{D_i^{(0)} = \mathbf{1}_{\{S_i \geq K_i\}}}. \quad (941)$$

The transition

$$\boxed{\text{loss distribution} \longrightarrow \text{realized loss} \longrightarrow \text{capital impairment} \longrightarrow \text{default}} \quad (942)$$

therefore separates tail risk from default risk.

An extreme event is a probabilistic statement about losses. Default is a balance-sheet outcome produced by the interaction between those losses and loss-absorption capacity.

15.4 From Default Risk to Systemic Risk

The introduction of financial networks changes the nature of the problem once again.

An institution's failure may affect its creditors through bilateral exposures. If W_{ij} denotes the exposure of institution i to institution j , and R_j is the corresponding recovery rate, default of j generates the loss

$$\Delta_{i \leftarrow j} = (1 - R_j)W_{ij}. \quad (943)$$

At cascade step t , institution i therefore faces total losses

$$\Lambda_i^{(t)} = S_i + \sum_{j=1}^n (1 - R_j)W_{ij}D_j^{(t)}. \quad (944)$$

The first component represents the original disturbance, while the second is generated endogenously by the financial network.

This decomposition provides one of the central distinctions of the study:

$$\underbrace{S_i}_{\text{direct loss}} + \underbrace{\sum_j (1 - R_j)W_{ij}D_j^{(t)}}_{\text{contagion loss}}. \quad (945)$$

The final state of the system is obtained when the cascade reaches a fixed point,

$$\mathbf{D}^* = \tilde{\mathcal{T}}(\mathbf{D}^*), \quad (946)$$

and the resulting systemic damage may be summarized by the final default fraction

$$\Phi = \frac{1}{n} \sum_{i=1}^n D_i^*. \quad (947)$$

Systemic risk is therefore not determined by the magnitude of the initial shock alone. It depends jointly on shock severity, institutional resilience, recovery assumptions, bilateral exposures, and network structure.

15.5 The Role of Network Amplification

A central implication of the network framework is that direct losses and final systemic losses need not coincide.

A relatively localized disturbance may remain localized when institutions possess sufficient buffers or when the exposure structure limits transmission.

The same initial disturbance may generate a larger systemic outcome when exposures create pathways through which losses propagate.

Thus,

$$\text{initial shock size} \neq \text{final systemic damage}. \quad (948)$$

The financial network acts as a transmission structure.

This observation also explains why individual risk measures alone cannot fully characterize systemic vulnerability. Two systems may contain institutions with similar stand-alone loss distributions while producing different cascade outcomes because their exposure networks differ.

Accordingly,

$$\text{individual vulnerability} \neq \text{systemic importance}. \quad (949)$$

15.6 Integration of Statistical and Network Risk

The principal contribution of the framework is the integration of the statistical and network perspectives.

The probabilistic layer generates potential losses:

$$\boxed{F_{\mathbf{L}} \longrightarrow \mathbf{L}(\omega).} \quad (950)$$

The balance-sheet layer translates those losses into institutional distress:

$$\boxed{\mathbf{L}(\omega) \longrightarrow \mathbf{S} \longrightarrow \mathbf{D}^{(0)}.} \quad (951)$$

The network layer determines whether the initial distress remains local or propagates:

$$\boxed{\mathbf{D}^{(0)} \longrightarrow \mathbf{D}^*} \quad (952)$$

Finally,

$$\boxed{\mathbf{D}^* \longrightarrow \Phi} \quad (953)$$

produces a system-level outcome.

The complete mechanism is therefore

$$\boxed{\begin{array}{ccccccc} \underbrace{F_{\mathbf{L}}}_{\text{uncertainty}} & \longrightarrow & \underbrace{\mathbf{L}(\omega)}_{\text{realized losses}} & \longrightarrow & \underbrace{\mathbf{S}}_{\text{financial shocks}} & \longrightarrow & \underbrace{\mathbf{D}^{(0)}}_{\text{initial defaults}} \\ & & & & & & \\ & \longrightarrow & \underbrace{W}_{\text{transmission structure}} & \longrightarrow & \underbrace{\mathbf{D}^*}_{\text{cascade outcome}} & \longrightarrow & \underbrace{\Phi}_{\text{systemic damage}} \end{array}} \quad (954)$$

15.7 Monte Carlo at the Systemic Level

This architecture also permits Monte Carlo simulation to be extended from tail-risk estimation to systemic-risk estimation.

For each simulation m ,

$$\boxed{\mathbf{L}^{(m)} \rightarrow \mathbf{S}^{(m)} \rightarrow \mathbf{D}^{(0,m)} \rightarrow \mathbf{D}^{(*,m)} \rightarrow \Phi^{(m)}} \quad (955)$$

Instead of obtaining only an estimate of VaR or Expected Shortfall, one therefore obtains a distribution of systemic outcomes:

$$\boxed{\Phi^{(1)}, \Phi^{(2)}, \dots, \Phi^{(M)}} \quad (956)$$

This makes it possible to estimate quantities such as

$$\boxed{\mathbb{E}[\Phi]} \quad (957)$$

and

$$\boxed{\mathbb{P}(\Phi \geq \gamma)}, \quad (958)$$

where γ represents a selected systemic-damage threshold.

The same simulation philosophy that estimates the tail of a loss distribution can therefore be extended to study the distribution of network-wide outcomes.

15.8 Scope and Future Development

The present framework deliberately isolates a specific contagion mechanism: losses transmitted through counterparty exposures following default.

This simplification makes the propagation mechanism transparent, but a financial system may contain additional channels of amplification.

Natural extensions include heterogeneous balance sheets, multiple and correlated shocks, endogenous recovery rates, dynamic exposure networks, common asset holdings, and fire-sale effects.

The Eisenberg–Noe clearing framework provides a natural extension from binary default states to endogenous payment vectors, while Cifuentes–Ferrucci–Shin introduces an important feedback between balance-sheet distress, asset liquidation, market prices, and mark-to-market losses.

These mechanisms suggest a broader multi-layer representation in which credit, market, liquidity, funding, and common-asset channels may interact.

Schematically,

$$\begin{array}{l}
 \text{Tail Risk} \longrightarrow \text{Balance-Sheet Stress} \\
 \longrightarrow \text{Default Risk} \\
 \longrightarrow \text{Counterparty Contagion} \\
 \longrightarrow \text{Market-Mediated Contagion} \\
 \longrightarrow \text{Multi-Layer Systemic Risk.}
 \end{array} \tag{959}$$

15.9 Final Conclusion

The main conclusion of this study is that financial risk changes meaning as the scale of analysis changes.

At the statistical level, the problem concerns the distribution and severity of losses.

At the institutional level, the problem concerns whether those losses can be absorbed by available financial buffers.

At the network level, the problem concerns whether distress at one set of institutions generates additional losses elsewhere.

These three levels are related, but they are not interchangeable:

$$\text{Tail Risk} \neq \text{Default Risk} \neq \text{Systemic Risk.} \tag{960}$$

Their relationship can instead be understood as a sequence:

$$\text{Extreme Losses} \longrightarrow \text{Capital Erosion} \longrightarrow \text{Defaults} \longrightarrow \text{Contagion} \longrightarrow \text{Systemic Outcomes.} \tag{961}$$

Consequently, a complete analysis of financial stability requires more than estimating the probability of an extreme loss. It also requires understanding the capacity of institutions to absorb that loss and the financial architecture through which distress can propagate.

The resulting perspective connects statistical risk measurement with network-based systemic-risk analysis and provides a natural foundation for more general models of financial contagion mechanisms.

Recent extensions of network-based contagion analysis have examined the interaction between market-derived shocks and default propagation under heterogeneous financial conditions [3, 4].

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